VENTURE PACKAGE

Backsight

Job tracking built for land-surveying firms — with a prior-work archive that remembers every parcel the firm has ever surveyed.

Complete SaaS venture package: research · tournament · adversarial review · business plan · local MVP
Produced by a multi-agent research & build process · 2026-07-10
Repository: github.com/ccas77/claude-code · branch claude/saas-product-build-yzyiib · `backsight-venture/`

Contents

Backsight — Venture Package

> **Job tracking built for land-surveying firms** — a pipeline that speaks surveying (fieldwork → drafting → licensed review → delivery), a client status link that kills "where's my survey?" calls, and a prior-work archive searchable by location, so every new request instantly shows what the firm already knows about that ground.

This folder is a complete, self-contained SaaS venture package: evidence-based market research, an adversarially tested business case, a full go-to-market plan, and a runnable local MVP. It was produced end-to-end by a multi-agent research and build process on 2026-07-10; every factual claim traces to a cited source, and every major decision is logged.

Executive summary

The problem. Small US land-surveying firms (1–5 field crews, 3–25 staff, $300K–$5M revenue) run 20–80 concurrent jobs across a whiteboard, a spreadsheet, and the owner's memory. Two costs follow: (1) *status chaos* — title companies and builders call the owner for updates, and jobs silently stall between field, drafting, and licensed review; (2) *prior-work amnesia* — the firm's most valuable asset, what it already knows about ground it surveyed before (control, prior boundary resolutions, plats), lives in the owner's head and paper files, so quoting ignores it and retirements erase it.

The product. Backsight is a vertical SaaS with three parts: a surveying-native job pipeline, a public tokenized client status page (the "pizza tracker" a title company refreshes instead of calling), and the wedge — **Prior-Work Radar**: import the firm's historical job list and every job is indexed spatially (address geocoding first; Section-Township-Range parsing where available). Every new request instantly surfaces prior jobs on or near that parcel: *"You have 3 prior jobs in this section — quote in minutes, win it on price, deliver at half the field time."*

Why this won. 8 parallel research scouts surfaced 34 evidenced problems across 8 hunting grounds; 12 became sharpened concepts scored by a 3-persona judge panel; the top candidates then faced dual adversarial skeptics doing fresh web research. The skeptics **killed 11 of 12 concepts** on verified structural defects (platform-native substitutes, free-tier floors from funded competitors, technically unsound wedges, evidence tracing to SEO farms, regulated deliverables). Backsight was the only concept whose market skeptic found **zero fatal flaws** — its objections were engineering-fixable and are incorporated into the PRD. The full autopsy record is in `research/ADVERSARIAL_REVIEW.md`.

The market, honestly. ~7,000 US surveying/mapping establishments (NAICS 541370, skeptic-corrected from an initial 25,000 estimate), mostly small firms; competitors are paid micro-vendors (Qfactor, SurveyOps, KudurruStone, Cyanic Job Book) with dated UX, per-user pricing, and no free tier — a fragmented category with no dominant, well-loved incumbent and no VC free-floor. At 5% penetration and ~$140/mo flat pricing, Backsight is a ~$590K-ARR bootstrapped business with adjacent-vertical expansion paths (septic design, forestry, geotech). This is deliberately an indie-scale, evidence-defensible opportunity rather than a venture-scale story with fatal flaws.

Pricing. Flat per-firm (anti add-on-creep, anti per-user): $79/mo (≤5 users), $149/mo (≤15), $249/mo (unlimited). Rationale and competitor anchoring: `business/PRICING_STRATEGY.md`.

Distribution. State survey societies and NSPS chapters, r/Surveying, conference booths measured in hundreds of dollars, a free "Prior-Work Audit" lead magnet, and a built-in referral loop: every client status link puts Backsight in front of title companies and builders who deal with many surveying firms.

Status. Local MVP in `product/app/` (Next.js + SQLite + Leaflet; `npm install && npm run seed && npm run dev`). Real: pipeline, radar with license-clean PLSS (Public Land Survey System) parser, client status pages, dashboards, seeded realistic data. Mocked and documented: auth, email (outbox), live geocoding, billing.

Folder guide

Path	Contents
`research/MARKET_RESEARCH_BRIEF.md`	Research method, all 8 hunting grounds, 34 problems, market context, citations
`research/PROBLEM_VALIDATION.md`	The surveyor pain: evidence chain, skeptic corrections, open interview questions
`research/COMPETITOR_ANALYSIS.md`	Verified competitor table incl. all 6 vertical micro-vendors
`research/GAP_ANALYSIS.md`	The residual gap Backsight occupies; claims labeled VERIFIED vs HYPOTHESIS
`research/TOURNAMENT.md`	12-candidate tournament, judge panel, weighted scores, full ranking
`research/ADVERSARIAL_REVIEW.md`	All 3 skeptic rounds, fatal flaws, evidence checks, the 7-pattern kill taxonomy

Path	Contents
`research/*.json`	Raw structured outputs of every research/judging/skeptic agent run
`business/FINAL_CONCEPT.md`	The selected concept with post-skeptic fixes (authoritative)
`business/CUSTOMER_PERSONA.md`	Buyer + daily-user personas with evidence traceability
`business/PRICING_STRATEGY.md`, `GO_TO_MARKET.md`, `CUSTOMER_ACQUISITION.md`, `SALES_ONBOARDING_FLOW.md`, `METRICS_SUCCESS_CRITERIA.md`, `RISK_REGISTER.md`, `LANDING_PAGE_COPY.md`	The full commercial plan
`product/PRD.md`, `MVP_SCOPE.md`, `TECH_ARCHITECTURE.md`	Product requirements, scope boundaries, architecture
`product/CLAUDE_CODE_IMPLEMENTATION_PLAN.md`	How a non-specialist continues building with Claude Code
`product/MVP_BUILD_SPEC.md`	The spec the MVP was built against
`product/app/`	**The runnable MVP** — see its README for the 3-command quickstart and demo script
`decisions/DECISION_LOG.md`	Every executive decision (D-001…) with reasoning
`decisions/FINAL_COMPLETENESS_REVIEW.md`	Independent 24-item audit against the Definition of Done (PASS) + fixes applied

Run the demo

```
cd product/app
npm install
npm run seed
npm run dev   # open http://localhost:3000
```

The landing page is at `/`, the product at `/app`, and a public client status page at `/status/<token>` (links are on each job). The seeded firm is "Whitfield Land Surveying" of Fort Collins, CO, with ~85 jobs from 2019–2026. Follow the 7-step demo script in `product/app/README.md` — step 5 (Prior-Work Radar search) is the aha moment.

What a stranger should take away

1. The pain is real and evidenced — and the parts of the original evidence that didn't survive verification are flagged, not hidden.
2. The category is buyable (firms already pay micro-vendors) but not owned; the wedge (prior-work monetization) is technically sound and demonstrated in the MVP.
3. The business is continuable by a non-specialist: no gated APIs, no licensing, no proprietary data; the implementation plan gives milestone-by-milestone Claude Code prompts.
4. The biggest risks are honest ones — TAM ceiling, incumbent response, archive-import quality — and each has a mitigation and an early-warning signal in the risk register.

Market Research Brief — Backsight

Venture: Backsight — job tracking + prior-work intelligence for small US land-surveying firms **Date:** 2026-07-10 **Status:** Final. Reflects the full research record: 8-scout evidence scan → 12-candidate tournament → three adversarial skeptic rounds → winner selection.

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1. Method

1.1 Evidence-first, ideas last

Before a single product idea was generated, eight research scouts ran **in parallel over eight mutually exclusive "hunting grounds"** chosen to span the three classic sources of software opportunity: underserved niches, newly created pain (regulation, AI), and incumbent failure (see `decisions/DECISION_LOG.md`, D-002):

#	Hunting ground	Thesis
1	SMB trades	Owner-operator field businesses squeezed between spreadsheets and enterprise suites
2	Niche professions	Licensed/credentialed small firms ignored by horizontal SaaS
3	New 2024–26 regulations	Compliance obligations created by law faster than tooling
4	AI-era ops pain	New operational problems created by AI adoption itself
5	Creators / solopreneurs	One-person media businesses running revenue ops in spreadsheets
6	E-commerce sellers	Margin leaks and compliance burdens on small DTC brands
7	Devtools / B2B ops	Engineering-adjacent operational pain in small software companies
8	Hated incumbents	Categories where the dominant vendor is actively resented

The scan returned **34 evidenced problems** (`research/raw-scan-full.json`).

1.2 Discipline rules enforced at research time

- **Citation discipline (D-004):** every problem carries structured evidence objects (`claim` + `source_url`); scouts were instructed never to invent sources. Claims were later spot-checked by adversarial skeptics with fresh web access, and several were flagged or refuted (Section 5).
- **Buildability filter (D-003):** no licensed professions, no gated APIs, no proprietary data, no hardware, no fragile scraping as core value — enforced as a hard filter on what scouts could return, then independently re-verified by implementability skeptics.

1.3 Tournament and adversarial review

Twelve of the 34 problems were shortlisted (D-006), merging two independent-scout convergences (COI tracking; per-client AI cost attribution). Three persona judges with deliberately conflicting priors (seed VC, bootstrapped indie hacker, pragmatic CTO) scored all 12 on eight criteria, weighted toward buildability and willingness to pay (D-007). The top four then faced **two independent skeptics each** (implementability and market), doing fresh web research including spot-checks of the original citations. When all four died (D-008), ranks 5–8 got the identical treatment (D-009). When those died too, the seven verified kill patterns were codified into a **kill taxonomy** and a final revival round ran four concepts engineered against it — from which Backsight emerged as the only concept of 12 whose market skeptic found **zero fatal flaws** (D-010).

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2. The landscape: 34 problems across 8 grounds

A compressed tour of what the scan found, with representative citations. (Full evidence: `research/raw-scan-full.json`.)

SMB trades (4 problems)

Maintenance-agreement tracking for small HVAC/plumbing shops (ServiceTitan at $250–$500/tech/mo with $5K–$50K implementation is the only "proper" fix — https://fieldcamp.ai/reviews/servicetitan/); COI collection for small GCs; backflow test-report filing (a 15-minute test drags ~30 minutes of per-city paperwork — https://flowcert.co/blog/how-to-become-certified-backflow-tester); quote-to-BEO workflow for independent caterers (~13,263 US catering businesses, no player over 5% share — https://www.ibisworld.com/united-states/industry/caterers/1682/).

Niche professions (4 problems)

COI tracking for small property managers (myCOI starts ~$500+/mo, designed for 200+ certificates — https://trackmyvendor.com/mycoi-alternatives); carrier commission reconciliation for small insurance agencies (claimed $10K–$16K/yr leakage — https://unlockedcrm.ai/blog/insurance-commission-reconciliation-guide — later flagged as vendor marketing); interpreter-agency back-office; and **job-status/workflow tracking for small land surveying firms** — the eventual winner's root problem (see Section 4).

New regulations (4 problems)

EAA/WCAG monitoring for SMB e-commerce (FTC hit accessiBe with a $1M penalty in April 2025 — verified by adversarial review, https://www.ftc.gov/news-events/news/press-releases/2025/04/ftc-approves-final-order-requiring-accessibe-pay-1-million); EU STR registration tracking (Regulation (EU) 2024/1028 applicable 20 May 2026 — https://eur-lex.europa.eu/EN/legal-content/summary/online-short-term-accommodation-rental-services-data-collection-and-sharing.html); US auto-renewal-law compliance (HelloFresh $7.5M consent judgment — verified by adversarial review, https://da.lacounty.gov/media/news/hellofresh-pay-75-million-deceptive-subscription-practices-consumer-protection-lawsuit); GPSR documentation for handmade sellers (https://help.etsy.com/hc/en-us/articles/28211364687383-What-is-the-General-Product-Safety-Regulation-GPSR).

AI-era ops (5 problems)

EU AI Act Article 50 disclosure workflow (obligations from 2 August 2026 — https://artificialintelligenceact.eu/transparency-rules-article-50/); support-AI answer QA (Cursor's bot invented a nonexistent policy, causing viral backlash — https://fortune.com/article/customer-support-ai-cursor-went-rogue/); knowledge-base freshness auditing; shadow-AI governance for SMBs (https://www.ibm.com/think/topics/shadow-ai); per-client LLM cost attribution (documented runaway bills up to $1.3M in 30 days — https://www.kucoin.com/news/flash/ai-token-bills-explode-500m-1-3m-and-18k-in-one-night).

Creators / solopreneurs (4 problems)

Brand-deal receivables chasing (30–90-day payment terms, still paid late — https://digiday.com/marketing/in-a-booming-influencer-economy-creators-seek-standardization-for-payment-terms/); sponsorship ops below Sponsy/Passionfroot pricing (https://www.sponsorflo.ai/blog/podcast-sponsorship-management-guide); usage-rights expiry tracking; affiliate link-rot monitoring for YouTubers (https://www.youfiliate.com/blog/broken-affiliate-links-costing-you-money).

E-commerce sellers (4 problems)

Landed-cost/HTS calculators after the end of de minimis (https://www.unicargo.com/de-minimis-ended-2026-tariffs-guide-amazon-fba/); 3PL invoice auditing (dim-weight overcharges framed as the most common billing error — https://www.seleryfulfillment.com/the-hidden-3pl-fees/); GPSR listing compliance; flat-fee chargeback evidence automation (chargebacks projected at $33.79B in 2025 — https://www.chargeback.io/blog/chargeback-statistics).

Devtools / B2B ops (4 problems)

API breaking-change radar; security-questionnaire answering for pre-SOC2 vendors (recurring founder complaint — https://news.ycombinator.com/item?id=36488436); EU Cyber Resilience Act readiness (reporting obligations from 11 September 2026 — https://digital-strategy.ec.europa.eu/en/policies/cyber-resilience-act); per-customer AI cost metering (top 10% of users drive ~60% of AI costs — https://www.drivetrain.ai/post/unit-economics-of-ai-saas-companies-cfo-guide-for-managing-token-based-costs-and-margins).

Hated incumbents (5 problems)

Wild Apricot refugees (20% Payment System Servicing Fee on orgs using outside payment processors — verified by adversarial review against Wild Apricot's own help center, https://support.wildapricot.com/hc/en-us/articles/24303136407821-Payment-System-Servicing-Fee); youth-sports orgs stranded by the SportsEngine→PlayMetrics consolidation (https://home.playmetrics.com/blog/playmetrics-acquires-sportsengine-versant); HOA treasurer tooling; pet-boarding software fleeing Gingr's rollup; day-camp registration priced out by CampMinder ($2,500–$10,000+ annual licenses — https://www.campnetwork.com/camp-registration-software-comparison).

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3. The tournament: how 11 concepts died and 1 survived

3.1 Judge ranking (weighted average of 3 persona judges, 12 concepts)

Rank	Concept	Score	Outcome
1	Chargeback evidence automation	72.2	Killed (round 1)
2	COI tracking	68.5	Killed (round 1)
3	3PL invoice audit	68.0	Killed (round 1)
4	Commission reconciliation	68.0	Killed (round 1)
5	EAA accessibility monitor	66.5	Killed (round 2)
6	HVAC agreements	65.0	Killed (round 2)
7	Membership refuge (Wild Apricot)	64.2	Killed (round 2)
8	Auto-renewal compliance	62.6	Killed (round 2)
9	AI cost/margin attribution	60.4	Not escalated
10	Sponsorship ops	60.3	Not escalated
11	Backflow compliance	55.8	Not escalated
12	Affiliate link-rot	55.3	Not escalated

3.2 Representative kill facts (all found by skeptics doing fresh web research)

- **Chargeback:** Stripe launched Smart Disputes (Nov–Dec 2025, enabled by default) doing the concept's core loop natively; ChargePay already sells flat-fee AI chargeback automation from $19.99/mo. The concept's foundational claim ("Stripe native forms are blank text boxes") was simply no longer true.
- **COI tracking:** TrustLayer Starter and bcs now give away free AI-parsing COI tiers to exactly the sub-200-certificate segment the concept claimed incumbents "structurally cannot serve."
- **3PL audit:** technically unsound wedge — billable dim weight comes from the 3PL's cubiscanned carton, not the merchant's SKU dimensions, so "recompute the correct charge" systematically produces false positives.
- **Commission recon:** zero-template AI parsing already commoditized (Fintary, unLocked with 332 carrier feeds); every one of the four cited evidence sources was content marketing by a vendor selling the same solution.
- **EAA monitor:** AudioEye sells self-serve continuous monitoring with statement generation at $49/mo — the exact price point the concept claimed honest players ignored.
- **HVAC agreements:** the standalone architecture contradicted its own wedge; Jobber Core already auto-schedules recurring visits (verified by adversarial review, https://help.getjobber.com/hc/en-us/articles/360048155913-The-Core-Plan); the anchor statistic traced to a programmatic-SEO content farm.
- **Membership refuge:** MemberDay is a near-clone at $29/mo; Zeffy is free; the "Wild Apricot alternative" SERP is saturated by funded players.
- **Auto-renewal compliance:** the recurring deliverable is legal judgment applied to a merchant's specific flows (unauthorized-practice-of-law exposure), and Recharge already sends the mandated reminders natively. The underlying legal facts were confirmed (California AB 2863 effective July 1, 2025 — verified by adversarial review, https://www.cooley.com/news/insight/2025/2025-06-04-california-automatic-renewal-law-amendments-take-effect-on-july-1-2025; FTC rulemaking restarted January 2026 — https://www.gibsondunn.com/ftc-restarts-negative-option-rulemaking-after-eighth-circuit-vacatur-enforcement-under-rosca-continues/) — the pain was real; the product was undeliverable by a non-law-firm.

3.3 The kill taxonomy and the revival round

Eight autopsies yielded seven verified kill patterns (D-009): platform-native substitutes; free-tier floors from funded players; technically unsound wedges; vendor-content-marketing evidence; regulated deliverables; saturated panic-SEO distribution; structural churn. Round 3 ran four concepts engineered against that taxonomy:

Round-3 concept	Verdict	Killing fact
Registry Radar (EU STR compliance)	Killed	Conforme.info already ships the identical "regulatory system of record" positioning (verified by adversarial review, https://conforme.info/); Spain's Supreme Court annulled the national registry, judgment 620/2026 of 21 May 2026 (https://skift.com/2026/05/22/spains-top-court-voids-national-short-term-rental-registry/); the launch markets' registrations mostly have **no renewal deadlines** — the retention engine tracked obligations that don't exist
TerpDesk (interpreter agencies)	Killed — the tournament's only outright "kill" verdict	The "empty $50–200/mo gap" is occupied by at least five purpose-built incumbents, e.g. Eclipse Scheduling from ~$125/mo all-inclusive (verified by adversarial review, https://interpreterscheduling.com/pricing/); HIPAA business-associate status for medical interpreting was never addressed; the pricing evidence traced to AI-generated listicle farms (Gitnux/ZipDo)
PlanPatrol (HVAC radar pivot)	Killed	Weekly manual CSV upload is a structural churn engine; Jobber/Housecall Pro already sync with QuickBooks Online, dissolving the "two systems no vendor will reconcile" premise
Backsight (surveyors)	**Winner — survives with fixes**	Market skeptic: "seriously weakened" but **zero fatal flaws** — the only concept of 12 with none; implementability skeptic: "survives with fixes," all engineering-addressable

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4. Market context: small US land-surveying firms

The winning segment. All figures below are either skeptic-verified or explicitly labeled as estimates.

4.1 Establishment counts (skeptic-corrected)

- The original scout claimed "~25,000+ small surveying establishments." **Refuted.** Census County Business Patterns data for NAICS 541370 (Surveying and Mapping, except Geophysical) shows **~7,000 firms / ~7,382 establishments** (2020 CBP), with a second directory source showing ~6,191 active companies (verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping). The 25,000 figure appears to conflate establishments with licensed individual surveyors.
- Serviceable market after segmenting to 1–5-crew firms in PLSS states (the wedge does not cover Texas or the colonial metes-and-bounds states — verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system): plausibly **2,000–4,000 firms** (skeptic estimate).
- Honest revenue ceiling (estimate, per `business/FINAL_CONCEPT.md`): 5% penetration of ~7,000 establishments at ~~$140/mo average ≈ **~~$590K ARR** — a strong bootstrapped outcome, honestly not a venture-scale one.

4.2 Firm profile and tech spend

- Target firm: 1–5 field crews, 3–25 staff, $300K–$5M revenue (estimate from concept research), running 20–80 concurrent jobs (boundary, ALTA, topo) at $500–$5,000 each.
- These are licensed professional firms that already spend heavily on technical tooling — the scout noted GNSS receivers run $10–30K (scout estimate; adjacent software price anchors cited were Jobber at ~$40–200/mo and BQE Core at ~$25–50/user/mo) — while business-operations software for the vertical barely exists (scout framing via https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/, itself flagged as a vendor listicle; see 5.2).

4.3 Buyer behavior (adversarial-review findings)

The market skeptic's fresh research materially reshaped the demand picture:

- **The vertical is occupied but fragmented.** At least six vertical practice-management products exist: Qfactor (founded 2016), KudurruStone, Cyanic Job Book, Info-Retriever by AGT, CQ, and SurveyOps (verified by adversarial review; SurveyOps confirmed at https://survey-ops.com/land-surveying-job-management-software). "No dominant player" is true; "no vertical solution" is false. All incumbents have near-zero review footprints — which is simultaneously evidence of a real gap and of a hard-to-reach buyer.
- **Whiteboard inertia is real and documented.** On the RPLS.com practice-management forum threads found by the skeptic, one surveyor abandoned KudurruStone (a purpose-built vertical tool) as "too slow," while others run their firms on ClickUp at ~$3/user/mo. A decade-old vertical vendor (Qfactor) remaining obscure suggests slow sales cycles and a buyer who satisfices on cheap generic tools.
- **Pricing reality check.** KudurruStone charges per-user by role (a 10-person firm ≈ $100/mo per the skeptic's review of its site) — below Backsight's original anchoring against BQE Core. This drove the post-skeptic flat-pricing reposition ($79/$149/$249 per firm).
- **The "free S-T-R lookup" lead magnet is a saturated space:** randymajors.org, Earth Point (verified by adversarial review, https://www.earthpoint.us/Townships.aspx), MapScaping, and BLM's own viewers are the tools surveyors already bookmark. Distribution must run instead through state societies under NSPS, r/Surveying, RPLS forums, surveying podcasts, and GNSS-dealer partnerships.

4.4 Implementability facts (skeptic-verified)

- **BLM PLSS CadNSDI data is genuinely public-domain and bulk-downloadable** — township/range/section polygons, no API key, no license fee (verified by adversarial review, https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons).
- **QuickBooks Online invoice push is a public OAuth API** with a fast, form-based production gate (verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ).
- **The only mature open-source legal-description parser, pyTRS, prohibits all commercial use** under a Modified Academic Public License (verified by adversarial review, https://github.com/JamesPImes/pyTRS). Backsight ships its own license-clean parser.
- Township America operates a free Township/Range lookup and a commercial PLSS geocoding API — proof the technical approach works, and a buy-vs-build option (verified by adversarial review, https://townshipamerica.com/).

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5. What survived adversarial review, and what died

5.1 Survived

1. **The pain itself.** Whiteboard-plus-email job tracking, status-call interruptions, delayed invoices, and prior-work amnesia are corroborated by the existence of six independent vertical vendors betting on the same problem — the strongest form of demand evidence the review produced.
2. **The buildability thesis.** Every required dataset and API is public and credential-free (BLM CadNSDI, US Census Geocoder, QBO OAuth); nothing in the core loop depends on scraping; no regulatory regime touches a job-tracking vendor (skeptic verified by absence, moderate confidence).
3. **Market fragmentation with no free-tier floor.** No VC-funded free tier exists in this vertical; competitors are paid micro-vendors — the exact opposite of the pattern that killed COI tracking and chargeback automation.
4. **The residual wedge.** No incumbent leads with automated parsing/spatial indexing of a firm's messy **historical** job spreadsheet — they map jobs going forward or require manual entry. Prior-work monetization at import time, plus flat non-per-seat pricing, is the defensible remainder.

5.2 Died or was corrected

1. **TAM: ~25,000 → ~7,000 establishments** (Section 4.1). All downstream economics were recomputed on the honest number.
2. **"No vertical solution exists" → false.** Six vertical products named; competitive framing rebuilt to acknowledge all of them (`business/FINAL_CONCEPT.md`, fix #4).
3. **Wedge uniqueness overclaimed.** Qfactor's Map View shows past-project locations; Cyanic Job Book offers legal-address search; SurveyOps markets S-T-R job organization on the very page the scout cited as gap evidence. Differentiation narrowed to historical-archive ingestion depth + client status links + flat pricing.
4. **All three original scout citations flagged as vendor content marketing** (CQ's blog, SurveyOps' landing page, QuoteIQ's listicle). They establish that vendors believe in the pain, not independent demand volume — a gap that only customer interviews can close (see `research/PROBLEM_VALIDATION.md`).
5. **pyTRS assumed usable → refuted** (license prohibits commercial use); build timeline re-planned from 6–8 weeks to 12–16 weeks for a sellable v1.
6. **Lead-magnet plan (free S-T-R map) dropped** — slot occupied by randymajors.org / Earth Point / MapScaping.
7. **Principal-meridian ambiguity and address-only spreadsheets** — two silent-failure modes the concept had not modeled — became mandatory product fixes (state→meridian resolution table; address-first geocoding via the free US Census Geocoder).

5.3 Why this winner, despite modest economics

Backsight's surviving risks are **execution risks** (micro-vendor competition, TAM ceiling, whiteboard inertia) rather than **structural falsehoods** (free-tier floors, platform-native substitutes, evidence built on SEO farms) — the trade the selection process was explicitly designed to make (D-010). It is also the strongest fit for the buildability constraint: the wedge (parse → geocode → spatial join → radar UI) is deterministic, demonstrable offline with sample data, and involves zero gated access, zero licensing exposure, and zero regulated deliverables.

Companion documents: `research/PROBLEM_VALIDATION.md` (problem deep-dive and interview plan), `business/CUSTOMER_PERSONA.md` (buyer and user personas), `business/FINAL_CONCEPT.md` (authoritative product direction), `decisions/DECISION_LOG.md` (full decision record).

Problem Validation — Small Land-Surveying Firms

Venture: Backsight **Date:** 2026-07-10 **Purpose:** State exactly what we know about the surveyor problem, how we know it, what adversarial review corrected, and what only customer interviews can settle.

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1. The painful workflow

A small US surveying firm (1–5 field crews, 3–25 staff) runs a high volume of small jobs — boundary surveys, ALTA surveys, topos at $500–$5,000 each, 20–80 concurrent — through a pipeline that crosses field and office: request → quote → scheduled → fieldwork → drafting → licensed review → delivered → invoiced.

Two failure modes, per the original scout research (`research/raw-scan-full.json` , problem "Job-status and workflow tracking for small land surveying firms"):

1. **Job-status chaos.** Status lives on whiteboards, in email folders, and in verbal updates from the field. Clients — title companies, builders, homeowners — call the owner asking "where's my survey?"; drafters sit idle waiting on field notes; finished jobs go uninvoiced for weeks (source: https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/ — **flagged by adversarial review as vendor content marketing**; the pain is plausible but this citation is not independent evidence).
2. **Prior-work amnesia.** The firm's most valuable asset is what it already knows about ground it has surveyed before — control points, prior boundary resolutions, plats. That knowledge lives in the owner's head and a shelf of file folders. Firms need to search past jobs by legal description (Section-Township-Range or subdivision), which generic tools cannot do (source: https://survey-ops.com/land-surveying-job-management-software — **flagged by adversarial review**: it is a competitor's landing page; it validates that a vendor ships S-T-R search, which cuts against uniqueness while confirming the need is real).

The structural context: the surveying software market is dominated by technical/CAD/GNSS vendors (Trimble, Leica, Carlson) with no interest in small-firm practice management, so firms cobble together spreadsheets, monday.com boards, or field-service apps built for landscapers (source: https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/ — **flagged**: a generic-SaaS vendor's listicle ranking itself; the underlying "no dominant player" claim held up under review, but "no vertical solution" did not).

2. Who feels it

- **The owner / licensed surveyor** (the buyer): quotes jobs, dispatches crews, personally fields status calls, personally remembers "we shot that section in 2019." Every hour of interruption is unbillable.
- **The office manager / drafter** (the daily user): chases field notes, shepherds jobs between stages, prepares invoices, answers the phone when the owner is in the field.
- **Secondary sufferers:** title companies and builders on deadline who cannot see status without calling; field crews re-shooting control that the firm already established on adjacent ground.

3. Frequency and cost-of-problem reasoning

All items below are **reasoned estimates from evidenced inputs**, labeled as such — no independent study of surveying-firm operational losses was found by any of the three research passes, which is precisely why interviews (Section 6) are the next gate.

Cost driver	Reasoning	Status
Status-call interruptions	20–80 concurrent jobs × client mix of title companies/builders on closing deadlines implies daily inbound status queries handled personally by the owner or office manager	Estimate; frequency must be measured in interviews
Delayed invoicing	"Finished jobs go uninvoiced for weeks" (scout evidence via CQ blog, flagged vendor source). On jobs of $500–$5,000, a handful of delivered-but-uninvoiced jobs is thousands of dollars of float at any moment	Directionally plausible; magnitude unverified
Lost repeat-work advantage	A request on ground the firm previously surveyed can be quoted faster and delivered at lower field cost — but only if prior work is found. Missed prior work = quoting like a stranger and either losing the bid or eating redundant fieldwork	Core wedge hypothesis; win-rate/field-time deltas are estimates until interviews
Software budget exists	Firms already pay heavily for technical tooling (scout estimate: GNSS receivers $10–30K; adjacent anchors Jobber ~$40–200/mo, BQE Core ~$25–50/user/mo); six vertical PM vendors sustain paid products in the niche	Vendor existence verified by adversarial review; individual firms' actual PM software spend unverified

Honest counter-evidence, surfaced by the market skeptic and retained deliberately: surveyors on the RPLS.com forum report running firms on ClickUp at ~$3/user/mo, and one abandoned the purpose-built KudurruStone as "too slow." The willingness to pay $79–249/mo flat is therefore **assumed, not demonstrated** — it must clear the interview gate.

4. The evidence chain

1. **Scout evidence (July 2026):** three citations establishing the pain narrative and the vertical's thin tooling —

    - https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/
    - https://survey-ops.com/land-surveying-job-management-software
    - https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/ All three were later **flagged as vendor content marketing** by the round-3 market skeptic. They prove vendors are betting on this pain; they do not prove demand volume.
2. **Adversarial market review (round 3):** fresh web research found six vertical competitors (Qfactor, KudurruStone, Cyanic Job Book, Info-Retriever, CQ, SurveyOps), corrected the TAM (below), mapped the real price floor, and identified the RPLS forum adoption evidence. Verdict: *seriously weakened, zero fatal flaws* — the only concept of 12 without one.
3. **Adversarial implementability review (round 3):** verified the wedge's data supply chain end-to-end — BLM PLSS CadNSDI is public-domain bulk data (verified by adversarial review, https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons); QuickBooks Online is a public OAuth API with a ~30-minute production assessment (verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ). Verdict: *survives with fixes*.

5. Corrections applied by skeptics

Every correction below is incorporated into `business/FINAL_CONCEPT.md` and the MVP spec.

#	Original claim	Correction	Source
1	"~25,000+ small surveying establishments in the US"	**~7,000 firms / ~7,382 establishments** in NAICS 541370 (2020 Census County Business Patterns); a second source shows ~6,191 active companies. The 25,000 figure conflated establishments with licensed individuals. Serviceable segment (3–25 staff, PLSS states): skeptic estimate 2,000–4,000 firms	verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping
2	"No dominant vertical solution exists" (read as: no vertical solution)	Six vertical products exist; the market is fragmented with no winner. Positioning rebuilt to name all competitors	adversarial review round 3 (Qfactor, KudurruStone, Cyanic Job Book, Info-Retriever, CQ, SurveyOps)
3	Wedge (spatial prior-work search) framed as unique	Qfactor's Map View shows past project locations; Cyanic offers legal-address search; SurveyOps markets S-T-R organization. Residual wedge: automated parsing/spatial indexing of **imported historical spreadsheets**, which none advertise	adversarial review round 3, incl. https://survey-ops.com/land-surveying-job-management-software
4	Legal-description parsing is a drop-in solved problem	**Refuted:** pyTRS, the only mature OSS parser, prohibits all commercial use; Backsight must ship an original, license-clean parser with a human-confirmation UI	verified by adversarial review, https://github.com/JamesPImes/pyTRS
5	(Unmodeled) principal-meridian ambiguity	"T2N R3W Sec 14" repeats across principal meridians; silent wrong-section matches would be the worst possible failure for a trust product. Fix: state→default-meridian table, explicit ambiguity flags, geocoded address preferred	adversarial review round 3
6	Demo assumes spreadsheets contain legal descriptions	Many firm spreadsheets carry only client + street address. Fix: address-first ingestion via the free US Census Geocoder; S-T-R parsing is enrichment, not a requirement	adversarial review round 3
7	PLSS wedge coverage	20 states are outside PLSS — the original 13 colonies plus TX, HI, KY, TN, VT, ME, WV; Texas uses its own GLO abstract/survey system. Initial GTM targets PLSS states (Midwest/Mountain West)	verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system
8	Pricing anchored against BQE Core	KudurruStone's per-user pricing ≈ $100/mo for a 10-person firm; ClickUp satisfices at ~$3/user. Repriced flat per-firm ($79/$149/$249) so the math beats per-seat at team size	adversarial review round 3
9	Free S-T-R lookup map as lead magnet	Slot occupied by randymajors.org, Earth Point, MapScaping, BLM viewers. Dropped in favor of an import-your-history onboarding funnel	verified by adversarial review, https://www.earthpoint.us/Townships.aspx
10	6–8 week build	Re-planned to 12–16 weeks for a sellable v1 (parser + meridian disambiguation + 30-state CadNSDI ingest + import UX + QBO OAuth)	adversarial review round 3

6. Open questions — closeable only by customer interviews

The evidence base establishes a plausible, buildable, competitively survivable thesis. It does **not** establish demand volume, willingness to pay, or data readiness. Target: 10–15 recorded interviews with owners of 3–25-staff firms in PLSS states before writing further code beyond the demo.

Interview questions:

1. Walk me through the last job that came in — from first phone call to invoice. What tools, boards, or paper touched it at each stage?
2. How many jobs are active right now, and how do you know, today, which ones are stuck and where?
3. How many "where's my survey?" calls or emails does your office handle in a typical week, and who answers them? What does each one interrupt?
4. When a request comes in for a parcel you might have surveyed before, how do you check? How long does it take, and when did you last discover prior work *after* quoting or fieldwork?

5. What does your historical job record physically look like — spreadsheet, folders, ledger? Do the rows contain legal descriptions, street addresses, both, or neither?
6. Which practice-management tools have you tried or evaluated — SurveyOps, Qfactor, KudurruStone, Cyanic Job Book, ClickUp, Jobber, monday? Why did you stop, stay, or never start?
7. What do you currently pay per month for office/business software versus field/CAD software, and who signs off on a new ~$150/mo subscription?
8. How often do delivered jobs sit uninvoiced for more than a week, and roughly how much delivered-but-unbilled work was outstanding at the end of last month?
9. If every new request instantly showed your prior jobs on and around that parcel, what would actually change — your quote price, win rate, or field hours? Can you put numbers on a recent example?
10. What would make you *distrust* a map of your firm's archive — and what happens to that trust the first time it shows a job in the wrong section?

Pass/fail heuristics (pre-registered, honest): the thesis strengthens if owners report ≥5 status interruptions/week, a prior-work check that takes >15 minutes or is skipped, and an existing software line item ≥$50/mo; it weakens if ClickUp-style satisficing is common, historical records are mostly address-only paper, or the prior-work advantage is priced at zero in owners' own quoting math.

Sources: `research/raw-scan-full.json` *(problem: land-surveying job tracking),* `research/revival-round3-results.json` *(Backsight concept + dual skeptic verdicts and evidence checks),* `business/FINAL_CONCEPT.md` *(post-skeptic fixes),* `decisions/DECISION_LOG.md` *(D-006, D-009, D-010).*

Backsight — Competitor Analysis

Date: 2026-07-10 **Product:** Backsight — job tracking + prior-work spatial intelligence for small US land-surveying firms (1–5 crews, 3–25 staff) **Status of this document:** All competitor facts below come from the venture's adversarial-review research (three rounds of skeptic verification with fresh web searches). Where the reviewing skeptic could not confirm a detail, that is stated explicitly rather than papered over.

> **Verification caveat (important, from the adversarial review itself):** `survey-ops.com`, `qfactor-llc.com`, `kudurrustone.com`, `agtcad.com`, `rpls.com`, and `apps.apple.com` all blocked automated fetching (403 via the research proxy). Competitor details therefore come from search-index snippets, indexed subpages, press releases (Morningstar/Newswire for Qfactor), and third-party directories (SoftwareAdvice, GetApp, Slashdot). The review judged these "directionally solid," but **exact SurveyOps, Qfactor, and Cyanic price points could not be confirmed** and should be re-verified by manual demos before any positioning goes to market.

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1. The competitive landscape in one view

The founding research initially claimed "no dominant vertical solution exists." Adversarial review **refuted that as stated**: at least six purpose-built land-surveying practice-management products are live today. What survived review is the narrower claim: the market is genuinely fragmented, **no vendor has won** (all incumbents have near-zero review footprints on Capterra/G2), and the pain itself is corroborated by six independent vendors betting on it (verified by adversarial review, https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/ — flagged as vendor content marketing for its recommendations, but substantively contradicted on its "no vertical solution" premise by the skeptic's own competitor discovery).

Competitor	Category	Target	Pricing (verified?)	Key strengths	Key weaknesses	Source
Qfactor (ex-BizWatt)	Vertical PM	Land-surveying firms	Annual contracts implied; **price not confirmed**	~10 years in market (founded 2016); Map View showing location + status of current **and past** survey projects	Near-zero review footprint after a decade — obscure despite tenure	Named in adversarial review of revival-round3-results.json (site 403'd; facts via press releases/snippets)
SurveyOps	Vertical PM	Land-surveying firms	Tiered; **exact tiers not confirmed** (site blocks bots)	S-T-R job organization ("by Section, Township, Range — not just addresses"); shipped iOS field app; aggressive programmatic SEO; free guided onboarding; markets eliminating status calls	New/unproven; feature depth unverified	https://survey-ops.com/land-surveying-job-management-software (own marketing page; existence and claims verified, depth not)
KudurruStone	Vertical PM	Land-surveying firms	**$25/$15/$5 per user/mo by role** — a 10-person firm ≈ $100/mo; no onboarding/import/training fees; free trial (verified by adversarial review from vendor site snippets)	Client portal (clients submit and track requests); mobile crew apps; cheapest vertical option at small headcounts	One RPLS.com forum surveyor tried it and abandoned it as "too slow" (verified by adversarial review)	Named in adversarial review (kudurrustone.com 403'd)
Cyanic Job Book	Vertical PM	Land-surveying firms	**Not confirmed**	"Searching jobs by legal address information and map-based job searching"; listed on SoftwareAdvice/GetApp	Thin public footprint	Named in adversarial review (getjobbook.com)
Info-Retriever by AGT	Vertical office mgmt	Surveying offices	**Not confirmed**	Long-standing; includes survey-control tracking	Legacy product; details unverifiable (agtcad.com 403'd)	Named in adversarial review
CQ	Vertical business mgmt	Surveyors	**Not confirmed**	Publishes the category's SEO content	Its "best software" blog is its own content marketing — flagged as non-independent evidence	https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/
BQE Core	Horizontal PSA/accounting	Engineering/A&E firms	~$25–50/user/mo (scout estimate from pricing anchors; treat as approximate)	Full accounting + time tracking; established	Accounting-heavy, engineering-firm priced; no surveying semantics	https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/ (roundup; vendor content)

Competitor	Category	Target	Pricing (verified?)	Key strengths	Key weaknesses	Source
ClickUp (and monday.com et al.)	Generic PM	Everyone	~$3/user/mo reported by surveyors on RPLS forum (verified by adversarial review)	Near-free; flexible; surveyors report "happily running their firms" on it	No legal descriptions, no map intelligence, no client status pages, no surveying pipeline semantics	RPLS.com PM-software threads, per adversarial review
Jobber-style FSM	Field-service mgmt	Lawn care/HVAC/trades	~$40–200/mo (scout price anchor)	Mature scheduling/dispatch/invoicing	Trade semantics, not surveying: no S-T-R, no plats/ALTA/licensed-review concepts	https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/ (roundup; vendor content)
Whiteboard + spreadsheet + email	Status quo	The actual majority	$0	Zero learning curve; owner already trusts it	Status calls interrupt the owner; drafters wait on field notes; invoices delayed; prior-work memory lives in one person's head	https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/ (pain claim; **flagged by adversarial review as vendor content marketing — plausible but not independent evidence**)

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2. The vertical micro-vendors in depth

These six were surfaced by the round-3 market skeptic's fresh research after the original scout named only SurveyOps and BQE Core. The skeptic's verdict language matters and is preserved here.

2.1 Qfactor (formerly BizWatt)

- Founded 2016 — roughly ten years in market — and **rebranded June 2026** claiming to be the "leading software for land surveying firms" (verified by adversarial review via press releases; site itself blocked fetching).
- Most important verified fact for Backsight: **Qfactor's Map View shows "the location and status of all your current and PAST survey projects" — a spatial prior-work archive.** This directly contradicts the naive framing that "no one will build" a prior-work map. Our wedge cannot be "we have a map of past jobs"; it must be what we do *with* it (see GAP_ANALYSIS.md).
- Equally important verified fact: after a decade of selling into this exact buyer, Qfactor has a **near-zero review footprint on Capterra/G2**. The skeptic's honest read: this pattern "suggests slow sales cycles, whiteboard inertia, and a buyer who is genuinely hard and expensive to reach relative to a $79–249/mo price point." That is a warning about the *market*, not a reassurance about the competitor.

2.2 SurveyOps

- Real, live, tiered pricing, **shipped iOS field app** (Backsight v1 is responsive web only — a verified feature deficit on our side).
- Its own marketing page — the very page the original scout cited as evidence of the gap — **names Section/Township/Range job search as a core need it addresses** ("by Section, Township, Range — not just addresses"), refuting the concept's early claim that SurveyOps "leads with generic job management" (verified by adversarial review, https://survey-ops.com/land-surveying-job-management-software).
- Executing **aggressive programmatic SEO** on exactly the keywords Backsight would want ("survey job tracking software," "land surveying job management software," "field-to-office workflow"), offers guided onboarding at no extra cost, and **markets elimination of status phone calls as a core benefit** — meaning two of Backsight's three headline features are already claimed in SurveyOps marketing.
- Feature depth and exact pricing unverified (site 403s bots). The adversarial review's explicit instruction: **manually demo SurveyOps before finalizing positioning.**

2.3 KudurruStone

- Idaho-built vertical tool with a **client portal** (clients submit and track requests — i.e., a client-facing status capability already exists in the vertical) and **mobile crew apps**.
- The only vertical competitor with verified pricing: **$25/$15/$5 per user per month by role**, so a 10-person firm ≈ **$100/mo** — cheaper than Backsight's Firm tier ($149) at that size. No onboarding/import/training fees, free trial. The adversarial review's blunt finding: Backsight's pricing "is positioned above, not below, the real vertical competition," because the original concept anchored only against BQE Core, the most expensive comparator.
- Counterweight, also verified: on the RPLS.com forum thread about PM software, **one surveyor tried KudurruStone and abandoned it as "too slow,"** with others defaulting to ClickUp or asking about MS Project. Latency and simplicity may matter more to this buyer than feature lists.

2.4 Cyanic Job Book

- Surveyor-specific project management (getjobbook.com), listed on SoftwareAdvice/GetApp.

- Verified capability claim: **"searching jobs by legal address information and map-based job searching."** Together with Qfactor's Map View and SurveyOps's S-T-R search, this is the third in-vertical contradiction of "nobody does spatial job search." Pricing and depth not confirmed.

2.5 Info-Retriever by AGT

- Long-standing office-management product for surveying offices including **survey-control tracking**. Site (agtcad.com) blocked fetching; treat as a legacy incumbent whose installed base is unknown. Its existence matters mainly as further proof the vertical has been repeatedly attempted.

2.6 CQ

- Sells "Business Management Software for Surveyors" and publishes the category's SEO content. The adversarial review flagged that the original scout's #1 pain citation was **CQ's own blog** — vendor content marketing whose job is to manufacture the pain narrative and rank CQ first. The pain is plausible; the citation is not independent. Backsight's evidence discipline now treats all such roundups as competitor intelligence, not demand evidence.

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3. The real incumbent: nothing

The status quo — whiteboard, spreadsheet, email folders, and the owner's memory — is the actual market leader, and the adversarial record supports treating it as the primary competitor:

- Six vertical vendors over ~10 years have collectively failed to produce a single product with a meaningful review footprint. The skeptic reads this as **whiteboard inertia and an expensive-to-reach buyer**, not as an open goal.
- Surveyors on RPLS.com report happily running firms on **ClickUp at ~$3/user/mo** — near-free generic tooling "evidently satisfices" (verified by adversarial review).
- Implication for Backsight (our assessment): the sales motion is not "switch from Qfactor," it is "retire the whiteboard." That demands day-one demonstrated value (the historical-import + Prior-Work Radar demo on the firm's own data) and an ROI visible inside a 14-day trial (fewer status calls, surfaced delivered-but-uninvoiced jobs). Any feature the whiteboard does better — glanceability, zero friction — is a real objection.

Willingness-to-pay context (scout signal, partially vendor-derived — treat as directional): these are licensed professional firms with $300K–$5M revenue that already spend heavily on technical gear (GNSS receivers $10–30K); adjacent tools anchor at Jobber ~$40–200/mo and BQE Core ~$25–50/user/mo (https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/, a roundup flagged as content marketing). The strongest *independent* WTP signal is simply that six paid vertical vendors exist and none is free.

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4. Adjacent surveying software (context, not competitors)

CAD and GNSS ecosystems (Trimble, Leica, Carlson). These vendors monetize hardware and CAD seats and have not built small-firm practice management (this is the concept's framing; the adversarial review did not contest it, but it was not independently verified — treat as a reasoned assumption). They define the firm's *technical* stack; Backsight deliberately does not touch it (no CAD integration, no data-collector integration in v1). They are a partner/channel surface (GNSS dealers talk to every small firm), not a competitive threat in job tracking — with the standing caveat that any of them could bundle a tracker if the segment ever looked lucrative.

Free PLSS/S-T-R map tools. The adversarial review killed the "free S-T-R lookup as lead magnet" plan: **randymajors.org, Earth Point (~85,000 townships, 3M sections, searchable by description/lat-lon; https://www.earthpoint.us/Townships.aspx), MapScaping's free PLSS viewer, and BLM's own ArcGIS viewers** are already the bookmarked defaults; **Township America** additionally operates a free Township/Range-to-GPS converter plus a **paid PLSS geocoding API covering 30+ states and all 37 principal meridians** (verified by adversarial review, https://townshipamerica.com/ — which also proves the geocoding component is buyable if building it stalls). These tools are not job trackers, but they own the free-utility mindshare Backsight once hoped to capture. The replacement distribution asset (per the review's recommendation): a one-shot "import your job history, get a map of everything your firm has ever surveyed" tool that doubles as the onboarding funnel.

Data/parsing infrastructure facts that shape the build (all verified by adversarial review):

- BLM National PLSS CadNSDI data is public-domain bulk data (geodatabase/shapefile, no key, no fee; ~2.6M section polygons fit in PostGIS) — https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons
- The only mature open-source legal-description parser, **pyTRS, prohibits all commercial use** under a Modified Academic Public License — Backsight ships its own license-clean parser — https://github.com/JamesPImes/pyTRS
- **20 states are outside PLSS** (the original 13 colonies plus TX, HI, KY, TN, VT, ME, WV); Texas runs its own abstract/survey-name system for which BLM data is useless — https://en.wikipedia.org/wiki/Texas_land_survey_system
- QuickBooks Online is a public OAuth API with a fast form-based production gate (~30-minute assessment) — https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ

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5. Market size (skeptic-corrected)

The original scout claimed ~25,000 US establishments. Adversarial review **refuted** this: Census County Business Patterns for NAICS 541370 shows **~7,000 firms / ~7,382 establishments (2020)**, with a second source verifying ~6,191 active companies; the 25,000 figure appears to conflate establishments with licensed individual surveyors (verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping). Filtering to 1–5-crew firms in PLSS states, the skeptic's serviceable estimate is **2,000–4,000 firms** being chased by 6+ vertical vendors plus horizontal tools. At 5% penetration of ~7,000 establishments × ~$140/mo average, the corrected ceiling is roughly **$590K ARR** — a strong bootstrapped outcome, honestly not a venture-scale one (per FINAL_CONCEPT.md and DECISION_LOG D-010).

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6. Where Backsight wins and loses — honest scorecard

vs.	Backsight wins on	Backsight loses on
Qfactor	Automated ingestion of the historical archive (no incumbent advertises parsing legal descriptions out of a firm's messy spreadsheet — the adversarial review's "most defensible residual claim"); quote-time adjacency intelligence, not just a map view; modern UX vs. a decade-old product with no review footprint	Ten years of installed base and references; a Map View of past projects that already exists and could be marketed harder tomorrow; unknown pricing may undercut us
SurveyOps	Prior-work monetization framing (they lead with tracking; our wedge is the archive's quoting value — hypothesis until their demo is inspected); flat per-firm pricing vs. their annual contracts; historical-import concierge	Shipped iOS app vs. our responsive web; head start on the exact SEO keywords; free guided onboarding blunts our concierge angle; they already market "no more status calls"
KudurruStone	Team-size economics: their per-user/role math beats us at 10 users ($100 vs $149) but crosses over as staff grows and field crews never count against a Backsight license; performance/UX (one verified "too slow" abandonment is a thin but real signal)	Price at small headcounts; they already ship a client portal AND mobile crew apps; zero onboarding fees
Cyanic Job Book	Same as Qfactor: backfill + quote-time intelligence vs. map-based search of entered jobs; UX modernity (hypothesis — undemoed)	Existing directory presence (SoftwareAdvice/GetApp); unknown pricing could be lower
Info-Retriever / CQ	Modern web product vs. legacy office software; focused wedge vs. broad "business management"	Incumbency in whatever installed base they hold; CQ owns category SEO content
ClickUp / monday / generic PM	Everything surveying-native: S-T-R and legal-description search, PLSS spatial join, licensed-review gate, client status links, stuck-job digests — none of which generic boards model	Price (~$3/user/mo is effectively free); zero migration risk; already adopted and "satisficing" per RPLS forum users
Jobber-style FSM / BQE Core	Surveying semantics vs. trade semantics; flat per-firm pricing vs. per-user creep; not accounting-first	Maturity, mobile apps, payments/scheduling depth we deliberately don't build
Whiteboard/spreadsheet	Institutional memory that survives the owner; status calls answered by a link; nothing falls between stages	Zero cost, zero learning curve, total trust; if Backsight's weekly value isn't felt, the whiteboard wins by default — this is the loss mode that has kept six vendors small

Bottom line

Backsight enters a fragmented, unwon micro-market where the pain is real (six independent vendors bet on it), the status quo is the main enemy, and the original differentiation claims ("no vertical solution," "an archive no one will build," "no cheap alternative") were each **factually wrong** and have been replaced (see DECISION_LOG D-010, FINAL_CONCEPT.md fix #4). The surviving, defensible position is narrow but real: **automated spatial ingestion of a firm's historical job archive, surfaced at quote time, at flat per-firm pricing, with a no-login client status link and modern UX** — validated next by manual demos of Qfactor, SurveyOps, KudurruStone, and Cyanic (the adversarial review's explicit precondition) and by measuring parse rates on 3–5 real firm spreadsheets before GTM spend.

Backsight — Gap Analysis

Date: 2026-07-10 **Purpose:** Define precisely which market gap Backsight occupies, which claims are adversarially verified versus our own hypotheses, and which gaps we deliberately refuse to chase. Companion to `COMPETITOR_ANALYSIS.md` (same evidence base and verification caveats: competitor sites blocked automated fetching; details rest on search snippets, directories, and press releases, per the round-3 adversarial review).

Labeling convention used throughout:

- **[VERIFIED]** — established by the tournament's adversarial skeptics through fresh web research, with source URLs from the research files.
- **[HYPOTHESIS]** — our positioning belief; plausible, consistent with the verified record, but not independently confirmed. Each carries its validation step.

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1. The gap in one paragraph

Six vertical micro-vendors (Qfactor, SurveyOps, KudurruStone, Cyanic Job Book, Info-Retriever by AGT, CQ) already sell job tracking to land surveyors, and several already put jobs on a map — so the gap is **not** "surveying job software" and **not** "a map of jobs." The verified residual gap is narrower and better: **no incumbent advertises automatically parsing a firm's messy historical job archive (legal descriptions, addresses) into a spatial index** — they map jobs *going forward* or require manual entry (adversarial review: "the most defensible residual claim found"). Backsight's wedge is to turn the archive a firm already owns into a quoting and profitability asset on day one, wrap it in a pipeline that speaks surveying, expose progress through a no-login client status link, and price flat per firm so field crews never count against a license. The rest of this document defends each element honestly.

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2. Gap #1 — Prior-work spatial intelligence as a first-class quoting/profit tool

What the record verifies

- **Competitors treat spatial as a view or a search, not as the product.** [VERIFIED] Qfactor's Map View shows "the location and status of all your current and PAST survey projects"; Cyanic Job Book offers "searching jobs by legal address information and map-based job searching"; SurveyOps organizes jobs "by Section, Township, Range — not just addresses" (adversarial review of the round-3 market skeptic; SurveyOps claim on its own page, https://survey-ops.com/land-surveying-job-management-software). The original claim "this is the value no platform will build" was **contradicted three times over inside the vertical** and has been withdrawn.
- **None of them advertise automated backfill of the historical archive.** [VERIFIED — by absence, per the skeptic's fresh search] The residual differentiator is "automated PLSS parsing of imported historical CSVs," which the skeptic sized honestly as "a real but thin onboarding convenience, not a moat" in isolation.
- **The ingredients are genuinely open.** [VERIFIED] BLM PLSS CadNSDI data is public-domain bulk data, no key, no fee (https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons). The one OSS parser (pyTRS) bans commercial use, so Backsight ships its own license-clean parser (https://github.com/JamesPImes/pyTRS); Township America's paid PLSS geocoding API (30+ states, all 37 meridians) proves the approach works and offers a buy-option (https://townshipamerica.com/).
- **The wedge has verified geographic limits.** [VERIFIED] 20 states — including Texas — sit outside PLSS (https://en.wikipedia.org/wiki/Texas_land_survey_system). Initial GTM targets PLSS states; address geocoding is the primary spatial index and S-T-R parsing is enrichment (FINAL_CONCEPT.md fixes #2–3).
- **Historical spreadsheets may not contain legal descriptions at all.** [VERIFIED as a risk] The implementability skeptic: in many firms the spreadsheet holds client + street address; the legal description lives in the deliverable PDF. Mitigation is architectural (address-first ingestion, map pin-drop fallback) and empirical: **measure parse rates on 3–5 real firm spreadsheets before believing the demo.**

What is our hypothesis

- **[HYPOTHESIS] The difference between "a map view of past jobs" and "prior-work intelligence at quote time" is commercially decisive.** Backsight surfaces adjacent past jobs (same section, within radius) *inside the quoting moment*, with year, type, deliverables, and original quote attached — "you have 3 prior jobs in this section; quote with confidence, win it on price, deliver it at half the field time." No verified fact says Qfactor or Cyanic *don't* do this; no evidence says they do. **Validation:** manual demos of all four named vertical products (the adversarial review's explicit precondition) before positioning ships.
- **[HYPOTHESIS] The archive compounds into a switching cost.** After a year, spatial search over hundreds of past jobs is institutional memory a firm won't re-enter elsewhere. Consistent with SaaS experience, unverified in this vertical.
- **[HYPOTHESIS] Concierge historical import converts trials.** SurveyOps already offers guided onboarding at no extra cost [VERIFIED], so the concierge must be measurably deeper (we parse and spatially index the backlog; they onboard forward workflow) — a claim to prove in the first ten trials, not assert.

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3. Gap #2 — Client-facing status links

- **The pain framing is plausible but the primary citation is tainted.** [VERIFIED flag] The "clients call for status" evidence traces to CQ's own SEO blog — vendor content marketing (https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/). We keep the feature because it is cheap to build and demo-able, not because the citation proves demand.

- **We are not first.** [VERIFIED] KudurruStone ships a Client Portal where clients submit and track requests; SurveyOps markets elimination of status phone calls as a core benefit (adversarial review).
- **[HYPOTHESIS] The tokenized no-login link beats a portal.** A title company will refresh a link embedded in an email; it will not create credentials in a small surveyor's portal. This UX distinction is Backsight's actual bet here, and the referral loop ("Tracked with Backsight" in the footer, seen by title companies working with dozens of surveyors) rides on it. **Validation:** count status-call reduction during trials; ask title-company recipients directly.

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4. Gap #3 — Flat per-firm pricing vs. per-user

- **The verified pricing landscape:** KudurruStone charges $25/$15/$5 per user/month by role — a 10-person firm ≈ $100/mo, below Backsight Firm at $149 [VERIFIED]; surveyors report running firms on ClickUp at ~$3/user/mo [VERIFIED, RPLS forum per adversarial review]; the skeptic's blunt finding was that Backsight is "positioned above, not below, the real vertical competition," since the original concept anchored only against BQE Core.
- **The honest version of the flat-pricing gap:** flat per-firm ($79/$149/$249, FINAL_CONCEPT.md) wins where per-user pricing punishes growth — field crews and role churn never trigger license math, and the crossover favors Backsight as headcount passes the mid-teens. At small headcounts we are **not** the cheap option and should not pretend to be. [Mix of VERIFIED inputs and our arithmetic on published tiers.]
- **[HYPOTHESIS] Flat pricing is also a message, not just math.** The buyer segment demonstrably resents add-on/per-seat creep (per the SMB research thread in the venture record); "one price, every crew" is a trust posture. **Validation:** the adversarial review's prescription — justify the premium with a concrete recovered-revenue calculator in the trial (e.g., delivered-but-uninvoiced jobs surfaced from the imported history on day one) or drop the Practice-tier anchor.

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5. Gap #4 — Modern UX vs. dated micro-vendor products

- **Verified signals, thin but consistent:** a decade-old vertical vendor (Qfactor, founded 2016) remains obscure with near-zero Capterra/G2 review footprint [VERIFIED]; an RPLS forum surveyor abandoned KudurruStone as "too slow" [VERIFIED]; the skeptic's inference is that **latency and simplicity may matter more than features** in this segment.
- **[HYPOTHESIS] A fast, modern, low-friction web app is itself a differentiator here.** No incumbent has been demoed yet; "dated UI" is inference from footprint and forum signals, not from screenshots. **Validation:** the mandated manual demos, plus trial-user session timing on core flows (create job, advance stage, radar search).

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6. Gaps that are NOT ours to win — and why that focus is a feature

The adversarial record shows categories adjacent to Backsight are either owned, hardware-gated, or kill-taxonomy violations. Declining them is deliberate scope discipline (FINAL_CONCEPT.md, MVP_BUILD_SPEC.md), not a roadmap deficit:

Gap we decline	Who owns it / why we decline	Backsight's posture
Accounting / time-and-billing	BQE Core and QuickBooks; accounting-heavy tools are exactly what this buyer finds oversized.	Push invoices to QuickBooks Online via its public OAuth API (30-minute production assessment, verified: https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ). Integrate, never replicate.
CAD / drafting	Carlson, Trimble, Leica ecosystems — decades of entrenchment, hardware-coupled (concept framing, uncontested by review but not independently verified).	None in v1. CAD files are attachments/metadata, not a canvas.
Field data collection / GNSS workflows	The hardware vendors' native turf; firms already spent $10–30K per receiver there (scout signal).	No data-collector integration in v1. The field crew's Backsight surface is a phone browser updating a stage and attaching photos.
Crew GPS tracking / scheduling optimization	Generic FSM (Jobber-class) does this well for trades; building it invites a feature war we can't win and the buyer didn't ask for.	Simple stage assignment + due dates + stuck-job digest.
Payments / estimating math	Kill-taxonomy item 5 (payment migration risk); estimating is judgment we must not fake.	Quote amounts are fields the owner fills; money moves in QuickBooks.
Free S-T-R lookup utilities	randymajors.org, Earth Point (https://www.earthpoint.us/Townships.aspx), MapScaping, BLM viewers, Township America already own the bookmark [VERIFIED].	Dropped as lead magnet; replaced by the one-shot "map everything your firm has ever surveyed" import tool that doubles as onboarding.

Why focus is a feature: every declined gap either (a) collides with an entrenched, well-capitalized owner (CAD/GNSS, FSM), (b) reintroduces a verified kill pattern (payments, regulated deliverables), or (c) dilutes the one wedge no incumbent advertises (archive backfill). A one-developer, Claude-Code-built product wins by being the *only* tool whose job is turning a surveying firm's past into quoting power — and by being unembarrassed that CAD, collection, and accounting live elsewhere. Integration posture also converts adjacent vendors (GNSS dealers with nothing to sell for the office) into a channel rather than a threat.

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7. What would close our gap, and the tripwires

Threat	Verified basis	Tripwire / response
SurveyOps adds historical-import concierge + archive search	They already ship S-T-R search, free onboarding, iOS app, and own the SEO keywords [VERIFIED]	Monitor their pages quarterly; keep the wedge moving toward what imports enable (quote-time adjacency, unbilled-job recovery), not the import itself
Qfactor markets its existing past-projects Map View as "prior-work intelligence"	The Map View exists [VERIFIED]	Same: differentiation must live in parse-quality, quote-moment UX, and flat pricing, which are harder to bolt on than a landing page
KudurruStone price pressure at small firms	$25/$15/$5 per user verified	Hold flat-pricing message; win on crews-don't-count and archive value, concede the 5-person price fight
The buyer stays on the whiteboard	Decade-old vendors still tiny; ClickUp satisfices [VERIFIED]	This is the dominant risk. If 10 concierge-imported trials don't convert ≥3 paying firms, the thesis — not the feature list — is wrong
TAM ceiling	~7,000 establishments NAICS 541370, serviceable 2,000–4,000 PLSS-state small firms [VERIFIED, https://siccode.com/naics-code/541370/surveying-mapping]	Accepted by design (~$590K ARR at 5%); adjacent expansion (septic designers, foresters, geotech field firms) shares the field→office→licensed-review shape [HYPOTHESIS]

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8. Summary

Backsight occupies a real but deliberately narrow gap: **the only surveying job tracker whose first act is to spatially ingest the firm's past, not just organize its future** — delivered with a no-login client status link, flat per-firm pricing, and modern UX, in a fragmented micro-market with no winner and a status quo of whiteboards. Every load-bearing fact above survived (or was corrected by) three rounds of adversarial review; every claim that did not survive has been relabeled as hypothesis with a named validation step. The two mandatory validations before GTM spend: manual demos of the four named vertical competitors, and parse-rate measurement on real firm spreadsheets.

The Backsight Candidate Tournament

Date compiled: 2026-07-10 **Inputs:** `research/raw-scan-full.json` (34 evidenced problems, 8 scouts) → `research/tournament-shortlist.json` (12 candidates) → `research/tournament-results.json` (12 concepts, 3 judge scorecards, weighted ranking, round-1 skeptic verdicts) → `research/skeptic-round2-partial.json` (round-2 verdicts) → `research/revival-round3-results.json` (round-3 concepts and verdicts). **Companion document:** `ADVERSARIAL_REVIEW.md` — the complete skeptic history that this tournament fed into.

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1. Pipeline overview

```
34 evidenced problems (8 parallel research scouts, every claim cited)
        │  shortlist: hard-dollar pain, evidence quality, buildability, thesis diversity
        ▼
12 tournament candidates (2 formed by merging independent-scout convergences)
        │  one product concept designed per candidate
        ▼
12 SaaS concepts, scored by a 3-persona judge panel (8 criteria × weighted aggregate)
        │  top 4 advance
        ▼
Round 1: dual adversarial skeptics per finalist (web-research-backed) → ALL 4 REJECTED
        ▼
Round 2: ranks 5–8 face the identical gauntlet → ALL 4 REJECTED
        │  seven-pattern kill taxonomy codified from the 8 autopsies
        ▼
Round 3: 4 revival candidates engineered against the kill taxonomy → dual skeptics again
        ▼
WINNER: Backsight (land-surveyor job tracking + prior-work intelligence)
— the only concept of 12 whose market skeptic found zero fatal flaws
```

The tournament was deliberately designed so that no concept could win on judge scores alone: every would-be winner had to survive two independent skeptics doing fresh web research, including spot-checks of the original scouts' citations (Decision Log, D-007/D-008).

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2. The 12 shortlisted candidates

The shortlist was drawn from the 34 scouted problems using four criteria (Decision Log, D-006): (a) pain denominated in hard dollars or legal risk, not convenience; (b) evidence quality — named prices, named incumbents, documented failures; (c) buildability without gated access; (d) diversity across the 8 hunting grounds.

The 8 scouts (hunting grounds): `smb-trades`, `niche-pros`, `reg-compliance`, `ai-era-ops`, `creators-solo`, `ecommerce-sellers`, `devtools-b2b`, `hated-incumbents`.

#	Candidate key	Title (abridged)	Scout(s)	Source problem(s) of the 34
1	`hvac-agreements`	Standalone maintenance-agreement tracker for small HVAC/plumbing shops	smb-trades	#0
2	`coi-tracking`	Affordable COI collection & expiry-chasing for small GCs and property managers	smb-trades **+ niche-pros (merge)**	#1 + #4
3	`backflow-compliance`	Backflow test report filing + annual retest reminder engine	smb-trades	#2
4	`commission-recon`	Carrier commission reconciliation for small independent insurance agencies	niche-pros	#5
5	`eaa-accessibility`	Continuous EAA/WCAG compliance monitoring for SMB e-commerce (anti-overlay)	reg-compliance	#8
6	`autorenewal-compliance`	Auto-renewal law (ARL) compliance auditor + annual-reminder engine	reg-compliance	#10
7	`ai-cost-margin`	Per-client LLM cost attribution and margin tracking for agencies	ai-era-ops **+ devtools-b2b (merge)**	#16 + #28
8	`sponsorship-ops`	Sponsorship-operations hub for indie newsletters and podcasts	creators-solo	#18
9	`affiliate-linkrot`	Affiliate link-rot monitor for YouTube creators	creators-solo	#20

#	Candidate key	Title (abridged)	Scout(s)	Source problem(s) of the 34
10	`threepl-audit`	3PL / fulfillment invoice auditing for small DTC brands	ecommerce-sellers	#22
11	`chargeback-evidence`	Flat-fee chargeback evidence automation for Shopify/Stripe merchants (anti-Chargeflow)	ecommerce-sellers	#24
12	`membership-refuge`	Membership management for small clubs fleeing Wild Apricot	hated-incumbents	#29

The two independent-scout merges. Two problems were discovered independently by two different scouts working disjoint hunting grounds — treated as a convergence signal and merged into single candidates:

- **COI tracking** — found by both `smb-trades` (problem #1, GC/property-manager angle, myCOI per-vendor pricing) and `niche-pros` (problem #4, enterprise-incumbent pricing / LLM-extraction feasibility angle).
- **Per-client AI cost attribution** — found by both `ai-era-ops` (problem #16, agency per-client billing) and `devtools-b2b` (problem #28, per-customer AI margin guardrails into Stripe).

Notable drops from the 34 (Decision Log, D-006): interpreter-agency back-office (ops burden), HOA treasurer-in-a-box (fiduciary/ACH exposure), pet-boarding software (crowded), support-AI answer QA (gated data), youth-sports league management (volunteer buyers, no budget), and **surveyor job tracking (#7 — dropped for "weaker evidence of software WTP vs hardware spend")**. That last drop matters: the eventual tournament winner, Backsight, is this problem revived in round 3 after every shortlisted finalist had been killed on evidence.

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3. The 12 concepts

Each shortlisted problem was turned into a concrete SaaS concept by a dedicated ideation agent (no web research at this stage — claims were tested later by the skeptics, who did have web access).

Candidate	Product concept	One-liner (abridged)	Pricing posture
`chargeback-evidence`	**DisputeKit**	Flat-fee chargeback evidence automation; 2-hour dispute response → 5-minute review-and-submit; keep 100% of recoveries	$49 / $99 / $199 per mo flat, "no success fees ever"
`coi-tracking`	**CertChase**	Auto-reads ACORD 25 certificates and chases subs' insurance agents for renewals	$49+ flat tiers by vendor count
`threepl-audit`	**MarginHawk**	Audits the 3PL's monthly invoice against your own rate card; drafts the dispute email inside the 30–90-day window	$99 / $199 / $349 by monthly fulfillment spend
`commission-recon`	**ShortPay**	Turns the pile of carrier commission statements into a reconciled ledger; flags every short-pay	$99 / $199+ per agency
`eaa-accessibility`	**Statement** (statement.eu)	Weekly WCAG scans + always-current EU accessibility statement + evidence file	$39 / $79+ per store
`hvac-agreements`	**TuneUp Keeper**	Standalone maintenance-agreement manager: auto-bills memberships, never lets a visit go unscheduled	$79 / $129 / $199 flat by agreement count
`membership-refuge`	**Clubhouse HQ**	One-click Wild Apricot import; bills only active members; never taxes your payment processor	$29 / $59 / $99 by active members
`autorenewal-compliance`	**RenewGuard**	Scores signup/cancel/reminder flows against 20+ state ARLs; runs the mandated annual reminders	Audit $99/mo; Operate $299/mo
`ai-cost-margin`	**Marginal**	Drop-in LLM proxy that tags every API call to a client; live margin per client; billable-usage statements	$49+ by clients tracked, not tokens
`sponsorship-ops`	**SlotDeck**	Slot calendar, sponsor portal, deliverable tracker; replaces the breaking spreadsheet	$29/mo flat, zero commission
`backflow-compliance`	**RetestHQ**	Gauge readings → the exact form each city wants + automatic 12-month retest chasing	Solo $59/mo; Crew $99/mo
`affiliate-linkrot`	**LinkPatrol**	Watches every affiliate link in the YouTube back catalog; ranks dead links by dollars; bulk-fixes via API	$19+ by channel/video count

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4. Judge panel design

Three judge personas with deliberately conflicting priors each scored **all 12 concepts** on 8 criteria (1–10). Persona diversity is a failure-mode net: the VC catches vitamin products, the indie catches unreachable niches and ops burden, the CTO catches gated-API fantasies (Decision Log, D-007). The personas, as prompted:

Judge	Persona	Stated priorities
VC	Seed-stage B2B SaaS investor	Market pull, willingness to pay, expansion potential, wedge-to-company path; "allergic to vitamin products and crowded categories"

Judge	Persona	Stated priorities
Indie	Successful bootstrapped indie hacker (Tyler Tringas / Rob Walling school)	Reachable niches, cheap distribution, low operational burden, time-to-first-dollar; avoids enterprise sales and 24/7 ops
CTO	Pragmatic CTO assessing technical risk	Whether a non-specialist can genuinely build **and maintain** this with Claude Code: API availability without gated credentials, data-model complexity, failure modes, ongoing accuracy/liability risk, integration fragility

Weighted scoring system

Each judge's concept score is a weighted sum of the 8 criterion scores; the final tournament score is the mean across the three judges. Weights encode the brief's hard constraint that Claude-Code buildability dominates (Decision Log, D-007). The weights below reproduce every published ranking score exactly (maximum possible weighted score: 97.0):

Criterion	Weight	Rationale
Buildability	×1.5	Hard constraint: non-specialist + Claude Code must be able to ship it
Claude Code fit	×1.5	Same constraint, maintenance included
Willingness to pay	×1.3	Revenue proof beats market theory
Pain	×1.2	Must be denominated in dollars or legal risk
Urgency	×1.1	Deadline-driven pain converts
Differentiation	×1.1	Wedge quality
Market size	×1.0	Useful, but indie scale acceptable
Distribution	×1.0	Channel realism

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5. Full ranking

5.1 Final ranking (weighted average across 3 judges, max 97.0)

Dimension columns are the 3-judge mean per criterion (1–10).

Rank	Concept	Weighted score	VC	Indie	CTO	Pain	Urg	WTP	Mkt	Diff	Build	CC-fit	Dist
1	DisputeKit (`chargeback-evidence`)	**72.20**	73.8	71.4	71.4	8.0	6.7	7.3	7.0	5.7	8.0	9.0	7.0
2	CertChase (`coi-tracking`)	**68.53**	69.3	67.0	69.3	7.7	5.7	7.0	7.0	5.0	8.0	9.0	6.0
3	MarginHawk (`threepl-audit`)	**68.00**	70.4	69.3	64.3	7.0	6.3	7.0	5.7	7.0	7.7	8.7	5.7
4	ShortPay (`commission-recon`)	**67.97**	70.8	67.4	65.7	7.7	5.0	8.0	6.0	7.0	7.7	8.7	4.7
5	Statement (`eaa-accessibility`)	**66.50**	66.3	66.9	66.3	6.0	7.3	6.0	7.0	6.0	7.7	8.0	6.3
6	TuneUp Keeper (`hvac-agreements`)	**64.97**	62.7	64.6	67.6	7.0	5.0	6.7	5.7	5.7	8.7	8.3	5.0
7	Clubhouse HQ (`membership-refuge`)	**64.17**	64.9	61.9	65.7	7.0	5.0	6.0	7.3	4.7	7.7	8.3	6.0
8	RenewGuard (`autorenewal-compliance`)	**62.63**	66.1	60.4	61.4	6.7	7.0	6.3	6.0	7.0	6.0	6.7	6.0
9	Marginal (`ai-cost-margin`)	**60.37**	67.8	56.6	56.7	6.3	5.3	5.3	5.7	4.7	6.7	8.3	6.7
10	SlotDeck (`sponsorship-ops`)	**60.27**	61.2	58.5	61.1	5.0	3.7	4.0	5.3	5.3	9.0	8.7	7.3
11	RetestHQ (`backflow-compliance`)	**55.77**	54.3	52.7	60.3	6.3	5.0	5.3	3.3	5.7	7.7	7.3	3.7
12	LinkPatrol (`affiliate-linkrot`)	**55.33**	59.7	53.7	52.6	5.7	4.7	5.3	5.3	6.3	5.3	6.3	6.7

Top 4 advanced to round-1 adversarial review: `chargeback-evidence` , `coi-tracking` , `threepl-audit` , `commission-recon` .

5.2 Per-judge raw scorecards (1–10 per criterion; weighted total per judge)

Judge: VC

Concept	Pain	Urg	WTP	Mkt	Diff	Build	CC-fit	Dist	Weighted
chargeback-evidence	8	7	8	7	6	8	9	7	73.8
commission-recon	8	6	8	6	7	8	9	5	70.8
threepl-audit	7	7	7	6	7	8	9	6	70.4
coi-tracking	8	6	7	7	5	8	9	6	69.3
ai-cost-margin	7	6	6	7	5	8	9	7	67.8
eaa-accessibility	6	7	6	7	6	8	8	6	66.3
autorenewal-compliance	7	7	6	6	7	7	8	6	66.1

Concept	Pain	Urg	WTP	Mkt	Diff	Build	CC-fit	Dist	Weighted
membership-refuge	7	5	5	7	5	9	9	5	64.9
hvac-agreements	7	5	6	5	5	9	8	5	62.7
sponsorship-ops	5	4	4	5	6	9	9	7	61.2
affiliate-linkrot	6	5	6	5	7	6	7	7	59.7
backflow-compliance	6	5	5	3	6	8	7	3	54.3

Judge: Indie

Concept	Pain	Urg	WTP	Mkt	Diff	Build	CC-fit	Dist	Weighted
chargeback-evidence	8	7	7	7	5	8	9	7	71.4
threepl-audit	7	6	7	6	7	8	9	6	69.3
commission-recon	7	4	8	6	7	8	9	5	67.4
coi-tracking	7	5	7	7	5	8	9	6	67.0
eaa-accessibility	6	8	6	7	6	7	8	7	66.9
hvac-agreements	7	5	7	6	6	8	8	5	64.6
membership-refuge	7	5	7	8	4	6	7	7	61.9
autorenewal-compliance	6	7	6	6	7	6	6	6	60.4
sponsorship-ops	5	3	4	5	5	9	8	8	58.5
ai-cost-margin	6	5	5	5	4	6	8	7	56.6
affiliate-linkrot	5	4	4	5	6	6	7	7	53.7
backflow-compliance	6	5	5	3	5	7	7	4	52.7

Judge: CTO

Concept	Pain	Urg	WTP	Mkt	Diff	Build	CC-fit	Dist	Weighted
chargeback-evidence	8	6	7	7	6	8	9	7	71.4
coi-tracking	8	6	7	7	5	8	9	6	69.3
hvac-agreements	7	5	7	6	6	9	9	5	67.6
eaa-accessibility	6	7	6	7	6	8	8	6	66.3
commission-recon	8	5	8	6	7	7	8	4	65.7
membership-refuge	7	5	6	7	5	8	9	6	65.7
threepl-audit	7	6	7	5	7	7	8	5	64.3
autorenewal-compliance	7	7	7	6	7	5	6	6	61.4
sponsorship-ops	5	4	4	6	5	9	9	7	61.1
backflow-compliance	7	5	6	4	6	8	8	4	60.3
ai-cost-margin	6	5	5	5	5	6	8	6	56.7
affiliate-linkrot	6	5	6	6	6	4	5	6	52.6

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6. Per-judge rationale highlights

Condensed from the three judges' written rationales in `tournament-results.json` (VC / Indie / CTO respectively).

1. DisputeKit (chargeback-evidence, 72.20)

- *VC:* Quantified, growing pain and documented incumbent resentment at a 25% take rate; "best pull-per-dollar-of-build in the batch." But "differentiation is a pricing model, not a moat — Chargeflow can ship a flat tier," and Stripe improving native dispute tooling is a standing platform threat.
- *Indie:* Best proof of WTP on the list (merchants paying Chargeflow 25% and publicly furious); Stripe's public Disputes API + test mode makes it the cleanest end-to-end indie build. Win-rate credibility must be earned fast "or flat-fee looks like paying for losses."
- *CTO:* The rare concept fully buildable and verifiable by a non-specialist on day one (Stripe test mode exercises the whole loop with zero real money). Shopify App Store review gates the best channel.

2. CertChase (coi-tracking, 68.53)

- *VC:* Seven-figure downside risk, proven budget (myCOI at $1.5–3k/yr), large horizontal market. "A visibly crowding category — TrackMyVendor, Billy, COI File are all racing to the same SMB tier and LLM extraction is no moat."

- *Indie:* Standardized public form (ACORD 25) + LLM extraction ≈ near-perfect indie build; but the low end is crowding, the pain is seasonal, and a false "compliant" grade "carries scary liability optics."
- *CTO:* Magic-link loop needs no gated credentials; but "the product is fundamentally an email robot whose deliverability is a fragile single point of failure," and the false-pass liability tail is carried forever.

3. MarginHawk (threepl-audit, 68.00)

- *VC:* Hard-dollar recovery with a built-in deadline (30–90-day dispute window) and a killer first-session "you were overcharged $1,214" moment; per-logo ceiling modest.
- *Indie:* No gated APIs at all; LLM-parse-then-deterministic-math is a near-ideal Claude Code build. Rate-card transcription is heavy onboarding friction; "churn looms after a few clean months."
- *CTO:* Everything arrives as the customer's own files, deterministic dim-weight math gives an instant dollar number; but multi-format invoice parsing is a brittle funnel where "one mis-flagged charge kills trust."

4. ShortPay (commission-recon, 67.97)

- *VC:* Hard-dollar recovery ($10–16k/yr claimed leakage) against a non-consuming Excel incumbent; free first-statement audit is a great conversion mechanic. Boomer-heavy buyer, distribution is a grind.
- *Indie:* "Upload one statement, see your leakage" is a genuine one-session conversion; but no deadline means chronic-not-acute urgency, and "the buyer... doesn't search for the category — distribution is the whole risk."
- *CTO:* Zero-template LLM parsing is a structural edge over legacy per-carrier-template tools; but "extraction accuracy on money is existential — one miscounted statement kills trust."

5. Statement (eaa-accessibility, 66.50)

- *VC:* Live regulatory forcing function (EAA enforceable, French suits filed) + smart anti-overlay wedge; urgency "may decay to vitamin status if lawsuits stay rare."
- *Indie:* The auto-maintained statement/evidence file is a retention hook nobody sells at SMB price; fear-driven demand converts fast but churns when the fear fades; FR/DE localization is real labor.
- *CTO:* Whole stack open-source and ungated (Playwright + axe-core); but axe-core's ~30–40% detection ceiling "means the promise can never be 'compliant.'"

6. TuneUp Keeper (hvac-agreements, 64.97)

- *VC:* Dollar-denominated pain with a demoable 30-second wedge; "suite gravity is brutal — this is a feature Jobber/Housecall Pro can fold into a tier... a nice $30–50k MRR indie business, not a venture outcome."
- *Indie:* Proven budget (shops pay $59–600/mo for FSM); but double-entry friction and the card-re-collection migration are retention/onboarding killers.
- *CTO:* Near-ideal technical profile (CRUD + Stripe + cron, zero gated APIs); the double-entry problem is one "software can't fully solve."

7. Clubhouse HQ (membership-refuge, 64.17)

- *VC:* Genuinely hated PE-degraded incumbent with documented price hikes and a petition; but volunteer buyers with board cycles are "the slowest, most inertia-bound purchasers in software."
- *Indie:* High-intent SEO ("Wild Apricot alternative") and sticky-once-migrated customers; but the feature surface is the biggest build here and low-tech volunteer admins generate margin-eating support load.
- *CTO:* Textbook indie stack, no gated APIs; but no tech moat beyond migration UX, and the support/feature-parity tail grinds a solo maintainer at $29–99 ARPU.

8. RenewGuard (autorenewal-compliance, 62.63)

- *VC:* Real settlements ($2.5–7.5M) create fear-driven conversion and the Operate tier is genuinely sticky; but "a solo-maintained 20-state legal matrix is both the moat and the existential liability."
- *Indie:* Genuinely unowned combination (matrix + flow audit + mandated-reminder ops); but one statute error is reputationally fatal and UPL exposure looms.
- *CTO:* "The software is trivial; the thing that's hard is exactly the thing Claude Code can't safely own."

9. Marginal (ai-cost-margin, 60.37)

- *VC:* Rides a fast-growing spend category with a genuinely underserved wedge; but small AI agencies are "the churniest, most mortal segment in SaaS" and LiteLLM/Helicone are one feature away.
- *Indie:* The founder can live where the buyer congregates; but running a production LLM proxy is 24/7 ops — "the opposite of low operational burden."
- *CTO:* Pure dev-tooling build; but sitting in the critical path of clients' production LLM traffic is a heavy availability/security burden for an indie.

10. SlotDeck (sponsorship-ops, 60.27)

- *VC:* The sponsor portal is a legitimately viral loop; but "classic vitamin economics with a hard ceiling... fun indie business; not a fund-returner."
- *Indie:* Superb distribution and the easiest build on the list; but $29 ARPU from high-mortality creators whose real competitor is "my sheet works fine."

- *CTO:* Fully mockable, easiest build of the twelve; the two-sided bet (sponsors must actually use the portal) can quietly collapse the core value.

11. RetestHQ (backflow-compliance, 55.77)

- *VC:* Real mandated workflow, clean per-metro land-and-expand; but "a lifestyle-business TAM... nothing here becomes a company at seed scale."
- *Indie:* Hyper-reachable niche (cities publish prospect lists); but tiny market, offline buyers sold one phone call at a time, and FlowCert already has a 1,700-city head start.
- *CTO:* Technically clean; but the per-jurisdiction form-template treadmill is a permanent maintenance tax where a stale form recreates the fine risk the product sells against.

12. LinkPatrol (affiliate-linkrot, 55.33)

- *VC:* Detect-plus-repair loop genuinely novel vs AMZ Watcher; personalized free leak scan is a great cold-outreach mechanic; triple platform dependency plus a built-in churn cliff.
- *Indie:* Best growth loop here; but the build is hostage to Google OAuth verification/write quotas and Amazon PA-API qualification.
- *CTO:* "The gated-credentials minefield of the batch" — the fallback edges into fragile ToS-adjacent scraping; value decays once the catalog is cleaned.

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7. What happened after the judging

The scores above decided **who got attacked first**, not who won. Full detail is in `ADVERSARIAL_REVIEW.md`; the outcomes:

Round	Candidates	Outcome
1	Top 4 (DisputeKit, CertChase, MarginHawk, ShortPay)	**All four rejected** — market skeptics found disqualifying facts (platform-native substitutes, free-tier floors, technically unsound wedge, commoditized moat + circular evidence)
2	Ranks 5–8 (Statement, TuneUp Keeper, Clubhouse HQ, RenewGuard)	**All four rejected** — occupied categories, self-contradicting architecture, near-clone competitors, regulated-deliverable exposure
3	Four revival candidates engineered against the kill taxonomy	**Backsight selected**; three others rejected

A telling pattern (Decision Log, D-008): the judges systematically rewarded "hard-dollar recovery" theses — but those same money-recovery categories attract fast-following competition and platform-native features, which is exactly what the skeptics' fresh web research exposed. This is why the tournament design refused to let a "seriously weakened" candidate win merely by out-scoring other weakened candidates.

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8. Round 3: the revival candidates

After eight consecutive kills, the seven verified kill patterns were codified into a **kill taxonomy** (see `ADVERSARIAL_REVIEW.md` §5) and four new candidates were engineered against it: three revived, previously unskepticized problems from the original 34, plus one skeptic-prescribed pivot of a killed candidate (Decision Log, D-009).

Revival concept	Origin in the 34 problems	What it proposed	Skeptic verdicts (impl / market)
Registry Radar (`str-eu-compliance`)	#9 (reg-compliance): EU short-term-rental registration compliance	Host-side compliance system of record for EU STR portfolios: curated registry-requirements DB, renewal deadlines, rule-change alerts, document vault. Per-unit pricing from €19/mo	seriously_weakened / seriously_weakened — **killed**
Backsight (`surveyor-jobtrack`)	#7 (niche-pros): land-surveyor job tracking — *dropped from the original shortlist*	Job pipeline that speaks surveying (fieldwork → drafting → licensed review → delivery), client status links, and a prior-work archive searchable by Section-Township-Range. Flat firm pricing $79/$149/$249	**survives_with_fixes / seriously_weakened (zero fatal flaws) — WINNER**
TerpDesk (`interpreter-backoffice`)	#6 (niche-pros): interpreter-agency back office — also dropped from the original shortlist	Scheduling + dual-rate billing for small onsite interpreter agencies at 1/5th incumbent price. $79–$249/mo flat	seriously_weakened / **kill** (the hardest verdict of the tournament) — **killed**
PlanPatrol (`hvac-radar-pivot`)	Pivot of #0 / `hvac-agreements`, re-architected per round-2 skeptic prescriptions: integration-first, read-only, no SMS, no payments	Read-only "profit-leak radar" over the CSV exports the shop already produces; flags sold-but-never-scheduled members, owed visits, imminent lapses. $79/mo flat	survives_with_fixes / seriously_weakened — **killed** (market fatal flaws)

Why Backsight won (Decision Log, D-010): it was the only concept in all three rounds whose **market skeptic returned zero fatal flaws**. Its market verdict was "seriously weakened" on correctable grounds — competitors exist and were named (Qfactor, SurveyOps, KudurruStone, Cyanic Job Book), the TAM claim was ~3× overstated (corrected to ~7,000 establishments per Census NAICS 541370), and the original evidence citations were vendor marketing — but nothing structural was false. Its implementability verdict was "survives with fixes," with a concrete, fully addressable fix list (license-clean S-T-R parser instead of pyTRS, principal-meridian disambiguation, address-first ingestion, repricing against KudurruStone). Every rejected competitor died on a structural falsehood: a free-tier floor, a platform-native substitute, a technically unsound wedge, a regulated deliverable, or an evidence base that collapsed under spot-checking.

The chosen trade is stated plainly in the Decision Log: preferring a concept with **surviving-but-modest economics** (~$590K ARR at 5% of ~7,000 firms) over concepts with bigger claimed TAMs and fatal flaws — execution risk over structural falsehood. The post-skeptic concept, with all mandatory fixes incorporated, is authoritative in `../business/FINAL_CONCEPT.md`; the build plan is in `../product/MVP_BUILD_SPEC.md`.

Adversarial Review — The Complete Skeptic History

Date compiled: 2026-07-10 **Inputs:** `research/tournament-results.json` (round-1 verdicts), `research/skeptic-round2-partial.json` (round-2 verdicts), `research/revival-round3-results.json` (round-3 concepts and verdicts), `decisions/DECISION_LOG.md`. **Companion document:** `TOURNAMENT.md` — the candidate shortlist, judge panel, and ranking that decided who got attacked.

This document is the proof that the winning concept, **Backsight**, was chosen by evidence rather than preference. Twelve concepts entered adversarial review across three rounds. Eleven were rejected on independently verified facts. Backsight was the only concept whose market skeptic returned **zero fatal flaws**.

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1. Methodology

Every finalist faced **two independent skeptic agents**, each explicitly instructed to *refute* the concept, not to evaluate it neutrally:

Lens	Mandate
Implementability skeptic	"Refute the claim that a non-specialist indie developer can build and run this SaaS with Claude Code and ordinary tools." Investigates with real web research: are the required APIs/data actually public and credential-accessible? Hidden regulatory/licensing exposure? Is core data only obtainable by fragile scraping?
Market skeptic	"Refute this SaaS opportunity." Hunts for well-executed competitors the research missed (app stores, G2, Capterra, ProductHunt, Reddit), evidence the pain is already solved or not worth paying for, unreachable/churning buyers — and **spot-checks 3–4 of the original cited evidence claims with fresh web research**.

Design properties (Decision Log, D-007/D-008/D-009):

- **Web-research-backed.** Skeptics ran live searches and fetches (typically 9–25 tool calls each). The ideation agents deliberately had *no* web access, so every load-bearing claim had to survive independent checking by an adversary who did.
- **Default-to-skepticism.** Verdict scale: `survives` / `survives_with_fixes` / `seriously_weakened` / `kill`. A concept needed *both* lenses to come back clean to win; a single verified fatal flaw was disqualifying.
- **Citation spot-checks.** Skeptics traced the original scouts' evidence to its sources. A recurring finding — vendor content-marketing masquerading as industry data — became kill-pattern #4 in the taxonomy (§5).
- **Equal scrutiny across rounds.** When round 1 killed the entire judge top-4, ranks 5–8 faced the identical prompts and effort rather than winning by default; when round 2 killed those too, four new candidates were engineered against the accumulated kill taxonomy and attacked again.
- **One gap, disclosed:** the round-2 *market* skeptic for `autorenewal-compliance` failed on a session rate limit and never returned a verdict (`skeptic-round2-partial.json` logs). The candidate was already disqualified by its *implementability* verdict (`seriously_weakened`, with two fatal flaws), so the missing run did not affect any decision.

Verdict summary — all three rounds

Round	Candidate	Implementability	Market	Disposition
1	DisputeKit (chargeback-evidence)	survives_with_fixes	**seriously_weakened** (3 fatal flaws)	rejected
1	CertChase (coi-tracking)	survives_with_fixes	**seriously_weakened** (4 fatal flaws)	rejected
1	MarginHawk (threepl-audit)	survives_with_fixes	**seriously_weakened** (2 fatal flaws)	rejected
1	ShortPay (commission-recon)	**seriously_weakened**	**seriously_weakened** (4 fatal flaws)	rejected
2	Statement (eaa-accessibility)	survives_with_fixes	**seriously_weakened** (3 fatal flaws)	rejected
2	TuneUp Keeper (hvac-agreements)	survives_with_fixes	**seriously_weakened** (4 fatal flaws)	rejected
2	Clubhouse HQ (membership-refuge)	survives_with_fixes	**seriously_weakened** (2 fatal flaws)	rejected
2	RenewGuard (autorenewal-compliance)	**seriously_weakened** (2 fatal flaws)	*(run failed — rate limit)*	rejected
3	Registry Radar (str-eu-compliance)	**seriously_weakened** (3 fatal flaws)	**seriously_weakened** (3 fatal flaws)	rejected
3	**Backsight (surveyor-jobtrack)**	**survives_with_fixes** (0 fatal flaws)	**seriously_weakened (0 fatal flaws)**	**WINNER**
3	TerpDesk (interpreter-backoffice)	seriously_weakened (2 fatal flaws)	**kill** (4 fatal flaws)	rejected
3	PlanPatrol (hvac-radar-pivot)	survives_with_fixes	**seriously_weakened** (3 fatal flaws)	rejected

Note the asymmetry that decided the tournament: several candidates earned "seriously_weakened" *with* fatal flaws (structural falsehoods), while Backsight's "seriously_weakened" market verdict contained **no fatal flaws** — only correctable overstatements. Verdict labels alone did not pick the winner; the content of the findings did.

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2. Round 1 — the judge top-4 (all rejected)

2.1 DisputeKit — flat-fee chargeback evidence automation

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-008)

Fatal flaws (market skeptic):

1. **The platform-native substitute already shipped.** Stripe launched Smart Disputes (rolled out Nov–Dec 2025, *enabled by default*): AI analyzes the dispute, auto-assembles evidence from Stripe's internal network data (IP/AVS/CVC/device signals the merchant's own account cannot match), pre-fills the packet, and auto-submits before the deadline if the merchant does nothing (verified by adversarial review, https://docs.stripe.com/disputes/smart-disputes).
2. **Shopify shipped the same thing natively.** Automated Dispute Response for Shopify Payments pulls evidence from order and shipping data and submits before the deadline — both halves of the ICP now have free/default platform automation (verified by adversarial review, https://help.shopify.com/en/manual/payments/chargebacks/resolve-chargeback).
3. **The exact concept already exists.** ChargePay sells flat-fee AI chargeback automation from $19.99/mo, explicitly "no success fees," on the Shopify App Store — the "unoccupied flat-fee wedge" was occupied (verified by adversarial review, https://apps.shopify.com/chargepay).

Key evidence checks (claim → result → source):

Original claim	Skeptic result	Source
Stripe Disputes API allows programmatic evidence submission with restricted keys	**CONFIRMED** — the core buildability claim held; test-mode disputes work end-to-end	https://docs.stripe.com/disputes/api
Chargeflow charges 25% of every recovered chargeback	**CONFIRMED** — uncapped success fee; the resentment wedge is documented	https://www.chargeflow.io/pricing
The angry reviews exist ($4,000+ fees, charged 25% on self-won cases, "lost 10 of 10")	**CONFIRMED verbatim** — but Chargeflow's aggregate rating is 4.6–4.7 across ~375 reviews; the resentful pool is a vocal minority, too small for a beachhead	https://apps.shopify.com/chargeflow/reviews
"Stripe could improve native dispute tooling" is a future hypothetical	**REFUTED — it already happened** (Smart Disputes, 30% success fee on wins, auto-enabled)	https://docs.stripe.com/disputes/smart-disputes
Stripe/Shopify native forms are "blank text boxes"	**OVERSTATED** — both platforms pre-fill evidence	https://help.shopify.com/en/manual/payments/chargebacks/resolve-chargeback
Headline stats ($33.79B, ~80% friendly fraud, $35/$100)	**CONFIRMED as widely repeated figures** — but nearly all sources are chargeback-industry vendors with an incentive to inflate	https://www.chargeback.io/blog/chargeback-statistics
Evidence packs can include "IP/device match" from the merchant's own accounts	**PARTIALLY REFUTED** — Visa CE 3.0 evidence requires data the merchant's integration must have captured; device fingerprints are not exposed by any merchant-accessible API	https://docs.stripe.com/disputes/api/visa-ce3

Additional implementability findings: Shopify protected-customer-data approval is weeks of gatekeeping (verified by adversarial review, https://shopify.dev/docs/api/admin-rest/latest/resources/dispute-evidence); irreversible one-shot submissions on 7–21-day deadlines are operationally unforgiving for a solo dev; Visa's Third Party Agent registration program is a monitorable gray area (https://usa.visa.com/dam/VCOM/download/merchants/tpa-registration-program-faqs.pdf).

2.2 CertChase — COI tracking for small GCs and property managers

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-008)

Fatal flaws (market skeptic):

1. **The core economic premise is false.** "Incumbents monetize human review so they can't serve the sub-200-cert tier below $500/mo" — but bcs offers a permanently **free** plan (25 vendors) that already includes AI-powered COI extraction, automated requests, deficiency notices, and no-login vendor submission, with paid self-service at $11.40/vendor/year (verified by adversarial review, https://www.getbcs.com/blog/free-coi-tracking-software-bcs-vs-competitors).
2. **The "funded player publishes free pricing tomorrow" risk already happened.** TrustLayer Starter gives away AI-powered COI tracking for up to 50 vendors — precisely CertChase's 10–75-vendor beachhead (same source set as above).
3. **The wedge is undifferentiated everywhere in the stack.** Magic-link submission and AI ACORD-25 extraction with instant pass/fail are shipped by bcs, TrackMyVendor, TrustLayer, COISimple, CertFocus/Vertikal — and myCOI's own new AI platform.
4. **Pricing was inverted relative to the market it claimed to undercut** — CertChase Growth ($99/mo for 75 vendors) is ~15× bcs self-service for the same vendor count and ~2.5× TrackMyVendor's $39/mo.

Key evidence checks:

Original claim	Skeptic result	Source
Cheap entrants are "mostly expiration-date spreadsheets in the cloud"	**REFUTED** — TrackMyVendor's $39/mo plan already ships AI COI parsing, magic-link uploads, 90/60/30/7-day alerts, W-9 collection, license checks	https://trackmyvendor.com/coi-tracking-software

Original claim	Skeptic result	Source
Blank ACORD forms are public and free to use	**REFUTED as stated** — ACORD forms are copyrighted and licensed (~$259/yr end-user license); parsing received certs is fine, redistributing blank forms is not	https://www.acord.org/forms-pages/forms-participation-programs
CertChase can verify additional-insured status from the ACORD 25	**PARTIALLY REFUTED** — the ACORD 25 "confers no rights"; AI status lives in the policy endorsement, so cert-only checks are a known compliance gap	https://mycoitracking.com/how-do-you-show-additional-insurance-on-coi/
myCOI runs $500+/mo, priced out below ~200 certs	**PARTIALLY CONTRADICTED** — sole source was a competitor's marketing page; Capterra lists myCOI from $29/user/month	https://www.capterra.com/p/234580/myCOI/
"$8M lawsuit over an expired COI" (Billy)	**NOT INDEPENDENTLY VERIFIABLE** — no court case or news coverage found; unnamed, uncited vendor-blog anecdote	https://billyforinsurance.com/resources/the-importance-of-tracking-insurance/
"$680K absorbed after a drywall sub's policy lapsed" (Struvia)	Appears only on an unsourced content-marketing blog; "treat as illustrative fiction, not evidence"	https://struvia.co/blog/subcontractor-insurance-requirements
"7 of 10 COIs non-compliant" (Jones)	Text exists, but it is an unsourced vendor-marketing statistic	https://getjones.com/blog/how-to-manage-subcontractor-certificates-of-insurance-cois/

Additional implementability findings: forged ACORD 25s can be produced in minutes and an LLM parser validates formatting, not existence of coverage (https://www.certificial.com/blog-post/how-to-detect-fraudulent-certificates-of-insurance-complete-coi-verification-guide); automated chase emails to non-opted-in insurance agents risk ESP suspension under Postmark's ToS (~0.1% complaint tolerance) (https://postmarkapp.com/terms-of-service).

2.3 MarginHawk — 3PL invoice auditing for small DTC brands

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-008)

Fatal flaws (market skeptic):

1. **The whitespace claim is false.** At least three direct competitors reconcile a brand's 3PL invoice against its rate card: Implentio (seed-funded, SKU-level 3PL invoice reconciliation), Mockly (upload-invoice-plus-rate-card audits with a free first review at 30% contingency), and 3plinvoiceaudit.com (contingency warehouse-invoice audits, free initial audit) (verified by adversarial review, https://www.implentio.com/).
2. **The wedge mechanic is technically unsound.** Dimensional weight is billed from the shipped carton's dimensions — the box the 3PL chose, cubiscanned, plus dunnage — **not** from the merchant's SKU dimensions; ShipBob's own docs show per-shipment package dims in the dashboard. "Deterministic dim-weight recalculation from SKU dims" cannot reproduce billable weight for multi-unit or multi-SKU orders (verified by adversarial review, https://support.shipbob.com/s/article/Billing-Overview-Page-in-the-ShipBob-Dashboard).

Key evidence checks:

Original claim	Skeptic result	Source
Core invoice data is user-accessible without gated APIs	**CONFIRMED** — ShipBob has downloadable invoices and a public billing API; ShipMonk has a billing portal	https://support.shipbob.com/s/article/Billing-Overview-Page-in-the-ShipBob-Dashboard
3PL contracts allow disputes within 30–90 days	**CONFIRMED** — the deadline-urgency hook is real (one source puts windows at 15–30 days, tighter than claimed)	https://invoicedataextraction.com/blog/3pl-invoice-reconciliation-guide
Dim-weight overcharges deterministically recomputable and disputable	**REFUTED in part** — carriers enforce their own dimensioner measurements; winning disputes requires timestamped measurement/photo evidence the merchant never possesses	https://packizon.com/ups-vs-fedex-dim-weight-rules-2026/
Refund Retriever charges 50% of refunds	**CONFIRMED** — but it audits carrier refunds, an adjacent market, not 3PL fulfillment fees	https://www.refundretriever.com/questions
Audit services recover 2–5% of spend	**CORROBORATED but weak** — all sources are audit vendors marketing their own services, and the figures refer to carrier parcel spend, not 3PL fees	https://www.darrigoconsulting.com/blog/parcel-audit-services-guide
"15–40% dim-weight inflation, single most common error"	**PARTIALLY CONFIRMED as a claim that exists** — the exact phrasing recycles across 3PL vendor SEO blogs with no primary data	https://shipdudes.com/blog/3pl-billing-audit-how-to-spot-overcharges-and-hidden-fees
The pain itself is real	**PARTIALLY CONFIRMED** — ShipBob Capterra reviews repeatedly cite overcharges, unexpected fees, and audit burden	https://www.capterra.com/p/166529/ShipBob/reviews/

Additional findings: ShipMonk "Period Adjustments" (carrier pass-throughs) are unauditable from the merchant's rate card alone (https://support.shipmonk.com/support/solutions/articles/9000099341-billing-faq); a confidently wrong "$1,214 overcharge" email to the 3PL is commercially catastrophic for trust; invoice lines contain recipient PII, creating data-processor obligations.

2.4 ShortPay — carrier commission reconciliation for small insurance agencies

Verdicts: implementability `seriously_weakened` · market `seriously_weakened` (4 fatal flaws) → **rejected** (D-008)

Fatal flaws (market skeptic):

1. **The claimed moat is commoditized table stakes.** Zero-template AI statement parsing is already sold by Fintary ($10M Series A, Nov 2025, SMB tiers $199–$899/mo) and Comulate ($25M raised), among others (verified by adversarial review, https://siliconangle.com/2025/11/05/fintary-lands-10m-modernize-insurance-commission-management/).
2. **The wedge already exists in superior form** — unLocked CRM advertises automated reconciliation across 332 carrier feeds (no file uploads), real-time underpayment detection, and carrier-ready dispute documentation — and it is the very vendor whose blog supplied the concept's headline $10K–$16K evidence (https://unlockedcrm.ai/blog/insurance-commission-reconciliation-guide).
3. **Every one of the four cited evidence sources is content marketing by a company selling commission reconciliation** (unLocked CRM, CommissionIQ, EnrollHere, AgencyBloc). The pain quantification justifying the ROI story is circular.
4. **Both halves of the ICP have existing cheap/free paths**: IVANS Direct Bill Commission Download feeds structured data into HawkSoft/AMS360/Epic for P&C; Medicare FMOs bundle commission tracking free for downline agents (https://www.ivans.com/globalassets/all-documents/resources/brochures-data-sheets/ivans-dbcs-data-sheet_en-us.pdf; https://enrollhere.com/products).

Key evidence checks:

Original claim	Skeptic result	Source
$10K–$16K/yr leakage	Traced to a direct competitor's SEO blog; **no independent source found anywhere**	https://unlockedcrm.ai/blog/insurance-commission-reconciliation-guide
8–12 hrs/month reconciliation	Appears only on CommissionIQ's own marketing site, unsourced — and CommissionIQ is not the "legacy template-based player" the concept claimed	https://mycommissioniq.com/
AgencyBloc requires full AMS migration; Applied Recon is enterprise-locked	**Substantially refuted** — AgencyBloc Commissions+ is standalone; Applied Recon runs $299–$799/mo and ingests any format	https://www.agencybloc.com/commissions-management/
LLM extraction reliably parses any statement with zero configuration	**Partially refuted** — published benchmarks show ~72–81% accuracy on complex financial tables, with hallucinated numeric values the dominant error mode; payment-grade accuracy needs validation layers far beyond the stated two-week build	https://doi.org/10.3390/computers13100257
Ecosystem dynamics are benign	Applied Systems won a preliminary injunction against Comulate (Feb 2026) in a trade-secrets fight — the dominant AMS vendor litigates against reconciliation challengers	https://www.insurancejournal.com/news/national/2026/02/13/857905.htm
Timing artifacts	EnrollHere's own guide notes many flagged "short-pays" are timing differences that self-resolve — undermining the recoverable-leakage math	https://enrollhere.com/blogs/insurance-commission-reconciliation-guide

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3. Round 2 — ranks 5–8 (all rejected)

Decision D-008 escalated ranks 5–8 to the identical dual-skeptic gauntlet rather than crowning a weakened round-1 survivor.

3.1 Statement — continuous EAA/WCAG compliance for SMB e-commerce

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-009)

Fatal flaws (market skeptic):

1. **The wedge is already shipped, repeatedly, at or below the proposed price.** The pitch's core claim — "every competitor scans; nobody at SMB price does the legal paperwork" — is false. Eye-Able offers an accessibility-statement generator "automatically kept up to date based on your audit reports" (verified by adversarial review, https://eye-able.com/accessibility-statement-generator); the Shopify App Store already lists WCAG Guard ("automated EAA audits, daily scans, visual history logs, generate legal compliance statements") and Avada Accessibility.
2. **The enterprise-vendor framing is factually wrong.** AudioEye sells self-serve at $49/mo with continuous monitoring and a statement included — the exact price point and buyer the concept claimed the "honest players" ignore; Equalize Digital's Accessibility Checker is a free WordPress plugin with a statement generator built in (verified by adversarial review, https://equalizedigital.com/accessibility-checker/pricing/).
3. **The urgency premise decayed in real time.** The first EAA court ruling (Auchan, Tribunal judiciaire de Paris, May 2026) was dismissed — it went to the *defendant*; the Carrefour ruling (June 2026) was an injunction against a €90B retailer, not a fine, and not an SMB. One year post-deadline, no documented enforcement against any small business (verified by adversarial review, https://silktide.com/blog/eaa-auchan-court-ruling/).

Key evidence checks:

Original claim	Skeptic result	Source
FTC $1M penalty against accessiBe	**VERIFIED** (announced Jan 2025; final order April 2025)	https://www.ftc.gov/news-events/news/press-releases/2025/04/ftc-approves-final-order-requiring-accessibe-pay-1-million
French notices July 2025, first EAA lawsuits Nov 2025	**VERIFIED** (ApiDV and Droit Pluriel; référés filed 12 Nov 2025)	https://eye-able.com/en/blog/supermarket-accessibility-lawsuit-france-2025

Original claim	Skeptic result	Source
"No grace period for existing sites"	**CONTRADICTED / overstated** — legal consensus distinguishes services vs products and transitional provisions exist	https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/disability/union-equality-strategy-rights-persons-disabilities-2021-2030/european-accessibility-act_en
800+ overlay users sued, ~25% of cases	**CONFIRMED in substance** via UsableNet's independent 2024 report (1,023 lawsuits, 25%)	https://info.usablenet.com/2024-year-end-report
Automated scanning catches ~30–40% of failures	**PARTIALLY VERIFIED / conservative** — Deque claims axe-core detects ~57% by volume	https://www.deque.com/automated-accessibility-coverage-report/
French statement generation can be automated from scan data	**REFUTED as pitched for France** — RGAA guidance requires the déclaration to result from an effective conformity evaluation	https://accessibilite.numerique.gouv.fr/obligations/evaluation-conformite/
Audit costs $1,500–$5,500	**PLAUSIBLE BUT ONLY BLOG-SOURCED**	https://www.digitala11y.com/how-much-does-a-web-accessibility-audit-cost/

3.2 TuneUp Keeper — standalone HVAC maintenance-agreement manager

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-009)

Fatal flaws (market skeptic):

1. **The standalone architecture contradicts its own wedge.** The "unscheduled visit" alarm only works if every booking is manually mirrored from Jobber/Housecall Pro/Google Calendar into TuneUp Keeper — double-entry by the very overloaded admin the product claims to save.
2. **The wedge is not unique.** Jobber Core ($39/mo) already auto-schedules recurring visits for agreements set up as recurring jobs; Jobber Connect ($119/mo — cheaper than TuneUp Keeper's mid tier) adds automatic payments inside the system of record.
3. **Payments migration blocks the biggest accounts.** Moving 100–800 members' auto-billing to Stripe requires re-collecting card/ACH details from every member; QuickBooks will not export stored card data (last-4 only) (verified by adversarial review, https://docs.stripe.com/get-started/data-migrations/pan-import).
4. **The anchor evidence is a programmatic SEO farm.** The Simply Connected Systems "pain" page supplying the $30k/yr figure is one of hundreds of templated `/pain/...` pages (flagged by adversarial review, https://help.simplyconnectedsystems.com/pain/service-contracts).

Key evidence checks:

Original claim	Skeptic result	Source
ServiceTitan $250–$500/tech/mo, heavy onboarding	**CORROBORATED** by multiple independent 2026 sources ($245–$500/tech/mo)	https://myquoteiq.com/servicetitan-pricing/
$30k/yr renewal-slide math, 2:1 pull-through	**SOURCE EXISTS BUT IS LOW QUALITY** — programmatic SEO pain-page farm; figures are illustrative arithmetic	https://help.simplyconnectedsystems.com/pain/service-contracts
HCP Service Plans add-on +$1,200/yr; add-on creep is #1 churn reason	**TIER-GATING CORROBORATED, CHURN CLAIM UNVERIFIABLE** — sole source is a competitor's review page	https://fieldcamp.ai/reviews/housecall-pro/
SMS is "Twilio behind an adapter"	**UNDERSTATED** — A2P 10DLC requires registering each end-customer shop as its own TCR brand + campaign	https://help.twilio.com/articles/1260800720410-What-is-A2P-10DLC-
Automated SMS renewal outreach carries no special legal exposure	**REFUTED IN PART** — TCPA statutory damages are $500/violation ($1,500 willful), uncapped, private right of action	https://activeprospect.com/blog/tcpa-text-messages/
Jobber squeezes small shops (QuoteIQ source)	**SOURCE HEAVILY FLAGGED** — QuoteIQ is a direct competitor; the article is an explicit hit piece	https://myquoteiq.com/jobbers-biggest-problem-exposed/

3.3 Clubhouse HQ — membership management for clubs fleeing Wild Apricot

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-009)

Fatal flaws (market skeptic):

1. **"None of them owns the migration moment" is factually false.** MemberDay is a near-clone of the exact concept: $29/mo flat, unlimited members *and* contacts (beating Clubhouse HQ's 250-member cap at the same price), free team-done Wild Apricot migration within a week, and a 45-day trial explicitly framed around "run real club work before your board decides" (verified by adversarial review, https://memberday.com/compare/wild-apricot-alternative).
2. **"Wild Apricot alternative" is not low-competition SEO.** Those SERPs are saturated by Zeffy (VC-backed, free, no transaction fees), Mighty Networks, Join It ($29/mo), MembershipWorks ($35/mo), Somiti, MemberDay, and Wild Apricot's own counter-pages.

Key evidence checks:

Original claim	Skeptic result	Source
20% Payment System Servicing Fee for non-AffiniPay processors	**CONFIRMED** via Wild Apricot's own help center	https://gethelp.wildapricot.com/en/articles/1661-payment-system-servicing-fee-pssf
Member data exportable to CSV — no scraping needed	**CONFIRMED**	https://gethelp.wildapricot.com/en/articles/152-exporting-members-and-contacts
Recurring payment profiles can be migrated out	**REFUTED** — switching processors requires cancelling and re-initiating every recurring payment	https://support.wildapricot.com/hc/en-us/articles/360054085394-Switching-recurring-members-to-a-different-payment-system
Price hikes 20% (2021) and 25% (2023); support degradation	**CONFIRMED** (hikes) / **DIRECTIONALLY CONFIRMED** (support — matching complaints on Trustpilot rather than the cited Capterra)	https://www.kesslerfreedman.com/2023/02/another-odd-numbered-year-another-wild-apricot-price-increase/ ; https://www.trustpilot.com/review/wildapricot.com
Change.org petition against the PSSF	**PARTIALLY CONFIRMED — but it dates to February 2019**, seven years ago; Wild Apricot retained customers through two hikes since, evidence of severe switching inertia	https://www.change.org/p/palmeto-corp-wild-apricot-stop-20-payment-system-servicing-fee-on-4-2-19
MembershipWorks $35/mo comparison	**FLAGGED** — sole source is the competitor's own comparison page	https://membershipworks.com/wildapricot-alternative/

Additional weaknesses: ClubExpress already bills by *active* members only (the concept's claimed "direct strike"); Zeffy offers membership management at literally $0; website-hosting lock-in shrinks the addressable wedge; a California AB 488 fundraising-platform tripwire exists if donation features are added (https://www.adlercolvin.com/blog/2024/05/07/california-issues-final-regulations-to-charitable-fundraising-platform-law-five-things-you-need-to-know/).

3.4 RenewGuard — auto-renewal-law compliance auditor

Verdicts: implementability `seriously_weakened` (2 fatal flaws) · market *(run failed — session rate limit; no verdict)* → **rejected** (D-009)

Fatal flaws (implementability skeptic):

1. **The differentiation claim is factually false for the core ICP.** Recharge natively sends 1+ year upcoming-charge notifications with AB 2863-updated default template language; Stripe Billing has built-in upcoming-renewal emails; Loop shipped a "United States ARL compliance" setting covering all states — the $299/mo Operate tier's core deliverable is a platform-native feature (verified by adversarial review, https://getrecharge.com/blog/what-subscription-merchants-should-know-about-californias-updated-automatic-renewal-law/).
2. **The recurring core deliverable is legal judgment, not software.** Red/yellow/green verdicts on a specific merchant's flows plus "compliant" copy is applying law to particular facts — the classic unauthorized-practice-of-law line — and FTC v. DoNotPay (final order Feb 2025, $193K) establishes that claims a product substitutes for lawyer review must be substantiated (verified by adversarial review, https://www.ftc.gov/news-events/news/press-releases/2025/02/ftc-finalizes-order-donotpay-prohibits-deceptive-ai-lawyer-claims-imposes-monetary-relief-requires). The concept's own pitch contained an apparent statutory misstatement (the Maryland/Utah notice-window examples were imprecise per https://www.wiley.law/alert-Automatic-Renewals-and-Risks-State-Negative-Option-Legislation-and-Enforcement-is-Trending) — a live demonstration of the maintenance risk.

Key evidence checks:

Original claim	Skeptic result	Source
HelloFresh $7.5M ARL settlement	**CONFIRMED** (approved Aug 14, 2025, Santa Clara Superior Court)	https://da.lacounty.gov/media/news/hellofresh-pay-75-million-deceptive-subscription-practices-consumer-protection-lawsuit
AB 2863 effective July 1, 2025	**CONFIRMED**	https://www.cooley.com/news/insight/2025/2025-06-04-california-automatic-renewal-law-amendments-take-effect-on-july-1-2025
FTC rule vacated July 2025; rulemaking restarted Jan 2026	**CONFIRMED** — a finalized federal rule could flatten the state-matrix complexity the product sells	https://www.gibsondunn.com/ftc-restarts-negative-option-rulemaking-after-eighth-circuit-vacatur-enforcement-under-rosca-continues/
Legal matrix maintainable from free public sources	**PARTIALLY CONFIRMED** — statutes are public and firms publish free trackers, but the synthesis burden and error liability remain the vendor's	https://www.kelleydrye.com/viewpoints/blogs/ad-law-access/auto-renewal-laws-2025-round-up

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4. Round 3 — the revival candidates

Four candidates engineered against the kill taxonomy (§5), each attacked by the same dual skeptics (D-009).

4.1 Registry Radar — EU short-term-rental compliance (the a-priori favorite)

Verdicts: implementability `seriously_weakened` · market `seriously_weakened` → **rejected** (D-010)

Fatal flaws (both skeptics, converging):

1. **The retention engine tracked obligations that don't exist.** Italy's CIN has no expiry/renewal; Portugal abolished the AL 5-year renewal (Decree-Law 76/2024); Greece's AMA does not expire — "renewal deadlines" largely do not exist in the launch markets (verified by adversarial review, https://www2.gov.pt/en/fichas-de-enquadramento/alojamento-local).
2. **The core evidence was falsified by events.** Spain's Supreme Court (judgment 620/2026, 21 May 2026 — one day after EU Regulation 2024/1028 took full effect) annulled the national Registro Único/NRUA as ultra vires (verified by adversarial review, https://www.poderjudicial.es/cgpj/es/Poder-Judicial/Tribunal-Supremo/Oficina-de-Comunicacion/Notas-de-prensa/El-Tribunal-Supremo-anula-el-Registro-Unico-de-arrendamientos-de-corta-duracion-por-considerar-que-el-Estado-carece-de-competencia-para-su-creacion ; https://skift.com/2026/05/22/spains-top-court-voids-national-short-term-rental-registry/). The skeptic's observation: the concept's own evidence base going stale within weeks *demonstrates* the curation-liability problem it claimed as a moat.
3. **The whitespace claim is false.** Conforme (conforme.info) is live with near-identical positioning — "the regulatory system of record for European STR operators" — tracking NRUA/RNAL/CIN registrations with PMS integrations and a free no-account audit (verified by adversarial review, https://conforme.info/).

Also refuted: the listing-number presence check required scraping Airbnb, rated "Hard" behind Cloudflare Enterprise (https://scraperly.com/scrape/airbnb) — a violation of kill-pattern #3. Confirmed: EU Regulation 2024/1028 itself and Italy CIN fines (https://eur-lex.europa.eu/eli/reg/2024/1028/oj/eng ; https://italianbusinesslawyers.com/cin-compliance-for-italys-tourist-accommodations/).

4.2 Backsight — surveyor job tracking (THE WINNER — detail in §6)

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` — **zero fatal flaws on either lens.** The only such result in the tournament.

4.3 TerpDesk — interpreter-agency back office

Verdicts: implementability `seriously_weakened` · market **`kill`** — the hardest verdict of the tournament → **rejected** (D-010)

Fatal flaws (market skeptic):

1. **The "evidenced $50–200 empty gap" — the concept's entire rationale — does not exist.** At least five purpose-built interpreter-agency tools sit in or below that band, led by Eclipse Scheduling ($125/mo, all plans include scheduling, payroll, billing/invoicing, QuickBooks integration) (verified by adversarial review, https://interpreterscheduling.com/pricing/).
2. **Core differentiation claims are factually false** — Anolla offers a free interpreter-scheduling tier; roundups list Aqua Schedules, Eclipse, MigiHub, Anolla, Caretap, MyInterpreter, Slated — all purpose-built.
3. **The competitive framing was four years stale** — Boostlingo acquired Interpreter Intelligence in March 2022; there are not two ~$500/mo incumbents leaving a hole.
4. **The evidence base collapsed under spot-checking** — the $500/$499 pricing anchors trace solely to Gitnux and ZipDo, sibling AI content farms that cross-publish articles vouching for each other (flagged by adversarial review, https://gitnux.org/best/interpreter-scheduling-software/ ; https://zipdo.co/best/interpreter-scheduling-software/). Boostlingo's actual pricing is quote-only (https://boostlingo.com/resources/pricing/).

Fatal flaws (implementability skeptic): HIPAA business-associate status for the medical half of the ICP was never mentioned — scheduling records for medical interpreting routinely contain PHI, and the specified stack (e.g., SendGrid, which does not sign BAAs for this use per https://www.twilio.com/docs/sendgrid/ui/account-and-settings/hipaa-compliant) is unsellable to hospitals as designed.

4.4 PlanPatrol — HVAC maintenance-agreement leak radar (the skeptic-prescribed pivot)

Verdicts: implementability `survives_with_fixes` · market `seriously_weakened` → **rejected** (D-010)

Fatal flaws (market skeptic):

1. **The premise that membership billing and scheduling "live in two systems no vendor will reconcile" is false for most of the ICP** — Jobber (Connect and up) and Housecall Pro sync invoices natively to QuickBooks Online.
2. **Weekly manual CSV upload is a structural churn engine** — the product is worthless the week the office manager stops exporting three CSVs, and the stated escape hatch (Jobber API sync + marketplace listing) converts the product back into the integration-dependent architecture the pivot was meant to avoid.
3. **$79/mo is priced against alternatives that deliver strictly more** — Housecall Pro's Service Plans add-on (~$1,200/yr per the concept's own evidence) or Jobber Grow give full agreement management, not a read-only list.

Implementability findings (survives, but with teeth): Jobber Core is single-user, so the stated ICP could not even run on it (verified by adversarial review, https://help.getjobber.com/hc/en-us/articles/360048155913-The-Core-Plan); Housecall Pro's API is gated to the MAX plan, making the CSV-forever risk structural (https://docs.housecallpro.com/); QBO Simple Start lacks the line-item report the reconciliation needs (https://quickbooks.intuit.com/learn-support/en-us/help-article/purchase-orders/reports-included-quickbooks-online-subscription/L0s4KrGgr_US_en_US); Jobber report CSVs arrive by email, chunked at 1,500 rows (https://help.getjobber.com/hc/en-us/articles/360043189194-One-Off-Jobs-Report).

The evidence-recheck also re-flagged the entire original HVAC evidence base as vendor content marketing (https://help.simplyconnectedsystems.com/pain/service-contracts ; https://fieldcamp.ai/reviews/servicetitan/ ; https://myquoteiq.com/jobbers-biggest-problem-exposed/ ; https://www.velappity.com/how-do-hvac-companies-track-maintenance-contracts-with-software/).

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5. The seven-pattern kill taxonomy

Distilled from the eight round-1/round-2 autopsies and codified verbatim as design constraints for the round-3 ideators (Decision Log, D-009). Every pattern below killed at least one candidate that had been verified by fresh web research.

#	Kill pattern	Definition	Candidates it killed (examples)
1	**Platform-native substitute**	The platform that owns the data ships the core loop as a free/default feature	DisputeKit (Stripe Smart Disputes, Shopify Automated Dispute Response); RenewGuard (Recharge/Stripe native reminders); PlanPatrol (native QBO sync)
2	**Free-tier floor**	A funded player gives the wedge away below any viable price	CertChase (TrustLayer Starter free/50 vendors; bcs free/25); Clubhouse HQ (Zeffy at $0); TerpDesk (Anolla free tier)
3	**Technically unsound wedge**	The demo-able mechanic doesn't survive contact with how the domain actually works	MarginHawk (dim-weight billed from cubiscanned cartons, not SKU dims); Registry Radar (listing checks require scraping Cloudflare-protected Airbnb); Statement (French déclaration can't be auto-generated from scans)
4	**Evidence built on vendor content-marketing**	Load-bearing statistics trace to competitors' SEO blogs or content farms, never to primary sources	ShortPay (all four sources were competing vendors); TuneUp Keeper (programmatic SEO pain-farm); TerpDesk (Gitnux/ZipDo content farms); CertChase ($8M/$680K anecdotes unverifiable)
5	**Regulated deliverables**	The recurring deliverable is legal judgment, regulated messaging, or a payments migration a solo vendor can't safely own	RenewGuard (UPL / FTC v. DoNotPay); TuneUp Keeper (TCPA/A2P 10DLC SMS, card re-collection); TerpDesk (HIPAA BAA chain)
6	**Saturated panic-SEO distribution**	The distribution plan is high-intent keywords already blanketed by 6+ competitors running the same playbook	CertChase ("myCOI alternatives"); Clubhouse HQ ("Wild Apricot alternative"); Statement (EAA panic queries); MarginHawk ("3PL billing audit")
7	**Structural churn**	The product's success extinguishes its own subscription, or the buyer graduates/dies quickly	MarginHawk (clean months = cancel); LinkPatrol-style catalog cleanup (judge-flagged); PlanPatrol (CSV-habit decay); TuneUp Keeper (graduation to full FSM suites)

The taxonomy is itself a research asset: round 3 candidates were *engineered* against it, and the winner had to survive attacks that killed eight predecessors — the tournament equivalent of a regression suite.

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6. The Backsight verdict in detail

6.1 What the market skeptic found (`seriously_weakened`, zero fatal flaws)

Every finding was a correction, not a structural falsehood:

Finding	Detail	Source
Vertical competitors exist	At least six vertical land-surveying practice-management products missed or understated: Qfactor (est. 2016, rebranded June 2026), KudurruStone, Cyanic Job Book, Info-Retriever, CQ, SurveyOps	https://survey-ops.com/land-surveying-job-management-software
The wedge is not unique as claimed	Qfactor's Map View shows past projects spatially; Cyanic offers legal-address job search; SurveyOps names S-T-R search on its own landing page	https://survey-ops.com/land-surveying-job-management-software
Pricing positioned above the real competition	KudurruStone ≈ $100/mo for a 10-person firm (per-user by role) vs Backsight Firm $149 / Practice $249; the concept anchored only against BQE Core	(skeptic fresh research; KudurruStone site blocked direct fetch — corroborated via directories)
TAM overstated ~3×	Census County Business Patterns, NAICS 541370: ~7,000 firms / ~7,382 establishments (2020), not the claimed 25,000+	https://siccode.com/naics-code/541370/surveying-mapping
Lead magnet slot occupied	randymajors.org, Earth Point, MapScaping, and BLM's own viewers already own free S-T-R lookup	https://www.earthpoint.us/Townships.aspx
All three original evidence citations were vendor marketing	CQ's SEO blog, SurveyOps's own landing page, QuoteIQ's self-ranking listicle	https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/ ; https://myquoteiq.com/top-8-softwares-for-land-surveying-businesses-in-2026/
Adoption friction is real	An RPLS.com forum surveyor abandoned KudurruStone as "too slow"; Qfactor has sold into this buyer since 2016 with a near-zero review footprint — the segment is winnable but slow	(skeptic fresh research, RPLS.com thread)

The skeptic's residual — and decisive — observation: **the most defensible verified gap is that none of the vertical competitors advertise automated parsing of legal descriptions from a firm's messy historical spreadsheet.** They map jobs going forward or require manual entry; nobody monetizes the *back catalog*. That is exactly the wedge Backsight kept.

6.2 What the implementability skeptic found (`survives_with_fixes`)

Core claim survived: "every required dataset and API is genuinely public and credential-accessible... nothing in the core loop depends on scraping, and no regulatory/licensing regime touches a job-tracking vendor."

Check	Result	Source
BLM PLSS data is public-domain, bulk-downloadable, no credentials	**VERIFIED**	https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons
QuickBooks Online invoice push is publicly accessible	**VERIFIED** with a manageable hurdle (Intuit app-assessment questionnaire before production keys)	https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ
S-T-R parsing is a drop-in solved problem	**REFUTED AS STATED** — pyTRS, the one mature library, prohibits all commercial use under a Modified Academic Public License	https://github.com/JamesPImes/pyTRS
PLSS excludes Texas + colonial states	**VERIFIED and slightly worse than framed** — 20 states outside PLSS; Texas uses its own GLO abstract system	https://en.wikipedia.org/wiki/Texas_land_survey_system
SurveyOps is a generic-leading entrant to out-position	**PARTIALLY REFUTED** — it claims S-T-R search itself and ships an iOS app	https://survey-ops.com/land-surveying-job-management-software
No regulatory/licensing exposure	**VERIFIED by absence** (moderate confidence) — the "licensed review" stage is a workflow label, not a professional act by the vendor	(skeptic search of state-board regimes)
Free S-T-R lookup is a novel lead magnet	**PARTIALLY REFUTED** — Township America already operates one (its paid geocoding API also proves the component is buyable)	https://townshipamerica.com/

Named weaknesses: pyTRS license; principal-meridian ambiguity ("T2N R3W Sec 14" repeats across 37 principal meridians and firm spreadsheets never record the meridian); many firm spreadsheets contain only client + street address, degrading the S-T-R demo; archive-of-record data-loss exposure (contractual, needs real backups); and a 6–8-week build estimate judged optimistic (realistically 12–16 weeks to a sellable v1).

6.3 The fix list, and where each fix landed

The skeptics' improvements were adopted as **mandatory fixes**, incorporated into the authoritative concept (`../business/FINAL_CONCEPT.md` , "Mandatory fixes from the skeptics") and the build plan (`../product/MVP_BUILD_SPEC.md`):

#	Skeptic finding	Fix	Where incorporated
1	pyTRS commercially unlicensed	Ship an original, license-clean S-T-R parser; never use or port pyTRS	FINAL_CONCEPT.md fix #1; MVP_BUILD_SPEC.md `lib/plss.ts` — "LICENSE-CLEAN, written from scratch... Do NOT use or port pyTRS (license-restricted); write original code," with ≥12 vitest cases
2	Principal-meridian ambiguity silently matches the wrong ground	State→default-meridian table; explicit `ambiguous: true` flag; prefer geocoded address when present	FINAL_CONCEPT.md fix #2; MVP_BUILD_SPEC.md `lib/plss.ts` (meridian table + ambiguity flag) and `jobs.plss_meridian` column in the data model
3	Spreadsheets often lack legal descriptions	Address-first ingestion: geocoding (US Census Geocoder, free, no key) is the primary spatial index; S-T-R parsing is enrichment, not a requirement	FINAL_CONCEPT.md fix #3; MVP_BUILD_SPEC.md Radar page accepts address **or** S-T-R search; `lib/geocode.ts` stub with documented Census Geocoder TODO
4	"No competition" framing false	Name Qfactor, SurveyOps, KudurruStone, Cyanic Job Book in the competitor analysis; differentiate on prior-work monetization + client status links + flat pricing, not on whitespace	FINAL_CONCEPT.md fix #4 and its "Honest competitive framing" paragraph
5	TAM overstated ~3×	Recompute from Census NAICS 541370: ~~7,000 establishments; honest outcome math (~~$590K ARR at 5% penetration) presented as a strong bootstrapped outcome, not venture scale	FINAL_CONCEPT.md "Honest market sizing (post-skeptic)"
6	Pricing above KudurruStone's per-user reality	Flat per-firm tiers ($79/$149/$249) deliberately positioned so the math beats per-user pricing at team size	FINAL_CONCEPT.md "Pricing (post-skeptic reposition)"; MVP_BUILD_SPEC.md landing page 3-tier pricing
7	Saturated S-T-R-lookup lead magnet	Distribution redirected to state survey societies, NSPS chapters, r/Surveying, and conference booths instead of the occupied SEO slot	FINAL_CONCEPT.md "Why it dodges the kill taxonomy" #6

Skeptic recommendations tracked but not yet fully implemented in the MVP demo (honestly labeled): live Census Geocoder integration is stubbed (`lib/geocode.ts` TODO — the MVP's "What's real vs. mocked" settings panel discloses this); QuickBooks sync is scoped to the Practice tier post-MVP with the Intuit assessment noted; automated offsite backups/data-export and the 12–16-week v1 schedule are execution items for the build phase, recorded via the skeptic verdicts in `revival-round3-results.json` .

6.4 Why this verdict, and not a higher-scoring one, chose the winner

Backsight ranked nowhere in the original judged tournament — its source problem (#7 of the 34) had been *dropped from the shortlist* for weak willingness-to-pay evidence. It won because it was the only concept, out of twelve attacked across three rounds, for which two adversaries armed with live web research **could not find a single disqualifying fact**: no platform-native substitute (kill #1), no free-tier floor found by adversarial search (kill #2), a deterministic and locally demonstrable wedge (kill #3), evidence re-based on Census data and public pricing pages after the vendor-blog citations were flagged (kill #4), no regulated deliverables (kill #5), distribution through societies and communities rather than contested SEO (kill #6), and a compounding archive whose value grows with every job (kill #7).

The Decision Log (D-010) states the trade explicitly: a surviving concept with honest, modest economics (~7,000-firm TAM, ~$590K ARR at 5% penetration) was preferred over concepts with bigger claimed markets and verified structural falsehoods. Eleven kills, each on cited evidence, are the proof.

Final Selected Concept: Backsight

> **One-liner:** Job tracking built for land-surveying firms — a pipeline that speaks surveying (fieldwork → drafting → licensed review → delivery), a client status link that kills "where's my survey?" calls, and a prior-work archive searchable by location so every new request instantly shows what the firm already knows about that ground.

(The term "backsight" is a surveying fundamental: the sighting a surveyor takes back to a known point to orient every new measurement — exactly what this product does with a firm's past jobs. Working title; trademark/name-collision check required before launch.)

Why this concept won (tournament summary)

Backsight was the **only concept out of 12** across three adversarial rounds whose market skeptic found **zero fatal flaws**. Every other finalist died on a verified structural defect:

Round	Concept	Fate	Killing fact (verified by skeptic web research)
1	Chargeback evidence automation	killed	Stripe Smart Disputes (Nov 2025, default-on) + Shopify native + ChargePay at $19.99/mo
1	COI tracking	killed	TrustLayer Starter & bcs give AI COI tracking away **free** at the low end
1	3PL invoice audit	killed	Wedge technically unsound: carriers bill from cubiscanned cartons, not SKU dims
1	Commission reconciliation	killed	AI parsing commoditized (Fintary $199/mo, unLocked 332 feeds); evidence = vendor blogs
2	EAA accessibility monitor	killed	Eye-Able/AudioEye ship scan+statement at $49/mo; first EAA rulings went to defendants
2	HVAC agreements (standalone)	killed	Standalone architecture contradicts wedge; Jobber Core auto-schedules recurring visits
2	Membership refuge	killed	MemberDay is a $29/mo near-clone; Zeffy is free; SERPs saturated
2	Auto-renewal compliance	killed	Deliverable is legal judgment (FTC v. DoNotPay exposure); Recharge sends mandated reminders natively
3	Registry Radar (EU STR)	killed	Conforme.info ships identical positioning; Spain's Supreme Court annulled the NRUA registry 21 May 2026; launch markets have no renewal deadlines to track
3	TerpDesk (interpreter agencies)	**killed** (hardest verdict of the tournament)	Five purpose-built incumbents in the "empty" $50–200 gap; HIPAA business-associate status unmentioned
3	PlanPatrol (HVAC radar pivot)	killed	Weekly manual CSV upload is a structural churn engine; QBO sync already reconciles the two systems
3	**Backsight (surveyors)**	**WINNER — survives with fixes**	Market skeptic found competitors but **no fatal flaw**; implementability skeptic's objections are all engineering-addressable

The concept (post-skeptic, fixes incorporated)

Customer: Owner/licensed surveyor of a US firm with 1–5 field crews and 3–25 staff ($300K–$5M revenue), running 20–80 concurrent jobs (boundary, ALTA/NSPS land title, and topographic surveys at $500–$5,000 each) on a whiteboard plus email and a spreadsheet. The owner personally fields status calls from title companies and builders, and personally remembers "we shot that section in 2019."

Painful workflow:

1. **Job-status chaos** — field crews, drafting, and licensed review live in different tools (or none); title companies and builders call the owner for status; deadlines slip silently between stages.
2. **Prior-work amnesia** — the firm's most valuable asset is what it already knows about ground it has surveyed before (control points, prior boundary resolutions, plats). That knowledge lives in the owner's head and a shelf of file folders. When a request comes in for a parcel the firm surveyed in 2019, quoting should take minutes and the job should be cheaper than any competitor's — but only if anyone remembers.

Wedge feature — Prior-Work Radar: import the firm's historical job list (CSV), and Backsight indexes every job spatially — by address geocode and, where available, by Section-Township-Range parsed with our own (license-clean) PLSS parser against public-domain BLM PLSS data. Every new request instantly shows prior jobs on and around that parcel: *"You have 3 prior jobs in this section — quote with confidence, win it on price, deliver it at half the field time."*

Retention features: the surveying-native pipeline board (Request → Quote → Scheduled → Fieldwork → Drafting → Licensed Review → Delivered → Invoiced) and the tokenized read-only **client status link** that eliminates "where's my survey?" calls.

Pricing (post-skeptic reposition): flat per-firm, not per-user — $79/mo (up to 5 users), $149/mo (up to 15), $249/mo (unlimited) — deliberately flat against KudurruStone's per-user/role pricing so the math wins at team size, and far under Qfactor/SurveyOps annual contracts. Flat pricing is also the anti-add-on-creep message this buyer segment (per the SMB research) demonstrably resents.

Honest market sizing (post-skeptic): ~7,000 US establishments in NAICS 541370 (Census County Business Patterns; the skeptic corrected the scout's 25,000 figure), of which the 3–25-staff segment is the majority. At 5% penetration × ~$140/mo average → ~$590K ARR: a strong bootstrapped outcome, honestly not a venture-scale one. Adjacent expansion: septic designers, foresters, geotechnical field firms share the field→office→licensed-review shape.

Mandatory fixes from the skeptics (all incorporated into the PRD):

1. **No pyTRS** — its Modified Academic Public License prohibits commercial use. We ship our own regex-based S-T-R parser (it's a constrained grammar; public-domain BLM PLSS CadNSDI shapefiles provide the spatial join).
2. **Principal-meridian ambiguity** — "T2N R3W Sec 14" repeats across principal meridians; we resolve with a state→default-meridian table, flag ambiguity, and always prefer geocoded address when present.
3. **Address-first ingestion** — many firm spreadsheets have client + street address only; geocoding (US Census Geocoder, free, no key) is the primary spatial index; S-T-R parsing is enrichment, not a requirement.
4. **Honest competitive framing** — Qfactor, SurveyOps, KudurruStone, Cyanic Job Book et al. exist and are named in the competitor analysis; differentiation is the prior-work-monetization wedge + client status links + modern UX + flat pricing, not "no competition exists."

Why it dodges the kill taxonomy:

1. *Platform-native substitute:* none — there is no platform in this workflow to subsume it.
2. *Free-tier floor:* none found by adversarial search; competitors are paid micro-vendors.
3. *Technically sound wedge:* parsing + geocoding + spatial join is deterministic and fully demonstrable locally with sample data.
4. *Primary-source evidence:* buyer economics verified against public pricing pages and Census establishment counts; no vendor-blog statistics load-bearing.
5. *Regulated deliverables:* none — no legal language generation, no SMS, no payments migration. (The licensed review stage is the customer's professional duty; Backsight only tracks it.)
6. *Saturated SEO:* the micro-vendor category has thin content; distribution runs through state survey societies, NSPS chapters, r/Surveying (~100k+ members), and conference booths measured in hundreds of dollars.
7. *Structural churn:* jobs keep flowing; the archive gets more valuable every month (the moat compounds); firms rarely "graduate" out of the segment.

Customer Personas — Backsight

Date: 2026-07-10

Two personas: the **primary buyer** (owner / licensed surveyor) and the **daily user** (office manager / drafter). Both are **illustrative composites** built strictly from the evidence in `research/raw-scan-full.json`, the round-3 adversarial reviews in `research/revival-round3-results.json`, and the post-skeptic concept in `business/FINAL_CONCEPT.md`. Names match the fictional demo firm in the MVP ("Whitfield Land Surveying," Fort Collins, CO — a PLSS state on the 6th Principal Meridian). No detail here is a claim about a real person; quantified traits are labeled estimates where they are not sourced.

Persona 1 — Primary buyer: "Dana Whitfield, PLS"

Owner and licensed professional surveyor, Whitfield Land Surveying

Attribute	Detail
Role	Owner, sole license-holder, chief rainmaker, final reviewer of every deliverable
Firm	3–25 staff, 1–5 field crews, $300K–$5M revenue (target-segment definition, `business/FINAL_CONCEPT.md`)
Workload	20–80 concurrent jobs — boundary, ALTA, topo — at $500–$5,000 each
Current stack	Whiteboard + email folders + a job spreadsheet; world-class field/CAD tools (GNSS, Carlson/Trimble-class software); scout-estimated $10–30K per GNSS receiver already spent on hardware
Geography (launch ICP)	PLSS states (Midwest / Mountain West) — the wedge does not cover Texas or the colonial metes-and-bounds states (verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system)

Day in the life

- **6:45** — dispatches two crews by text; one asks which control was set on the Hendersons' adjacent parcel "a few years back." Dana thinks it was 2019. Nobody is sure which folder.
- **8:30–11:00** — drafting reviews and stamping. Interrupted four times by status calls: two title companies with closings this week, a builder, a homeowner. Each answer requires reconstructing status from email and the whiteboard (pain pattern per the scout evidence — clients call for status, drafters wait on field notes, invoices slip: https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/, flagged as a vendor source; frequency is an estimate pending interviews).
- **11:30** — a quote request lands for a rural parcel. Dana suspects the firm surveyed the neighboring quarter-section but can't confirm without an hour in the file room, so the quote goes out priced like a stranger's job.
- **Afternoon** — field visit to an ALTA site; the whiteboard back at the office is now stale.
- **18:00** — remembers three delivered jobs still haven't been invoiced. Adds it to tomorrow's list. Again.

Jobs to be done

1. *Know where every job stands without being the human database* — so deadlines stop slipping silently between fieldwork, drafting, and review.
2. *Answer "where's my survey?" without answering the phone* — protect billable review hours.
3. *Turn the firm's history into quoting power* — when the firm has prior control/plats on or near a parcel, quote in minutes, win on price, deliver at lower field cost.
4. *Get delivered work invoiced* the week it ships.
5. *Get the archive out of my head* — the firm should survive Dana's memory (succession, illness, sale).

Buying triggers

- A lost bid on ground the firm had already surveyed — the prior-work advantage priced at zero.
- A blown title-company deadline discovered only when the client called.
- A month-end discovery of delivered-but-unbilled jobs.
- Hiring crew #2 or #3 — the whiteboard stops scaling.
- Seeing "Tracked with Backsight" on a competitor's client status page (built-in referral loop per concept distribution plan).

Objections (evidence-grounded)

Objection	Basis	Counter Backsight must actually deliver
"We tried a vertical tool; it was too slow"	An RPLS.com forum surveyor abandoned KudurruStone as "too slow" (adversarial review, round 3)	Speed and simplicity over feature count
"ClickUp costs $3 a user"	RPLS forum users report running firms on ClickUp (adversarial review, round 3)	Value must come from what generic tools can't do: legal-description search, prior-work radar, client status links

Objection	Basis	Counter Backsight must actually deliver
"My data is my business — what if you disappear?"	Archive-of-record data-loss exposure flagged by implementability skeptic	Full CSV+files export in v1, documented backups/restore drill
"I don't want per-seat pricing that punishes field crews"	Segment's documented resentment of add-on/per-seat creep (SMB-trades research, e.g. https://myquoteiq.com/jobbers-biggest-problem-exposed/); KudurruStone charges per-user	Flat per-firm pricing: $79 / $149 / $249
"Wrong-section match once and I'm done"	Principal-meridian ambiguity failure mode (adversarial review, round 3)	Explicit ambiguity flags, human map-confirmation on import, address-first indexing
"Six other vendors already call me"	Qfactor, KudurruStone, Cyanic, Info-Retriever, CQ, SurveyOps all exist (adversarial review, round 3)	Lead with historical-import concierge: day-one value from the firm's existing spreadsheet

Watering holes

- **State survey societies** and their newsletters/conferences under **NSPS** — every state has one; sponsorships priced in the hundreds of dollars (concept distribution plan).
- **r/Surveying** (~100k+ members per `business/FINAL_CONCEPT.md`) and **RPLS.com forums** — where the practice-management threads the skeptic mined actually live.
- Surveying podcasts (gear-and-practice shows).
- **GNSS equipment dealers and survey-supply houses** — trusted vendors who talk to every small firm and have nothing to sell them for the office.

What Dana already pays for

- GNSS receivers and data collectors (scout estimate $10–30K per unit) and CAD/technical software from Trimble/Leica/Carlson-class vendors.
- Possibly a generic PM tool (ClickUp ~$3/user/mo per RPLS forum evidence) or nothing — the whiteboard is free.
- Reference price anchors in the vertical: KudurruStone ≈ $100/mo for a 10-person firm (per-user, per adversarial review); BQE Core ~$25–50/user/mo (scout anchor); SurveyOps and Qfactor sell paid annual contracts (pricing not publicly verifiable — their sites blocked automated review; noted as a verification gap in round 3).

Persona 2 — Daily user: "Marcus Lee"

Office manager / drafting technician, Whitfield Land Surveying

Attribute	Detail
Role	Runs the office: intake, scheduling, chasing field notes, drafting support, invoicing, and the phone
Reports to	Dana (owner). Not the economic buyer, but the veto: if Marcus won't live in the tool, it dies in the trial
Current tools	The job spreadsheet (his creation), email, the whiteboard, QuickBooks Online for invoicing
Technical comfort	High on office software, allergic to "systems" that add data entry without removing work

Day in the life

- **8:00** — reconciles the whiteboard with reality: calls crew A for yesterday's status, discovers a job moved to drafting Friday and nobody updated anything.
- **9:00–12:00** — the phone. Title companies calling about closings; Marcus digs through email threads to answer, interrupting his drafting queue each time (idle-drafter/status-call pattern per scout evidence, vendor-flagged source).
- **13:00** — assembles field notes for the drafting backlog; one job stalls because the crew's notes are on a device in a truck.
- **15:00** — invoicing hour that usually evaporates; delivered jobs from two weeks ago still unbilled.
- **16:30** — updates the spreadsheet, knowing it's already wrong somewhere.

Jobs to be done

1. *One source of truth* that field crews actually update from a phone browser — no re-keying from texts and calls.
2. *Deflect the phone*: send every repeat caller a status link instead of playing detective (the tokenized read-only client status page).
3. *Never let a delivered job sit unbilled*: a "stuck jobs" digest with delivered-but-uninvoiced first, and one-click invoice push to QuickBooks Online (public OAuth API, verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ).
4. *Find prior work without asking Dana*: search by address or Section-Township-Range instead of relying on the owner's memory.
5. *Painless import*: the historical spreadsheet must load without weeks of cleanup — address-first geocoding, map pin-drop fallback (mandatory fix from adversarial review; many firm spreadsheets have client + street address only).

What makes Marcus champion it — and what makes him kill it

Champion: the trial imports the firm's existing spreadsheet and immediately shows (a) a map of everything the firm ever surveyed, (b) the delivered-but-unbilled list, (c) a status link he can paste to the next title-company caller. Value on day one, using data he already maintains.

Kill: double data entry (tool + whiteboard both maintained), a slow UI (the KudurruStone "too slow" abandonment is the cautionary tale), field crews refusing the mobile flow, or a single wrong-section search result that makes Dana distrust the archive.

Persona-to-evidence traceability

Persona element	Evidence
Firm size, job volume/pricing, whiteboard status quo	Scout problem record, `research/raw-scan-full.json` (land-surveying job tracking)
Status calls / idle drafters / delayed invoices	https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/ (vendor source, flagged; to be validated in interviews per `research/PROBLEM_VALIDATION.md`)
S-T-R search need; vertical entrants	https://survey-ops.com/land-surveying-job-management-software (competitor page; confirms need and competition)
ClickUp satisficing; KudurruStone "too slow"; six vertical competitors; per-user price floor	Round-3 market skeptic, `research/revival-round3-results.json`
TAM ~7,000 establishments (NAICS 541370)	verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping
PLSS coverage limits shaping launch geography	verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system
QBO invoicing integration feasibility	verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ
Flat-pricing resentment of per-seat/add-on creep	SMB-trades scout evidence, e.g. https://myquoteiq.com/jobbers-biggest-problem-exposed/

Caveat: interruption counts, hours lost, and dollar magnitudes in the day-in-the-life narratives are illustrative estimates, not measured data. The interview plan in `research/PROBLEM_VALIDATION.md` exists to replace them with numbers from 10–15 real firm owners before further build investment.

Backsight — Pricing Strategy

Date: 2026-07-10 · **Status:** Launch pricing per `FINAL_CONCEPT.md`; test plan below governs changes.

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1. The tiers

Tier	Price	Users	Included	Who it's for
Solo	**$79/mo** ($790/yr — 2 months free)	Up to 5	Pipeline, Prior-Work Radar, historical import, client status links, full data export	1-crew shops, 3–5 people
Firm	**$149/mo** ($1,490/yr)	Up to 15	Everything in Solo + QuickBooks Online invoice push	2–3 crews, 6–15 people — the center of the ICP
Practice	**$249/mo** ($2,490/yr)	Unlimited	Everything in Firm + client-branded status pages	4–5 crews, 16–25 people

All tiers: 14-day free trial, no credit card to start, no setup/onboarding/import fees, no per-user charges beyond the tier cap, self-serve full export on every tier.

Blended ARPA assumption for planning: ~$140/mo (per `FINAL_CONCEPT.md`; assumes a mix skewed toward Firm — this is an assumption until real tier mix data exists).

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2. Rationale anchored to the verified competitive landscape

The adversarial review (round 3, market lens) corrected our original pricing posture. The original concept anchored only against BQE Core — the most expensive comparator — and the skeptic's blunt finding was that Backsight as first drafted was *"positioned above, not below, the real vertical competition."* This strategy is rebuilt against the verified field.

2.1 KudurruStone — the per-user math we price against

KudurruStone charges **$25/$15/$5 per user/month by role**, so a 10-person firm lands at roughly **$100/mo**, with no onboarding/import/training fees and a free trial (verified by adversarial review; kudurrustone.com blocked automated fetch, figures obtained via the skeptic's search-index research and flagged for manual re-confirmation before being used in sales collateral).

What this means for us, honestly:

- **We do not win a 5-person price fight against KudurruStone.** A 5-person firm on KudurruStone might pay $75–125/mo depending on role mix; Backsight Solo at $79 is parity, not a discount. We concede the "cheapest" claim at the small end (this concession is recorded in `research/GAP_ANALYSIS.md`).
- **We win the math as the team grows, and we win it predictably.** Per-user pricing creates a hiring tax and a role-classification chore. Flat tiers mean the owner knows the number, and adding a crew chief or a summer field hand costs $0. At 10–15 people, Firm at $149 is in the same band as KudurruStone's ~$100–200 but with zero marginal seat cost and no incentive to under-license field staff.
- **The relevant KudurruStone datapoint isn't only price:** an RPLS forum surveyor abandoned it as *"too slow"* (verified by adversarial review). Speed and simplicity are part of what the price buys; we treat product latency as a pricing feature.

2.2 Qfactor and SurveyOps — the posture we price under

- **Qfactor** (founded 2016, rebranded June 2026 claiming "leading software for land surveying firms") has been selling into this buyer for ~10 years and remains obscure with a near-zero review footprint (verified by adversarial review). Its Map View already shows past-project locations. Exact pricing could not be confirmed — qfactor-llc.com blocked automated fetch (per skeptic notes).
- **SurveyOps** ships an iOS field app, S-T-R job organization, guided onboarding at no extra cost, and runs aggressive programmatic SEO (verified by adversarial review, https://survey-ops.com/land-surveying-job-management-software; site blocked deeper automated fetch, so tier prices are unconfirmed). `FINAL_CONCEPT.md` characterizes Qfactor/SurveyOps as annual-contract postures we deliberately undercut with monthly, published, no-quote pricing.

Our posture against both: published prices on the website, monthly billing available, no sales call required, no annual lock-in required. In a segment where a decade-old vendor is still obscure, transparency itself is differentiation. **Action item (from skeptic):** manually demo SurveyOps and Qfactor and re-confirm their pricing before any comparison collateral goes out — their exact price points are a verification gap.

2.3 The floor below everyone: ClickUp and the whiteboard

Surveyors on the RPLS forum report happily running firms on ClickUp at **~$3/user/mo** (verified by adversarial review), and the modal "competitor" is a whiteboard plus email — cost $0. This is the real pricing constraint: **Backsight's $79–249 must be justified against near-free satisficing, not against other vertical vendors.** The justification is the wedge: ClickUp cannot spatially index a job archive or parse an S-T-R string, and neither can a whiteboard. Every trial must therefore surface a dollar-denominated artifact fast — a radar hit on ground already surveyed, or a delivered-but-uninvoiced job found in the import. That artifact, not feature count, is what carries the price.

2.4 Why flat per-firm (not per-seat)

1. **Field crews must never count against a license** (`FINAL_CONCEPT.md`). Per-seat pricing punishes exactly the behavior we need — crews updating stages from the field. Under-licensing field staff is the known SMB failure mode of per-seat tools.
2. **Anti-add-on-creep is a message this segment demonstrably resents being violated** (per the SMB research behind `FINAL_CONCEPT.md`). One price, everything included at the tier, is a sales line the owner can repeat to their partner in one sentence.
3. **It converts KudurruStone's model into our talking point:** "their price goes up when you hire; ours doesn't."
4. **It fits the buyer's mental model:** firms budget software like an insurance premium or a license fee — per firm, per year — not like headcount.

The known cost of flat pricing: we leave money on the table at the largest firms and can't monetize seat growth. Section 4 addresses expansion revenue without breaking the flat promise.

2.5 Value anchor (the line sales actually uses)

Jobs in the segment run $500–$5,000 each (`FINAL_CONCEPT.md`). At $149/mo:

- **One radar-assisted win** on previously-surveyed ground pays for 3–30 months of Backsight, before counting the field time saved by existing control.
- **One delivered-but-uninvoiced job caught** by the stuck-jobs digest typically pays for a year.
- These are illustrative arithmetic on our own price and the concept's stated job values — **not market statistics** — and should be presented as such.

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3. Discounting policy

Deliberately narrow. Published price integrity is a strategic asset against quote-only competitors.

Discount	Terms	Why
Annual prepay	2 months free (≈16.7%)	Cash flow + churn insulation; industry-standard, already in `FINAL_CONCEPT.md`
Design partner (first 10 firms)	50% off for 12 months, in writing, in exchange for: 2 feedback calls/quarter, permission to use anonymized usage data, and a quote if (and only if) they're genuinely happy	Buys the social proof and interview access the validation plan requires; capped and time-boxed
State society member	10% off first year with a society partnership code	Makes society partnerships (see `GO_TO_MARKET.md`) concretely valuable to the society and trackable for us
Everything else	No	No haggling, no "founder's special" beyond the above, no free tier (the audit lead magnet is the free taste — see `CUSTOMER_ACQUISITION.md`), no discounts to close a quarter

No free tier, ever, at launch: the adversarial review across 12 concepts showed free-tier floors killing paid products; no funded free tier exists in this vertical (per `FINAL_CONCEPT.md`), and we will not create the precedent ourselves.

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4. Expansion revenue paths (without breaking "flat")

Ordered by likelihood; none are committed roadmap, all preserve the flat-per-firm promise.

1. **Tier upgrades (organic):** Solo → Firm on hiring or QuickBooks need; Firm → Practice on the 16th user or the first client who should see a branded status page. This is the primary expansion motion; the tier caps are set so growth naturally crosses them.
2. **Annual conversion:** monthly → annual is +0 revenue but −churn; treat it as expansion for LTV purposes.
3. **Archive services (one-time fees):** paid concierge digitization of paper archives (scan → index → pin) beyond the free CSV import — priced per job or per box, one-time, explicitly not a subscription. Labor-intensive; only offer once there's demand pull and capacity.
4. **Adjacent-vertical editions** (septic designers, foresters, geotechnical field firms — `FINAL_CONCEPT.md`): same field→office→licensed-review shape, new TAM rather than new price. 2027 question, not 2026.
5. **Later candidates, explicitly not promised:** county-recorder data enrichment, title-company-side portal (multi-surveyor status dashboard for the title co. that already sees our links). Both monetize a *different buyer*, so they don't violate firm-flat pricing.

Anti-path (do not do): per-crew add-ons, SMS packs, storage tiers, "premium support." Each contradicts the anti-add-on-creep positioning that is load-bearing in the sales story.

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5. Price-test plan

Principles: never test on existing customers (grandfather everything); test one variable at a time; this market is small (~2,000–4,000 serviceable firms per the skeptic-corrected TAM — verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping), so tests are sequential cohorts, not A/B splits.

Phase	When	Test	Decision rule
P0 — WTP interviews	Pre-launch (part of the 15-owner interview gate in `research/PROBLEM_VALIDATION.md`)	Van Westendorp-style questions on $79/$149/$249 + "what do you pay today?"	If >½ of interviewed owners anchor below $79 for the whole category, revisit tiers before launch — willingness to pay at these levels is **assumed, not demonstrated** (per `PROBLEM_VALIDATION.md`)
P1 — Launch cohort	Months 1–4	Ship $79/$149/$249 as-is; measure objection rate in founder-led demos, tier mix, trial→paid	If price is cited as primary objection in >30% of lost demos, test P2a; if tier mix is >70% Solo, test P2b
P2a — Firm-tier probe	Months 5–7	New-cohort test of Firm at $129 vs $149 (sequential months)	Keep the higher price unless conversion delta >25% relative
P2b — Cap adjustment	Months 5–7	Move Solo cap from 5 → 3 users for new signups (pushes 2-crew shops to Firm)	Only if Solo cannibalization is proven, and never retroactively
P3 — Annual push	Months 6–9	At-activation annual offer vs. at-renewal offer	Optimize for % of base on annual by month 12 (target ≥40% — assumption)
P4 — Practice validation	By month 9	If <5% of customers are on Practice, interview why	Either add a genuine Practice-only capability or fold branded pages into Firm and retire the tier (skeptic explicitly warned the $249 anchor may be unearned)

Repricing tripwire: if SurveyOps or KudurruStone publish pricing that undercuts us with feature parity on the radar (see `RISK_REGISTER.md`, R-1), do not reflex-discount — the segment evidence says speed, simplicity, and the compounding archive retain better than price. Respond on the wedge, cut price only with a specific win/loss dataset showing price as the verified loss reason.

Source notes

- Tier structure, flat-pricing logic, ARPA, and market sizing: `business/FINAL_CONCEPT.md` (authoritative).
- KudurruStone per-user pricing, ClickUp ~$3/user RPLS evidence, "too slow" abandonment, Qfactor obscurity, SurveyOps posture: round-3 market skeptic in `research/revival-round3-results.json` (verified by adversarial review; kudurrustone.com / qfactor-llc.com / survey-ops.com / rpls.com blocked automated fetch — figures from search-index research, flagged for manual re-confirmation).
- SurveyOps existence + S-T-R positioning: https://survey-ops.com/land-surveying-job-management-software (note: this is the competitor's own page).
- TAM correction (~7,000 firms / ~7,382 establishments, NAICS 541370): verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping

Backsight — Go-to-Market Plan

Date: 2026-07-10 · Assumes a sellable v1 at launch (skeptic-corrected build estimate: 12–16 weeks, per `research/revival-round3-results.json`).

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1. Positioning statement

> **For** owners of small US land-surveying firms (1–5 crews, 3–25 staff) **who** run 20–80 concurrent jobs on a whiteboard, email, and memory, **Backsight** is job-tracking software that speaks surveying — **unlike** generic tools (ClickUp, Jobber, monday.com) that don't know what a plat or a licensed-review gate is, and unlike the vertical micro-vendors (Qfactor, KudurruStone, Cyanic Job Book, SurveyOps) that track jobs going forward, **Backsight** also imports the firm's entire job history and indexes it spatially — so every new request instantly shows what the firm already knows about that ground, at one flat price per firm where field crews never count against a license.

Three disciplines baked into this positioning (all skeptic-mandated, per `research/revival-round3-results.json`):

1. **Never claim "no competition."** Six vertical vendors exist and are named in our own materials (verified by adversarial review). The differentiator is the *historical import → spatial archive* wedge — the one capability none of them advertise (skeptic's residual-gap finding) — plus flat pricing and speed.
2. **Never lead with S-T-R search as unique.** SurveyOps advertises S-T-R job organization (https://survey-ops.com/land-surveying-job-management-software) and Cyanic offers legal-address search. Uniqueness lives in *automated parsing of the messy historical spreadsheet* and quote-time surfacing.
3. **Sell to the owner, ROI the office manager.** The owner buys the radar and fewer interruptions; the office manager lives in the pipeline daily. Demos address both.

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2. Beachhead: PLSS states with dense small-firm counts

Why PLSS-first: the wedge's strongest demo (section-level prior-work hits) only works in the 30 PLSS states; Texas and the colonial metes-and-bounds states — about 20 states — fall back to address/pin search (verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system). Launching where the flagship demo is strongest is the skeptic's explicit recommendation.

Market sizing (corrected): ~7,000 firms / ~7,382 establishments in NAICS 541370 nationally (verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping — this corrected the scout's 25,000 figure). Serviceable segment after filtering to 3–25-staff firms in PLSS states: **~2,000–4,000 firms** (skeptic estimate; treat as a planning assumption).

Beachhead wave 1 (months 1–6): Colorado, Wisconsin, Minnesota, Missouri, Kansas. Rationale (assumption-based, to be validated against state licensing-board rosters in week 1 of GTM):

- All fully PLSS; strong sectionalized rural + exurban survey demand (boundary/ALTA on section-referenced land).
- Colorado is where the seeded demo lives (Fort Collins / 6th PM, per `product/MVP_BUILD_SPEC.md`) — demos are geographically native.
- Each has an active state society (PLSC, WSLS, MSPS, MSPS-MO, KSLS) with newsletters and annual conferences where sponsorships cost hundreds, not thousands (`FINAL_CONCEPT.md`).
- Deliberately avoids Texas (no PLSS) and California (large-firm skew) in wave 1.

Wave 2 (months 6–12): remaining Midwest/Mountain West PLSS states (IA, NE, OK, MT, ID, UT, WY, ND, SD, plus MI/OH/IN/IL). **Wave 3 (2027):** Texas via GLO abstract/survey-name data as a first-class second system (skeptic's phase-2 recommendation), then East Coast on subdivision/address search.

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3. Channel plan

No saturated-SEO plays: the obvious keywords are being blanketed by SurveyOps's programmatic SEO (verified by adversarial review), and the free S-T-R-lookup lead magnet slot is occupied by randymajors.org, Earth Point, and Township America (verified by adversarial review, https://www.earthpoint.us/Townships.aspx, https://townshipamerica.com/). Distribution runs through people, not SERPs.

#	Channel	Motion	Cost profile	Expected role
1	**State survey societies** (wave-1 states)	Newsletter sponsorships, annual-conference booths/talks, member-discount partnership (10% code — see `PRICING_STRATEGY.md`), offer a free CE-friendly talk: "Turning your job archive into a quoting asset"	$200–800 per newsletter slot; $500–1,500 per state conference booth (estimates)	Primary trust channel — the buyer trusts the society more than any ad
2	**NSPS chapters**	National visibility layer over the state societies; exhibit at the NSPS annual meeting once wave-1 traction exists	Moderate	Air cover, not lead volume

#	Channel	Motion	Cost profile	Expected role
3	**r/Surveying** (~100k+ members per `FINAL_CONCEPT.md`) + RPLS.com forums	Genuine participation under the founder's real name; answer PM-software threads honestly (including "a whiteboard is fine at your size"); share the free Prior-Work Audit when contextually invited. **Never astroturf** — the skeptic found these forums are where tools get abandoned in public ("too slow"); they're also where reputations are made	Founder time	Credibility + steady trickle of high-intent trials
4	**Conference booths**	Wave-1 state conferences (typically fall/winter); live demo of the seeded Whitfield firm + "bring your spreadsheet, leave with your coverage map" on-the-spot audits	Hundreds per event (`FINAL_CONCEPT.md`)	The founder-led sales floor; target 5 booths in year 1
5	**Client status link loop** (built-in referral)	Every status page footer: "Tracked with Backsight." Title companies and builders refresh these pages weekly and work with *dozens* of surveyors — each becomes a passive referral surface. Add a low-key "Are you a surveyor? See how this works" link	$0	The compounding channel; slow to start, structural once ~30 firms are active
6	**Referral program**	Give-a-month/get-a-month for firm-to-firm referrals; surveyors in a region know each other and subcontract to each other	~$140 per conversion	Word-of-mouth accelerant from month 4
7	**Podcasts / YouTube**	Surveying practice podcasts and channels; the founder tells the honest tournament story ("we researched 34 problems; here's why surveying won") — this audience respects rigor	Founder time	Trust + long-tail

Deliberately excluded: paid Google Ads on category keywords (SurveyOps SEO war, bad CAC math at flat pricing), generic software directories as a primary play (the vertical's near-zero review footprints show buyers don't shop there — verified by adversarial review), cold email at scale (small community; reputation risk exceeds yield).

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4. Partnership angles

1. **GNSS equipment dealers & survey-supply houses** (`FINAL_CONCEPT.md`): they talk to every small firm in their territory and have nothing to sell them for the office. Offer a white-glove referral fee (one-time bounty) rather than a reseller margin — dealers won't do SaaS support. Start with wave-1-state dealers.
2. **State societies as partners, not just venues:** member discount + a society-branded webinar per year + sponsorship of the society's map/GIS committee work where one exists.
3. **Title companies (demand-side pull):** once ≥5 firms in a metro use status links, approach the title companies seeing them: "ask your other surveyors for a Backsight link." A title-side dashboard is a later monetizable product (see `PRICING_STRATEGY.md` §4); in year 1 it's an endorsement loop only.
4. **Township America (buy-vs-build option):** they sell a PLSS geocoding API covering 30+ states and all 37 principal meridians (verified by adversarial review, https://townshipamerica.com/). A licensing conversation is both a fallback if our parser hits edge-case walls and a possible co-marketing channel to their surveyor users. Not a dependency — our parser is license-clean and in-house (pyTRS is prohibited for commercial use: verified by adversarial review, https://github.com/JamesPImes/pyTRS).
5. **QuickBooks ecosystem:** complete the Intuit app assessment in week 1 (a ~30-minute form with near-immediate approval — verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ); a QBO App Store listing is a small but warm discovery surface among firms that already run QBO.
6. **Adjacent-vertical associations (2027):** septic designers, foresters, geotechnical field firms (`FINAL_CONCEPT.md`) — parked until the surveying beachhead holds.

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5. 90-day launch plan

Assumes day 0 = sellable v1 (post the 12–16-week build) **and** the interview gate passed (15 owner interviews per `research/PROBLEM_VALIDATION.md` ; if WTP fails there, this plan does not execute — see kill criteria in `METRICS_SUCCESS_CRITERIA.md`).

Days 1–30 — Foundation & first friendlies

- Complete the **trademark/name-collision check** on "Backsight" *before* any paid marketing (flagged as required in `FINAL_CONCEPT.md` ; see `RISK_REGISTER.md` R-10). Have a fallback name.
- Manually demo SurveyOps, Qfactor, KudurruStone, and Cyanic Job Book; write the honest comparison one-pager (skeptic-mandated; their sites blocked automated research).
- Submit Intuit app assessment; stand up landing page (`LANDING_PAGE_COPY.md`) + free Prior-Work Audit funnel (`CUSTOMER_ACQUISITION.md`).
- Recruit **5 design-partner firms** from interviewees at 50% off (per `PRICING_STRATEGY.md`): concierge-import their real spreadsheets personally. Target: 5 active firms, 3 with radar hits on their own data.
- Join wave-1 state societies as an affiliate/vendor member; book newsletter slots and the first two conference booths.
- Begin genuine r/Surveying + RPLS participation (no pitching for 30 days — build history first).

Days 31–60 — Public launch in wave-1 states

- Announce in 5 state-society newsletters + r/Surveying launch post (transparent founder story + free audit link).

- Publish 3 proof artifacts from design partners (with permission): a coverage-map screenshot, a radar-hit-to-won-quote story, a status-link/title-company story. **Real ones only** — social-proof placeholders stay empty until then (`LANDING_PAGE_COPY.md`).
- First conference booth (whichever wave-1 society meets first).
- Ship the referral program; turn on the "Tracked with Backsight" footer link.
- Target by day 60: **75 audit runs, 25 trials, 8–12 paying firms** (assumptions; see funnel math in `CUSTOMER_ACQUISITION.md`).

Days 61–90 — Repeatability

- Founder-led sales cadence: every audit completion gets a personal 15-minute "walk your coverage map" call offer (the demo script in `SALES_ONBOARDING_FLOW.md`).
- Second booth; first GNSS-dealer partnership signed; first society webinar delivered.
- Win/loss review on every closed-lost demo (price? inertia? competitor? missing feature?) → feeds pricing P1 gate (`PRICING_STRATEGY.md`) and roadmap.
- Instrument and review the metrics stack (`METRICS_SUCCESS_CRITERIA.md`); day-90 checkpoint against the month-3 thresholds there, including the explicit slow-start tripwire.
- Target by day 90: **20 paying firms, ≥$2,500 MRR, ≥60% of trials completing an import** (assumptions, reconciled with `METRICS_SUCCESS_CRITERIA.md`).

Source notes

- ICP, pricing, channel seeds (societies, NSPS, r/Surveying, booths, dealer partners, status-link loop): `business/FINAL_CONCEPT.md` .
- Competitor set, TAM correction, PLSS coverage limits, lead-magnet saturation, SEO warning, build-timeline correction: round-3 skeptic verdicts in `research/revival-round3-results.json` (URLs cited inline above).
- All booth/newsletter cost figures and all day-30/60/90 targets are **estimates/assumptions**, not researched facts.

Backsight — Customer Acquisition Plan

Date: 2026-07-10 · Companion to `GO_TO_MARKET.md` (channels) and `SALES_ONBOARDING_FLOW.md` (conversion mechanics).

Ground rules inherited from the research: the serviceable market is **~2,000–4,000 firms** (skeptic-corrected from NAICS 541370's ~7,000 firms — verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping), buyers are slow-adopting field professionals (a decade-old vertical vendor, Qfactor, remains obscure — verified by adversarial review), and category SEO is contested by SurveyOps's programmatic pages (verified by adversarial review, https://survey-ops.com/land-surveying-job-management-software). So: high-trust channels, patient funnel, founder-led early.

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1. The lead magnet: the free Prior-Work Audit

Offer: *"Import your job list. Get a map of everything your firm has ever surveyed — free."*

Upload the spreadsheet the firm already has → Backsight geocodes/parses/pins it → the firm gets an interactive **coverage map** plus a one-page audit: jobs indexed, sections with repeat work ("you've been in Section 14, T7N R69W four times"), clusters that suggest quotable ground, and rows the importer couldn't place. Converting to a trial keeps the imported archive; walking away gets a PNG of the map and a clean CSV back.

Why this and not an S-T-R lookup tool: the lookup slot is already owned by randymajors.org, Earth Point, and Township America (verified by adversarial review, https://www.earthpoint.us/Townships.aspx, https://townshipamerica.com/). The skeptic's explicit recommendation — adopted here — was to replace it with *"a free 'import your job history, get a map of everything your firm has ever surveyed' one-shot tool that is itself the onboarding funnel"* (`research/revival-round3-results.json`). The audit **is** the product's first-run experience wearing a marketing hat: it demos the exact wedge no competitor advertises, and its output is inherently shareable (owners show the coverage map to partners and peers).

Mechanics:

- No credit card; email required (the map is emailed as well as shown).
- Runs on address geocoding first (US Census Geocoder, free — `FINAL_CONCEPT.md` fix #3); legal descriptions are bonus, never required.
- Every completed audit triggers a personal founder email offering a 15-minute "walk your map" call — the top of the founder-led motion.
- Guardrail: the audit is one-shot and read-only. Ongoing radar-on-new-requests, pipeline, and status links require the trial — the free taste must not become a free tier (`PRICING_STRATEGY.md` §3).

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2. Funnel math (all rates are assumptions — no cohort data exists yet)

Model for months 1–12, blended across channels. **Every number in the "assumed rate" column is an explicit assumption to be replaced by observed data at the day-90 and month-6 checkpoints.**

Stage	Definition	Assumed rate	Year-1 volume
Reached	Sees Backsight (newsletter, booth, forum, status-link footer, referral)	—	~6,000 impressions of the ~2,000–4,000-firm SAM (multiple touches per firm)
Engaged lead	Runs the Prior-Work Audit **or** books a demo	5% of reached →	**300 audits/demos**
Trial	Starts the 14-day trial	50% of engaged →	**150 trials**
Activated	Meets activation definition (`SALES_ONBOARDING_FLOW.md` §4: import + radar hit + pipeline live + status link shared)	55% of trials →	**~82 activated**
Paying	Converts to any paid tier	75% of activated (≈41% of trials) →	**~62 paying firms**
Retained @12mo	Still subscribed	97.5%/mo logo retention (assumption) →	**~55 net**

Cross-checks on plausibility:

- 62 paying firms ≈ **1.6–3.1% of SAM** in year 1 — aggressive but not absurd for a founder-led vertical launch; `FINAL_CONCEPT.md` 's long-run scenario is 5% penetration.
- The 41% trial → paid assumption is high for horizontal SaaS but reflects a concierge-touched, high-intent funnel (audits are self-selected owners with their own data already in). **If observed trial → paid is <20% with ≥50 trials, the funnel model is wrong — re-plan** (tripwire mirrored in `METRICS_SUCCESS_CRITERIA.md`).
- Whiteboard inertia is the named adoption risk (`FINAL_CONCEPT.md`); the audit exists precisely to convert inertia into curiosity with the firm's own data.

Channel mix assumption for the 300 engaged leads

Channel	Share	Leads	Basis
State societies (newsletters, webinars, member code)	30%	90	5 wave-1 states × newsletter reach (assumption)
Conference booths (5 events)	20%	60	~12 booth audits/demos per event (assumption)
r/Surveying + RPLS participation	15%	45	steady trickle, high intent (assumption)
Referrals + status-link footer	20%	60	back-loaded to H2 as installed base grows (assumption)
Podcasts, QBO App Store, dealers, misc.	15%	45	assumption

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3. CAC targets vs LTV at flat pricing

LTV model (all assumptions labeled):

- Blended ARPA: **$140/mo** (`FINAL_CONCEPT.md` planning figure).
- Monthly logo churn: **2.5%** (assumption → ~40-month average lifetime; no vertical benchmark exists in our research — the compounding-archive moat argues for lower churn over time, per `FINAL_CONCEPT.md` , but that is a thesis, not data).
- Gross margin: **~85%** (assumption; hosting + geocoding are cheap, support is founder time).
- **Gross-margin LTV ≈ $140 × 40 × 0.85 ≈ $4,760.**

CAC targets:

Guardrail	Target	Meaning at $140 ARPA
Blended CAC (all spend + valued founder time ÷ new customers)	**≤ $1,200**	LTV:CAC ≥ ~4:1
Cash CAC (out-of-pocket only, excl. founder time)	**≤ $500**	Payback ≤ ~4 months of revenue
Per-channel kill line	Any channel with cash CAC > $2,000 after 2 quarters gets cut	—

Year-1 sanity check: planned cash spend ≈ $15–20K (5 booths ~$5–7K, newsletters/sponsorships ~$4K, referral credits ~$3K, tools/design-partner subsidies ~$4K — all estimates) ÷ 62 customers ≈ **$240–320 cash CAC**. The binding constraint is founder time, not dollars — which is correct for a bootstrapped niche play. **Honest caveat:** if founder time is priced at even $75/hr, blended CAC in the founder-led phase will exceed $1,200; that is the deliberate, temporary cost of building the reference base and does not scale past customer ~50 (see `SALES_ONBOARDING_FLOW.md` §6).

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4. Content strategy (explicitly not saturated-SEO-dependent)

The kill taxonomy bans panic-SEO distribution (D-009, `decisions/DECISION_LOG.md`), and the category's commercial keywords are SurveyOps's programmatic turf (verified by adversarial review). Content therefore exists to build *trust and referability in community channels*, not rankings:

1. **The coverage map as content.** Every audit produces a shareable artifact. Anonymized "firm archaeology" posts — "this 3-person Wisconsin firm didn't know it had surveyed the same section 11 times" — are the flagship recurring format for newsletters, talks, and forums (with permission, always).
2. **Practice-economics essays, not keyword pages.** One well-researched piece per month aimed at owners: quoting repeat ground, the real cost of status calls, succession and the retiring owner's memory (the archive-as-succession-asset angle), delivered-but-uninvoiced leakage. Distributed via society newsletters and forum posts, not optimized for Google.
3. **The honest-research story.** The tournament itself — 34 problems, 12 concepts, adversarial review, why surveying won — is a credibility asset for podcasts and conference talks with this rigor-respecting audience.
4. **Long-tail SEO as a byproduct only.** Publish the essays on our site with clean titles and let non-contested long-tail queries accrue; spend zero effort on the contested head terms.
5. **What we don't do:** listicles ("Best surveying software 2026" — the format the skeptics repeatedly exposed as vendor slop, e.g. the flagged CQ and QuoteIQ pages in our own original evidence), comparison-page warfare, AI-generated volume content.

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5. First-100-customers playbook

Sequenced, cumulative. Numbers are targets (assumptions), reconciled with `GO_TO_MARKET.md` and `METRICS_SUCCESS_CRITERIA.md` .

Customers 1–10 (months 1–2): design partners. Recruited personally from the 15 interview-gate owners and their referrals. 50% off for 12 months, founder does the import concierge-style on a screen-share, weekly check-ins. Goal is not revenue — it's 3 permissioned proof stories, real-spreadsheet parser hardening (the skeptic's "stress-test the wedge on real data" instruction), and the first radar-hit-won-a-job anecdote.

Customers 11–35 (months 2–5): the audit engine + societies. Public launch per `GO_TO_MARKET.md` days 31–60. Every audit → personal email → 15-min map walk → trial with concierge import. Founder closes every deal personally (`SALES_ONBOARDING_FLOW.md` demo script). Newsletter mentions in all 5 wave-1 states; first 2 booths. Watch metric: audit→trial ≥40%, else the audit output isn't compelling enough — fix the artifact before buying more reach.

Customers 36–70 (months 5–9): repeatability + referral flywheel. Referral program live (give-a-month/get-a-month); status-link footers now on hundreds of client-facing pages; 2–3 more booths; first GNSS-dealer bounty deals; QBO App Store listing live. Founder still demos but onboarding is now self-serve-first with concierge as fallback. Watch metric: ≥25% of new trials citing a referral/status-link/word-of-mouth source — the compounding channels must be visible by now or the model is booth-dependent and caps out.

Customers 71–100 (months 9–12+): second wave. Wave-2 PLSS states via the same society playbook; webinar-per-society cadence; the proof library (10+ real stories) carries conversion so founder time shifts toward product. Begin Texas/GLO scoping only if wave-1 retention is healthy (`METRICS_SUCCESS_CRITERIA.md` month-12 gate).

Playbook-wide rules:

- Every closed-lost gets a 5-question win/loss note (tool used today, price reaction, competitor evaluated, missing feature, would-they-audit-anyway).
- No customer counts as "acquired" until activated (`SALES_ONBOARDING_FLOW.md` §4) — a paying-but-empty account is churn on a delay.
- If months 1–4 close <15 customers total, trigger the slow-start review in `METRICS_SUCCESS_CRITERIA.md` rather than spending more on reach.

Source notes

- SAM correction and competitor/SEO/lead-magnet landscape: round-3 skeptic verdicts, `research/revival-round3-results.json` (verified by adversarial review; URLs inline).
- Audit-as-lead-magnet design: skeptic improvement adopted verbatim from the same file.
- ARPA, penetration scenario, whiteboard-inertia risk, channel seeds: `business/FINAL_CONCEPT.md`.
- All conversion rates, churn, margin, spend, and volume figures in §2–§3 and §5 are **assumptions**, labeled as such, pending real cohort data.

Backsight — Sales & Onboarding Flow

Date: 2026-07-10 · Covers: self-serve trial, the 10-minute aha, onboarding checklist, activation definition, expansion triggers, and the founder-led motion for the first 50 customers.

Design constraints inherited from research: the buyer is a field professional with whiteboard inertia (`FINAL_CONCEPT.md` risk), a purpose-built competitor was abandoned as *"too slow"* by an RPLS forum user (verified by adversarial review, `research/revival-round3-results.json`) — so every step below is optimized for speed-to-value — and import must assume legal descriptions are absent (skeptic fix: address-first, per `FINAL_CONCEPT.md` fix #3).

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1. Self-serve trial flow

```
Landing page ─┬─> "Start 14-day trial" ──────────────┐
              └─> "Free Prior-Work Audit" ─> audit ──┤ (audit account converts in place;
                                                     ▼  imported data carries over)
                                    Create account (email + firm name + state)
                                                     ▼
                                    Seeded demo firm loads INSTANTLY
                                    (Whitfield Land Surveying, ~85 jobs, per MVP spec)
                                                     ▼
                                    Guided 3-stop tour (≤3 min):
                                    ① radar search that hits ② pipeline board ③ a live status page
                                                     ▼
                                    "Now do it with YOUR jobs" → CSV import wizard
                                                     ▼
                                    Coverage map + first radar search on own data
                                                     ▼
                                    Day-14: pick a tier (card required only here)
```

Rules:

- **No credit card to start; card only at conversion.** No feature gates inside the trial (trial = Practice-tier capabilities).
- **The demo firm is pre-seeded, never empty.** An empty state kills field-professional patience; the seeded firm (per `product/MVP_BUILD_SPEC.md`) has deliberate hooks — a request-stage job with 3 prior jobs in the same section, overdue jobs, a stuck-in-review job, unbilled delivered work — so every screen makes its point immediately.
- **Demo data is clearly labeled and one-click removable** once real data is in.
- Trial extension: one no-questions 7-day extension self-serve (owners disappear into fieldwork weeks; punishing that is churn).

2. The 10-minute aha

The aha is scripted as three beats, in order — seeded proof, own data, own radar hit:

Minute	Beat	What the user sees
0–3	**Seeded radar hit.** Tour opens on the Radar page with a pre-wired example search ("try these" row, per MVP spec).	An incoming request pinned in a section with 3 prior jobs highlighted — years, types, original fees. The concept lands: *the archive quotes the job.*
3–8	**Import own CSV.** Wizard: upload → column-map (saved templates for common layouts) → geocode/parse progress → review queue for unplaced rows (pin or skip).	Their own firm's coverage map materializing. This is the emotional peak — owners recognize thirty years of work appearing on one map.
8–10	**Radar on own ground.** Prompt: "Type the address or section of the last request you quoted."	Either a hit (**the aha**: "we've been there before — and now the software knows it") or a clean miss with adjacent-work shown (still credible; prompts a second search).

Instrumentation: time-to-import-complete, % rows auto-placed, first-own-radar-search rate, and hit rate on first search are the four numbers reviewed weekly (feeds `METRICS_SUCCESS_CRITERIA.md`). If median time-to-own-coverage-map exceeds 10 minutes, that's a product bug, not a marketing problem.

3. Onboarding checklist (in-app, 7 items)

Shown as a progress card on the dashboard until complete; each item deep-links to the action.

1. ☐ Import your job history (or finish the audit import) — *target: ≥50 jobs in*
2. ☐ Resolve the import review queue (pin or skip every unplaced row)

3. ☐ Run a radar search on real ground you're quoting
4. ☐ Put your active jobs on the pipeline board (importer flags likely-active rows)
5. ☐ Share one client status link (email it to yourself first if you like)
6. ☐ Invite your office manager / a crew chief
7. ☐ Turn on the weekly stuck-jobs digest (on by default; confirm recipients)

Concierge overlay (first 50 customers): the founder does items 1–2 *with* the customer on a 30-minute screen-share — see §6. Post-50, a "book a free import session" link remains but self-serve is the default path.

4. Activation definition

A trial firm is **activated** when, within 14 days of signup:

1. **≥50 historical jobs imported** and ≥60% spatially placed (auto or manual), AND
2. **≥1 radar search run on the firm's own data** by a firm user (not the founder), AND
3. **≥5 active jobs being tracked** with ≥3 stage advancements performed, AND
4. **≥1 client status link shared externally** (link opened by a non-firm viewer).

Rationale: (1) is the archive moat being seeded, (2) is the wedge experienced, (3) is the daily-use habit, (4) is the retention feature reaching the client side and priming the referral loop. All four are measurable in-product. Firms with fewer than 50 historical jobs (young firms) activate on 25 imported + the other three criteria.

Activation is the unit of acquisition (`CUSTOMER_ACQUISITION.md` §5): a paying, non-activated firm is treated as at-risk from day one.

5. Expansion triggers (instrumented, human-followed-up)

Trigger (automatic detection)	Action
Solo firm adds 5th user or invite #6 bounces off the cap	In-app: "Firm tier removes the cap — crews never count extra"; founder email if design partner
Firm marks jobs `delivered` but no invoice within 7 days, repeatedly	Surface QBO push (Firm tier) — "delivered-but-uninvoiced" is the dollar-denominated upgrade story
Status links opened >50 times/month by external viewers	Practice-tier prompt: client-branded status pages ("your title companies are already watching — put your logo on it")
Monthly subscriber passes month 4 with healthy usage	Annual offer (2 months free) — churn insulation, per `PRICING_STRATEGY.md` P3 test
Firm's radar hit converts to a job (user marks "won — prior work used")	Ask for the story (proof library) + referral nudge (give-a-month/get-a-month)
Usage drops to zero for 14 days (any tier)	Counter-expansion: personal "did fieldwork season eat you?" email — retention save, not upsell

6. Founder-led sales motion — first 50 customers

Posture: the founder is a researcher who built the tool the research demanded, not a rep. Every demo doubles as a validation interview (win/loss questions logged per `CUSTOMER_ACQUISITION.md` §5).

Cadence: every completed Prior-Work Audit or demo request → personal email within 24h offering a 15-minute call → 30-minute concierge import session for trials → day-7 check-in → day-13 conversion call. Max ~10 concurrent trials in concierge mode (founder capacity — assumption).

Demo script (15–20 minutes, screen-share)

[0–2 min] Open with their world, not the product.

> "Before I show anything — how do you track jobs today, and who fields the 'where's my survey?' calls?" (Log answers verbatim. If they say the whiteboard works fine and they're 2 people with one crew, say so honestly: "You might not need us yet — run the free audit anyway and keep the map." That honesty is the brand, and this market talks.)

[2–6 min] Seeded radar hit.

> "This is a demo firm in Fort Collins. A request just came in on this parcel — watch." *(Run the wired search: 3 prior jobs light up in the section.)* "Three prior jobs on that ground: control likely set, boundary already resolved once, research paid for. You'd quote this in minutes and could win it on price because your cost is genuinely lower. That's the whole idea: your firm already knows things — this makes the knowing searchable."

[6–10 min] Their data, live.

> "Send me the spreadsheet you already keep — whatever columns it has. No legal descriptions needed; addresses are enough, and anything we can't place we pin by hand together." *(Import live if they brought a file — the coverage map appearing is the close. If not, book the 30-min import session before ending the call.)*

[10–13 min] Pipeline + status link.

> "Day to day it's this board — your actual stages, fieldwork through licensed review. Crews advance jobs from a phone browser; nobody counts against your license. And every job gets this —" *(open a status page)* "— a read-only link your title company bookmarks instead of calling you. The footer says 'Tracked with Backsight,' which is how your clients' other surveyors find us — fair warning."

[13–15 min] Price, straight.

> "Flat per firm: $79, $149, or $249 by team size — the price is on the website, it doesn't move, there are no add-ons, and your full data exports any time. For your size that's the $X tier. One radar-assisted win or one caught uninvoiced job covers a year. Fourteen-day trial, no card — worst case you keep the coverage map. Want me to import your history this week?"

Objection handling (from verified research):

- *"We use ClickUp / a spreadsheet and it's fine"* (verified real behavior, RPLS forum) → "It is fine — for tracking. It can't search your archive by ground. Run the audit; if the map doesn't show you money, don't buy."
- *"How are you different from SurveyOps / KudurruStone?"* → Name them respectfully (never claim they don't exist — skeptic-mandated): "They track jobs going forward. We also import your history and make it quotable, and we price flat per firm instead of per user." (Comparison one-pager only after manual competitor demos — `GO_TO_MARKET.md` day-1 task.)
- *"We're in Texas / no PLSS here"* → Honest answer per landing FAQ 3: address/subdivision/pin search works; section search doesn't; "you're wave 3, and I'll tell you when that's real rather than sell you a weak version now."
- *"What if you disappear?"* → Full self-serve export on every tier + documented offsite backups (skeptic-mandated, `research/revival-round3-results.json` improvements).

Exit criteria for founder-led mode (at ~customer 50): proof library ≥10 permissioned stories, self-serve activation ≥50% without concierge, demo script converted into a 6-minute recorded walkthrough, import wizard handles the 10 most common spreadsheet layouts unassisted. Founder time then rebalances toward product and the wave-2 society circuit (`GO_TO_MARKET.md`).

Source notes

- Seeded-demo contents and hooks: `product/MVP_BUILD_SPEC.md` .
- Address-first import, meridian-ambiguity confirmation UX, export/backup promises: skeptic-mandated fixes in `FINAL_CONCEPT.md` and `research/revival-round3-results.json` .
- "Too slow"/ClickUp objection evidence and never-claim-no-competitors rule: round-3 market skeptic (verified by adversarial review; rpls.com blocked automated fetch, per skeptic notes).
- All conversion cadences, capacity numbers, and thresholds are **assumptions** pending live data.

Backsight — Metrics & Success Criteria

Date: 2026-07-10 · Clock convention: **M0 = first day of public availability** (post 12–16-week build and post interview gate). All targets below are **planning assumptions**, not benchmarks from research — no comparable cohort data exists for this vertical. The kill criteria, however, are commitments.

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1. North star

> **Engaged firms:** the number of firms that, in a given week, both (a) advance at least one job through the pipeline and (b) run at least one Prior-Work Radar search on their own data.

Why this metric: it combines the retention feature (pipeline as daily habit) with the wedge (archive being consulted). A firm doing both is getting the two things it pays for; a firm doing neither is churn on a delay regardless of billing status. Revenue follows engaged firms almost mechanically at flat pricing (MRR ≈ engaged firms × tier price, with a lag).

2. Input metrics (the tree under the north star)

Layer	Metric	Why it matters
Acquisition	Prior-Work Audits completed / week	Top of funnel; the lead magnet working
	Audit → trial rate	Quality of the audit artifact (guardrail: ≥40%, per `CUSTOMER_ACQUISITION.md`)
	% of new trials from referral / status-link footer / word-of-mouth	The compounding channels becoming visible (target ≥25% by M9)
Activation	Trial → activated rate (definition: `SALES_ONBOARDING_FLOW.md` §4)	The 10-minute aha converting
	Median time-to-own-coverage-map	Must stay ≤10 min; the segment abandoned a competitor for being "too slow" (verified by adversarial review)
	% of imported rows auto-placed (geocode or S-T-R parse)	Parser/geocoder quality on real-world data — the skeptic's central implementability concern
Engagement	Radar searches per firm per week; radar hit rate	Wedge usage; hit rate also measures archive density
	Radar hits marked "used in quote" / "won"	The dollar story — feeds proof library and pricing defense
	Status-link views by external viewers per firm	Retention feature reaching clients + referral surface size
	Stage advancements per active firm per week	Pipeline as habit
Revenue	MRR; ARPA; tier mix; % annual	Business health; tier mix feeds pricing tests P2/P4 (`PRICING_STRATEGY.md`)
Retention	Monthly logo churn; M3 cohort retention; net revenue retention	The compounding-archive thesis is testable here: churn should *fall* with archive age
Efficiency	Cash CAC (≤$500) and blended CAC (≤$1,200) per `CUSTOMER_ACQUISITION.md` §3	Channel discipline

3. Definitions (binding)

- **Activated:** per `SALES_ONBOARDING_FLOW.md` §4 — ≥50 historical jobs imported (≥60% placed) + ≥1 own-data radar search + ≥5 active jobs with ≥3 advancements + ≥1 externally-opened status link, within 14 days. (25-job threshold for young firms.)
- **Retained:** a paying firm counted in the month it renews; a firm is **churned** on the day its subscription lapses without renewal (dunning grace: 14 days). Paused/seasonal accounts count as churned unless on a prepaid annual plan.
- **Revenue:** MRR = normalized monthly value of active subscriptions (annual ÷ 12). One-time concierge/archive-service fees (if ever offered, `PRICING_STRATEGY.md` §4) are excluded from MRR.
- **Engaged (weekly):** north-star definition above; measured on firm level, not user level.

4. Success thresholds

Month 3 checkpoint (early signal, per `GO_TO_MARKET.md` day-90)

Metric	On track	Concern	Trigger review
Paying firms	≥20	10–19	<10
Audits completed (cumulative)	≥100	50–99	<50
Trial → activated	≥50%	30–50%	<30%
MRR	≥$2,500	$1,200–2,500	<$1,200

6-month success thresholds (M6)

Metric	Target	Minimum acceptable
Paying firms	35	20
MRR	$4,500	$2,500
Trial → paid	≥35%	≥20%
Monthly logo churn (M3+ cohorts)	≤2.5%	≤4%
Weekly engaged firms / paying firms	≥60%	≥45%
Radar hits marked used-in-quote (cumulative)	≥40	≥15
Permissioned proof stories	≥6	≥3

12-month success thresholds (M12)

Metric	Target	Minimum acceptable
Paying firms	90	55
MRR	$12,000	$7,000
ARR run-rate	~$145K	~$85K
Monthly logo churn	≤2%	≤3%
% trials from referral/status-link/WOM	≥30%	≥20%
% base on annual plans	≥40%	≥25%
Cash CAC	≤$500	≤$900
Wave-2 states opened	≥4	≥2

Context for scale expectations: `FINAL_CONCEPT.md`'s honest ceiling is ~$590K ARR at 5% of ~7,000 firms (TAM verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping). M12 target of ~$145K ARR ≈ 2.2% penetration of the 2,000–4,000-firm SAM — a bootstrapped-healthy trajectory toward that ceiling, not a venture curve. Anyone evaluating this business against venture-scale expectations should stop at `FINAL_CONCEPT.md`'s own admission.

5. Kill criteria (specific numbers, specific dates — these are commitments)

A **kill** means: stop acquisition spend, stop feature build, honor existing subscriptions through term with export support, and either (a) pivot per the documented alternatives or (b) wind down. "Keep trying harder" is not a listed option because the adversarial tournament showed this category punishes wishful persistence.

#	Gate	Kill condition	Rationale
K-0	**Pre-build interview gate** (before M0)	<5 of 15 interviewed owners express concrete WTP at ≥$79/mo (would trial with intent, or pre-order)	WTP is *assumed, not demonstrated* per `research/PROBLEM_VALIDATION.md`; building without this signal repeats the mistakes the tournament killed 11 concepts for
K-1	**Wedge-on-real-data gate** (design partners, ~M1)	On ≥5 real firm spreadsheets, <40% of rows can be spatially placed even with manual pinning assistance, or 0 of 5 firms get a radar hit they call useful	The skeptic's core warning: if real archives are unmappable or hits don't matter, the wedge degrades to a pin map Qfactor already offers (verified by adversarial review)
K-2	**M4 traction gate**	<10 paying firms AND <75 cumulative audits by end of M4	Neither demand nor funnel exists; channels this cheap not producing this little means the buyer isn't reachable at viable cost
K-3	**M6 conversion gate**	Trial → paid <10% with ≥50 trials, after at least one funnel iteration	Interest without purchase = ClickUp/whiteboard satisficing wins (the verified $3/user floor); price/value story failed
K-4	**M9 retention gate**	Monthly logo churn >6% for 3 consecutive months (M3+ cohorts), or weekly-engaged <30% of paying base	The compounding-archive retention thesis is false; flat-price LTV math collapses at this churn (LTV < ~$1,700 → CAC ceiling unworkable)
K-5	**M12 scale gate**	<30 paying firms or <$4,000 MRR at M12 despite passing K-2/K-3	The SAM is real but penetration velocity means decade-to-meaningful; redeploy effort (this mirrors the verified Qfactor pattern: 10 years in market, still obscure)
K-6	**Competitive event (any time)**	A funded or platform player ships free historical-import + spatial archive AND our win rate in head-to-heads drops below 25% for a quarter	The free-tier-floor kill pattern (taxonomy item 2, `DECISION_LOG.md` D-009) arriving in-category; fight only with evidence we're winning

Soft pivots short of kill (documented for completeness): K-1 partial failure → lean the product toward pipeline + status links with the archive as secondary (weaker moat — flag honestly); K-5 near-miss with strong retention → lifestyle-business mode: cut acquisition spend, run profitably at small scale, which flat pricing and low COGS permit.

6. Reporting cadence

- **Weekly (founder, 30 min):** north star, funnel counts, activation, time-to-map, parser auto-place rate, at-risk firms (zero-usage 14 days).

- **Monthly:** full metric tree vs. thresholds; win/loss log review; channel CAC table; pricing-test gates (`PRICING_STRATEGY.md` §5).
- **Gate reviews:** at M4, M6, M9, M12 — written verdict against K-gates appended to `decisions/DECISION_LOG.md`, pass or fail, no silent slippage.

Source notes

- TAM/SAM and ceiling: `FINAL_CONCEPT.md`; NAICS 541370 correction verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping.
- "Too slow" latency evidence, Qfactor-obscurity pattern, ClickUp floor, pin-map degradation risk: round-3 market skeptic, `research/revival-round3-results.json` (verified by adversarial review; competitor sites blocked automated fetch per skeptic notes).
- Kill-taxonomy grounding for K-6: `decisions/DECISION_LOG.md` D-009.
- All targets in §4 are assumptions; all kill numbers in §5 are commitments.

Backsight — Risk Register

Date: 2026-07-10 · Owner: founder. Reviewed at every gate review (M4/M6/M9/M12 per `METRICS_SUCCESS_CRITERIA.md`). Likelihood/Impact scale: Low / Medium / High. Every risk raised by the round-3 skeptics (`research/revival-round3-results.json`) is included and marked **[skeptic]**; facts marked *(verified)* were confirmed by adversarial review with the cited source.

ID	Risk	Category	Likelihood	Impact	Mitigation	Early-warning signal
R-1	**Competitor micro-vendors respond** [skeptic]: six vertical products exist — Qfactor (Map View already shows past-project locations), KudurruStone (client portal, $25/$15/$5 per-user pricing), Cyanic Job Book (legal-address + map search), Info-Retriever, CQ, SurveyOps (S-T-R search, iOS app, free guided onboarding, aggressive programmatic SEO) *(verified, https://survey-ops.com/land-surveying-job-management-software; other sites blocked automated fetch per skeptic notes)*. Historical-import concierge is "easy for them to copy" per the skeptic.	Market	High	Medium	Lead with the one gap none advertise: automated spatial indexing of messy historical spreadsheets (skeptic's residual-gap finding); ship faster than annual-contract vendors iterate; hold flat pricing + speed as structural contrasts; manually demo all six before finalizing positioning (GTM day-1 task); never claim "no competition"	A competitor announces CSV historical import or coverage-map features; head-to-head loss rate rises in win/loss log; their SEO pages start targeting "import job history" terms
R-2	**TAM ceiling** [skeptic]: ~7,000 firms / ~7,382 establishments in NAICS 541370 (skeptic corrected the scout's 25,000) *(verified, https://siccode.com/naics-code/541370/surveying-mapping)*; SAM ~2,000–4,000 PLSS-state small firms chased by 6+ vendors; honest ceiling ~$590K ARR at 5% penetration (`FINAL_CONCEPT.md`) — a bootstrapped outcome, not venture-scale.	Market	Certain (it's a fact, not a probability)	Medium	Accepted by design (D-010, `DECISION_LOG.md`); keep cost base bootstrap-shaped; adjacent verticals (septic, forestry, geotech) share the field→office→licensed-review shape for later expansion; Texas via GLO data as wave 3	Penetration velocity: if M12 lands at minimum-acceptable while CAC rises, the ceiling binds sooner — trigger K-5 review
R-3	**Spreadsheets without legal descriptions** [skeptic]: many firm archives hold client + street address only; legal descriptions live in deliverable PDFs/CAD. Radar degrades toward a geocoded pin map "Qfactor already offers" per the skeptic.	Product/wedge	High	High	Address-first ingestion is the primary path (US Census Geocoder, free — `FINAL_CONCEPT.md` fix #3); manual pin-drop with section snap; stress-test on ≥5 real firm spreadsheets before launch (kill gate K-1); demo script rebuilt so address-only is the normal case, not the degraded one	Auto-place rate <60% on design-partner imports; audit users abandoning at the review queue; "so it's just a pin map" appearing in win/loss notes
R-4	**Meridian ambiguity edge cases** [skeptic]: "T2N R3W Sec 14" repeats across principal meridians; firm spreadsheets essentially never record the meridian. A silent wrong-section match is "the worst possible failure mode" for a trust product.	Product/technical	Medium	High	State→default-meridian table + county-scoped resolution against CadNSDI; ambiguous parses flagged for human confirmation, never silently indexed (`FINAL_CONCEPT.md` fix #2); parser test corpus built from real spreadsheets; prefer geocoded address when present	Any customer-reported wrong-section match (treat first one as a sev-1 incident); ambiguity-flag rate >15% of parsed rows in a state (table gap)
R-5	**PLSS doesn't cover ~20 states incl. Texas** [skeptic]: flagship demo is dead in roughly 40% of states and a top surveying market *(verified, https://en.wikipedia.org/wiki/Texas_land_survey_system)*.	Market/product	Certain	Medium	Beachhead GTM restricted to PLSS states (wave 1: CO/WI/MN/MO/KS); subdivision/address search built as first-class; Texas as explicit wave 3 via GLO abstract data; honest landing-page FAQ so non-PLSS trials aren't mis-sold	Non-PLSS-state signups churning fast or rating the radar poorly; Texas inbound volume growing (that's also the wave-3 go signal)
R-6	**License-clean parser is real engineering** [skeptic]: pyTRS prohibits commercial use *(verified, https://github.com/JamesPImes/pyTRS)*; own parser + meridian handling realistically 2–3 weeks with a test corpus, and overall sellable-v1 timeline is 12–16 weeks, not 6–8.	Technical/schedule	High	Medium	Timeline re-planned to 12–16 weeks (skeptic figure adopted); never use/port pyTRS; Township America's commercial PLSS geocoding API (30+ states, all 37 meridians) exists as buy-vs-build fallback *(verified, https://townshipamerica.com/)*; email pyTRS maintainer re commercial license as a second fallback	Parser failing >20% of the real-spreadsheet corpus after week 3 of parser work; schedule slipping past week 16

ID	Risk	Category	Likelihood	Impact	Mitigation	Early-warning signal
R-7	**Map-tile dependency**: product UI leans on OpenStreetMap tiles; tile outage, rate-limiting, or usage-policy enforcement degrades the flagship map views.	Platform	Medium	Medium	Graceful fallback already specified — markers render on a plain coordinate grid if tiles fail (`product/MVP_BUILD_SPEC.md`); before scale, move to a paid tile provider or self-hosted tiles with usage headroom; cache tiles where licensing permits	Tile error rates in monitoring; OSM tile-usage-policy warnings; map-load latency complaints ("too slow" is this segment's documented abandonment trigger)
R-8	**Archive-of-record data loss** [skeptic]: a firm that loses 500 jobs of institutional memory has "a business-ending grievance" — contractual, not regulatory, exposure.	Operational/legal	Low	High	Automated offsite backups + documented, drilled restore *before first paying customer* (skeptic-mandated); full self-serve data export (CSV + files) on every tier — doubles as the anti-lock-in sales answer; ToS liability caps drafted by a lawyer	Backup job failures; restore drill not run in past quarter; export feature broken in QA
R-9	**Single-founder key-person risk**: sales, support, curation, and code are one person; illness, burnout, or the concierge load at customer ~30 stalls everything simultaneously.	Operational	Medium	High	Concierge capacity capped (~10 concurrent trials); self-serve import must replace concierge by customer ~50 (`SALES_ONBOARDING_FLOW.md` §6 exit criteria); runbooks + infrastructure-as-code so a contractor can operate the service; buffer of prepaid-annual cash before hiring	Founder support hours >25/week; onboarding queue >1 week; roadmap velocity at zero for a month
R-10	**Name/trademark check pending**: "Backsight" was adopted as a working title with trademark/name-collision check explicitly required before launch (`FINAL_CONCEPT.md`).	Legal/GTM	Medium	Medium	Run USPTO + state + domain + surveying-industry collision search *before* any paid marketing (GTM day-1–30 task); keep a fallback name and hold domains for it; don't print booth materials until cleared	Search reveals a live mark in software/surveying classes; C&D letter; confusingly-named product surfacing in the vertical
R-11	**Whiteboard inertia / unproven WTP** [skeptic-adjacent, `PROBLEM_VALIDATION.md`]: surveyors satisfice on ClickUp at ~$3/user/mo and abandoned a purpose-built tool as "too slow" (verified by adversarial review, RPLS forum per skeptic notes); $79–249 flat WTP is assumed, not demonstrated.	Market/GTM	High	High	Interview gate K-0 before build-out (15 owners, `METRICS_SUCCESS_CRITERIA.md`); trial must produce a dollar-denominated artifact in 10 minutes (radar hit / uninvoiced-job catch); speed as a product requirement; honest "you might not need us" posture in demos	K-0 interview results weak; trial→paid <20% at 50 trials (K-3 precursor); "we'll stick with the spreadsheet" dominating win/loss notes
R-12	**Slow, expensive buyer reach** [skeptic]: Qfactor sold into this exact buyer for ~10 years and stayed obscure (near-zero review footprint, verified by adversarial review) — evidence of long sales cycles and hard-to-reach owners relative to a $79–249/mo price.	GTM	High	Medium	Community-trust channels over ads (societies, forums, booths — `GO_TO_MARKET.md`); the audit lead magnet lowers first-touch friction to zero; referral + status-link loops to compound reach; per-channel CAC kill lines (`CUSTOMER_ACQUISITION.md`)	Audit volume <8/week by M3; any channel cash CAC >$2,000 for 2 quarters; conference pipeline not converting within 60 days of event
R-13	**Lead-magnet/SEO territory occupied** [skeptic]: free S-T-R lookup tools are saturated (randymajors.org, Earth Point, Township America — *(verified, https://www.earthpoint.us/Townships.aspx)*) and SurveyOps owns programmatic SEO on category keywords.	GTM	Certain (current fact)	Low–Medium	Lead magnet redesigned to the Prior-Work Audit (own-data import → coverage map), which no competitor offers (skeptic's own recommendation); content strategy is community-distribution-first, head-terms conceded (`CUSTOMER_ACQUISITION.md` §4)	Audit→trial <40%; organic search referrals dominated by unwinnable head terms; competitors launching a copycat free audit
R-14	**QuickBooks Online platform dependency**: invoice push (Firm tier value) rides Intuit's OAuth app regime — assessment questionnaire, refresh-token lifecycle, evolving compliance asks *(process verified, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ)*.	Platform	Low–Medium	Medium	Submit assessment week 1 (~30-minute form, near-immediate approval — verified); build token-refresh handling and re-auth UX defensively; QBO is an enhancement, not the core loop — pipeline/radar/status links all work without it	Intuit policy-change notices; token-refresh failure rate; assessment renewal friction

ID	Risk	Category	Likelihood	Impact	Mitigation	Early-warning signal
R-15	**Geocoder dependency**: US Census Geocoder (free, keyless) is the primary spatial indexer; outages, rate limits, or accuracy gaps on rural addresses hit import quality.	Platform	Medium	Medium	Batch + cache geocodes (one-time per row); manual pin-drop always available as fallback; abstraction layer so a commercial geocoder (or Township America's PLSS API) can be swapped in; rural-accuracy measured on the real-spreadsheet corpus	Geocode failure rate >15% on imports; Census API latency/outage alerts; rural design partners reporting misplaced pins
R-16	**Free-tier floor emerges in-category** (kill-taxonomy pattern 2, `DECISION_LOG.md` D-009): none found by adversarial search today (`FINAL_CONCEPT.md`), but a funded entrant or platform could give job tracking + import away free.	Market	Low	High	Speed to archive lock-in (the imported, enriched archive compounds and is painful to abandon — with export offered, retention must be earned by value); watch the vertical quarterly; K-6 kill gate defines the evidence-based response instead of panic	Funding announcements in surveying software; a competitor launching free import/audit; head-to-head win rate <25% for a quarter (K-6)

Notes for the reader

- **Skeptic coverage check:** every weakness in the round-3 implementability verdict (pyTRS license → R-6; meridian ambiguity → R-4; address-only spreadsheets → R-3; SurveyOps contradiction → R-1; occupied lead-magnet slot → R-13; PLSS/Texas exclusion → R-5; archive data-loss exposure → R-8; 6–8-week timeline optimism → R-6) and every weakness in the market verdict (six-competitor field → R-1; pricing above KudurruStone/ClickUp floor → R-11; TAM 3x correction → R-2; saturated free-tool space → R-13; Qfactor-obscurity adoption evidence → R-12; SurveyOps SEO → R-13; client portal non-uniqueness → R-1) is mapped to a register row.
- **Corrected claims are presented corrected** throughout: TAM is ~7,000 establishments (not 25,000); competitors exist and are named (the original "no vertical solution" claim was refuted); the S-T-R lead magnet was dropped for the audit; the build is 12–16 weeks (not 6–8). Sources: `research/revival-round3-results.json` , `business/FINAL_CONCEPT.md` , `decisions/DECISION_LOG.md` .
- Likelihood/impact ratings are the founder's judgment (assumptions), not researched probabilities.

Backsight — Landing Page Copy

> Voice guide for whoever implements this: plainspoken, technically respectful, zero hype. The reader is a licensed professional (PLS) who has been sold bad software before and distrusts adjectives. Every claim on the page must be one the product can demonstrate in the first ten minutes of a trial. No exclamation points anywhere on the page.

Meta

- **Meta title (58 chars):** `Backsight — Job Tracking & Prior-Work Search for Surveyors`
- **Meta description (152 chars):** `Job tracking built for land surveying firms. A pipeline that speaks surveying, a client status link, and a searchable map of every job you've ever done.`

Hero

Headline:

> ### Your firm has surveyed this ground before. Backsight remembers.

Subhead:

> Job tracking built for land surveying firms — a pipeline that runs fieldwork → drafting → licensed review → delivery, a status link that ends "where's my survey?" calls, and a searchable map of every job your firm has ever done.

Primary CTA button: `Start a 14-day trial` (no credit card required) **Secondary CTA link:** `Or run the free Prior-Work Audit — import your job list, see your coverage map`

Hero visual: screenshot of the Prior-Work Radar map — an incoming request pinned in a section, three prior jobs highlighted in the same section, dated 2019, 2021, 2024.

Trust line under CTA: `Flat per-firm pricing. Field crews never count against your license. Your data exports any time, in full.`

Pain Section 1 — The whiteboard doesn't scale past the parking lot

Header: Twenty-eight jobs, four crews, one whiteboard.

Every job in your shop moves through the same stages: request, quote, fieldwork, drafting, licensed review, delivery, invoice. But the whiteboard shows one word per job, the details live in email, and the due dates live in somebody's head. Nothing tells you a plat has been sitting in review for eleven days, or that three delivered jobs were never invoiced.

Backsight is a pipeline board with surveying's actual stages — not "lead / proposal / won" renamed. Advance a job from your desk or a crew's phone browser. A weekly stuck-jobs email flags anything idle too long at any stage, delivered-but-uninvoiced first.

Section CTA: `See the pipeline →`

Pain Section 2 — You are the status hotline

Header: Title companies don't need to call you. They need a link.

Every closing that's waiting on your survey generates a phone call, and the person who answers it is usually the licensed surveyor whose time bills highest. Multiply by every active job.

Every Backsight job gets a tokenized, read-only status page — no login, no app, no texting. The title company, builder, or attorney bookmarks it and watches the progress bar move: scheduled, fieldwork complete, in drafting, under review, delivered. When the stage changes, they see it before they think to call.

Section CTA: `See a live client status page →`

Pain Section 3 — Your most valuable asset is a memory

Header: "We shot that section in 2019" shouldn't depend on who's in the office.

The most valuable thing your firm owns isn't the GNSS gear — it's what you already know about ground you've surveyed before: control you set, corners you found, boundary resolutions you already worked out, plats already drafted. Today that asset lives in one person's memory and a shelf of folders. When that person is in the field, or retires, the asset is unreadable.

Backsight turns your job history into an index you can search by location — so the knowledge outlives the whiteboard and the memory.

Wedge Feature Story — Prior-Work Radar

Header: Import your job list. Backsight maps everything your firm has ever surveyed.

1. **Import your history.** Bring the spreadsheet you already have — job number, client, address, county, whatever's in it. No legal descriptions required: addresses geocode automatically, and anything the importer can't place, you drop on the map by hand.
2. **Backsight indexes it spatially.** Where your records carry Section-Township-Range, our parser reads it and joins it to public-domain BLM PLSS section data. Ambiguous descriptions get flagged for your confirmation, never silently guessed — a wrong-section match is worse than no match, and we built the product knowing that.
3. **Every new request checks the archive.** A request comes in; the radar instantly shows every prior job in that section and within a configurable radius — year, type, deliverables, original fee.

The payoff paragraph:

> When the radar hits, you're quoting a job your firm has already partly done. You know the ground, you may hold control there, and the research is already paid for. Quote it in minutes instead of hours, win it on price if you want to, and deliver it in a fraction of the field time. One radar hit that converts to a job can cover a year of Backsight.

Honesty note (keep on page — this audience respects it): PLSS section search covers the 30 public-land-survey states. In Texas and the metes-and-bounds states, the radar works from geocoded addresses, subdivision references, and map pins instead. If your archive is address-only, that's the normal case, not the degraded one.

Section CTA: `Run the free Prior-Work Audit on your own job list →`

Social Proof (placeholders — do not fabricate)

> **[PLACEHOLDER — do not ship fabricated quotes.]** Populate this section only with real, attributed quotes gathered from design-partner firms during the founder-led phase. Target format:
>
> - *"[Quote about a radar hit that won a job]"* — [Name], PLS, [Firm], [Town, State]
> - *"[Quote about status calls stopping]"* — [Name], Office Manager, [Firm], [State]
> - *"[Quote from a title company about the status link]"* — [Name], [Title Co.]
>
> Until three real quotes exist, replace this section with a factual strip: `Built for firms of 3–25 people · Runs on public-domain BLM PLSS data · Your data exports in full, any time`.

Pricing Section

Header: Flat per-firm pricing. Crews never count against a license.

	Solo — $79/mo	Firm — $149/mo	Practice — $249/mo
Users	Up to 5	Up to 15	Unlimited
Pipeline, radar, status links	✓	✓	✓
Historical import + coverage map	✓	✓	✓
QuickBooks Online invoice push	—	✓	✓
Client-branded status pages	—	—	✓
Full data export (CSV + files)	✓	✓	✓

- 14-day free trial on any tier. No credit card to start.
- Annual billing: 2 months free.
- No per-user math, no add-on creep, no "contact sales." The price on this page is the price.

Pricing CTA: `Start your trial`

FAQ

1. Can I import our existing job history, and how messy can it be? Messy is the expected case. The importer takes CSV or spreadsheet exports with whatever columns you have. Addresses are geocoded automatically; Section-Township-Range text is parsed where present; anything unplaceable lands in a review queue where you drop a pin or skip it. You don't need legal descriptions in your spreadsheet — most firms don't have them there, and the product is designed around that.

2. We're a three-person shop. Is this for us? Yes — you're who the Solo tier is for. The smaller the firm, the more the institutional memory lives in one head, and the more a searchable archive is worth. If a whiteboard is genuinely still working for you, keep it; the usual tipping point is the second crew or the first time a status call interrupts a boundary resolution.

3. What happens in Texas and other non-PLSS states? The Public Land Survey System covers 30 states. Outside them (Texas, the original colonies, and a few others), section search doesn't apply, so the radar runs on geocoded addresses, subdivision/lot-block references, and manual map pins. Job tracking and client status links work identically everywhere.

4. What if a legal description is ambiguous? "T2N R3W Sec 14" exists under multiple principal meridians. Backsight resolves township/range against your job's state and county, and when a description is still ambiguous it flags the row for your confirmation instead of guessing. We'd rather ask than silently index the wrong section.

5. Do my field crews need accounts? Do they count against my user limit? Crews update stage and attach field notes/photos from a phone browser — no app install. Pricing is flat per firm by tier; we will never charge per crew.

6. Can clients see my whole pipeline through the status link? No. Each status link is a tokenized, read-only page for one job: current stage, progress, expected delivery, and your office contact. No fees, no notes, no other jobs. You can revoke a link at any time.

7. What happens to our data if we leave? Full export — every job, event history, and attachment — in open formats (CSV + files), self-serve, at any time, on any tier. Your archive is your firm's asset; treating it as a hostage would be a bad way to run a company that sells to licensed professionals.

8. Does Backsight do CAD, adjust traverses, or talk to my data collector? No. Backsight is practice management: jobs, stages, clients, history, invoicing handoff to QuickBooks Online. Carlson, Civil 3D, and your data collector do what they do; Backsight tracks the business around them.

Closing CTA Block

Header: Find out what your firm already knows.

> Import your job list. In about ten minutes you'll be looking at a map of every parcel your firm has touched — and the next request that lands on ground you've already surveyed will tell you so.

Primary CTA: `Start a 14-day trial` **Secondary CTA:** `Run the free Prior-Work Audit`

Footer line (also appears on every client status page): `Tracked with Backsight — job tracking for land surveying firms.`

Copy sourcing notes (internal, not for page)

- ICP, stage names, pricing, and wedge framing per `business/FINAL_CONCEPT.md` (authoritative).
- PLSS covers 30 states / excludes Texas and colonial states: verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system — hence the honesty note and FAQ 3.
- BLM PLSS data is public domain: verified by adversarial review, https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons
- Meridian-ambiguity handling (FAQ 4) and address-first import (FAQ 1) are the skeptic-mandated fixes; the copy sells them as design choices because they are.
- The "no per-user math" line is a deliberate contrast with KudurruStone's per-user/role pricing (verified by adversarial review; competitor sites blocked automated fetch, figures per skeptic search-index research) — but we never name competitors on the page.
- Full-data-export promise (FAQ 7) implements the skeptic's lock-in/backup mitigation and is a sales asset.

Backsight — Product Requirements Document

Date: 2026-07-10 **Status:** Approved for MVP build (see `MVP_BUILD_SPEC.md` for the build-level spec, `MVP_SCOPE.md` for the in/out line) **Authoritative upstream documents:** `business/FINAL_CONCEPT.md` (product direction), `decisions/DECISION_LOG.md` (D-010, D-011)

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1. Problem statement

Small US land-surveying firms (1–5 field crews, 3–25 staff, $300K–$5M revenue) run 20–80 concurrent jobs — boundary, ALTA, topo at $500–$5,000 each — on a whiteboard plus email and a spreadsheet. Two failures follow:

1. **Job-status chaos.** Field crews, drafting, and licensed review live in different tools or none. Title companies and builders call the owner for status; deadlines slip silently between stages; delivered jobs sit unbilled. The pain pattern (status calls, idle drafters, delayed invoices) is documented in the scout evidence — with the caveat that the primary written source is a vendor blog and is flagged as such (https://www.cq-business-management-software.com/blog/best-business-management-software-for-surveyors-2025-complete-practice-management-guide/); it is scheduled for validation via owner interviews.
2. **Prior-work amnesia.** The firm's most valuable asset — control points, prior boundary resolutions, plats on ground it has already surveyed — lives in the owner's head and a shelf of folders. When a request arrives for a parcel the firm surveyed in 2019, quoting should take minutes and the bid should be unbeatable on price. It only works if someone remembers.

Backsight is job tracking that speaks surveying (Request → Quote → Scheduled → Fieldwork → Drafting → Licensed Review → Delivered → Invoiced), plus a **Prior-Work Radar** that indexes the firm's historical jobs spatially and surfaces everything the firm already knows about any new parcel, plus a tokenized read-only **client status link** that kills "where's my survey?" calls.

Market reality (post-adversarial-review, honest version)

- **TAM:** ~7,000 US establishments in NAICS 541370 — the adversarial review corrected the original scout's 25,000 figure, which conflated establishments with licensed individuals (verified by adversarial review, https://siccode.com/naics-code/541370/surveying-mapping). At 5% penetration × ~$140/mo average, ~$590K ARR: a strong bootstrapped outcome, not venture-scale, by design.
- **Competition exists and is named:** Qfactor, KudurruStone, Cyanic Job Book, Info-Retriever, CQ, and SurveyOps are real vertical vendors (round-3 market skeptic). SurveyOps's own marketing page claims Section-Township-Range job search (https://survey-ops.com/land-surveying-job-management-software). Differentiation is the prior-work-monetization wedge on *imported historical data*, client status links, modern UX, and flat per-firm pricing — not "no competition exists."
- **Geographic limit:** the PLSS wedge does not cover Texas or the ~20 colonial metes-and-bounds states (verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system). Launch ICP is PLSS states (Midwest/Mountain West); address-based search must carry non-PLSS ground.
- **Pricing anchor:** flat per-firm — $79/mo (up to 5 users), $149/mo (up to 15), $249/mo (unlimited) — against KudurruStone's per-user pricing (≈$100/mo for a 10-person firm per adversarial review) and the segment's documented resentment of per-seat/add-on creep (https://myquoteiq.com/jobbers-biggest-problem-exposed/).

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2. Personas

Full personas live in `business/CUSTOMER_PERSONA.md`. Summary:

Persona	Role	What they need from Backsight
Dana Whitfield, PLS (primary buyer)	Owner, sole license-holder, chief rainmaker, final reviewer	Know where every job stands without being the human database; deflect status calls; turn the firm's history into quoting power; get delivered work invoiced; get the archive out of her head
Marcus Lee (daily user, veto-holder)	Office manager / drafting technician	One source of truth crews actually update; a status link to paste at callers; a delivered-but-unbilled queue; prior-work search that doesn't route through Dana's memory; painless import of the existing spreadsheet

Design rule derived from the personas: **Marcus kills the tool if it adds data entry without removing work; Dana kills it after one wrong-section match.** Speed, import quality, and spatial-match trustworthiness are therefore acceptance criteria, not nice-to-haves (the KudurruStone "too slow" abandonment on the RPLS forum, surfaced by the round-3 market skeptic, is the cautionary tale).

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3. Skeptic-mandated requirements (non-negotiable)

These come from the round-3 implementability review (`research/revival-round3-results.json` , verdict: *survives_with_fixes*) and are incorporated as hard requirements:

ID	Requirement	Rationale
SM-1	**License-clean PLSS parser — no pyTRS.** The parser in `lib/plss.ts` is written from scratch. pyTRS's Modified Academic Public License explicitly prohibits commercial use (verified by adversarial review, https://github.com/JamesPImes/pyTRS). No code from pyTRS may be used, ported, or consulted during implementation.	Only mature OSS parser is legally unusable; the wedge's hardest component must be owned in-house.
SM-2	**Principal-meridian disambiguation.** "T2N R3W Sec 14" repeats across principal meridians. Every parse resolves against a state→default-meridian table (e.g., CO → 6th PM); if the state is unknown or the meridian cannot be resolved, the parse result carries `ambiguous: true` and the UI must display an explicit ambiguity flag rather than silently picking a match.	A wrong-section match, discovered by a customer, permanently destroys trust in the archive — the worst failure mode for this product.
SM-3	**Address-first ingestion.** Many firm spreadsheets contain client + street address only; the legal description lives in the deliverable PDF. Geocoded address (US Census Geocoder — free, no API key) is the *primary* spatial index; S-T-R parsing is *enrichment*, never a requirement. When both exist, the geocoded address wins. Manual map pin-drop is the fallback of last resort (post-MVP).	The "instant institutional memory" demo must not overpromise; import must degrade gracefully.

Two further skeptic directives shape the roadmap rather than the MVP: (a) full CSV+files **data export** ships in v1 production (mitigates archive-of-record data-loss liability and the lock-in objection); (b) the **Intuit app assessment** questionnaire is submitted in week 1 of production work, not after months of mocking (it is a ~30-minute form with near-immediate approval — verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ).

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4. MVP user stories and acceptance criteria

The MVP is a locally runnable demo (see `MVP_SCOPE.md` and `MVP_BUILD_SPEC.md`): SQLite, no external credentials, seeded with the fictional firm **Whitfield Land Surveying** (Fort Collins, Larimer County, CO — 6th PM), ~85 jobs 2019–2026.

4.1 Pipeline board (`/app/jobs`)

US-1 — As Marcus, I want a kanban board with surveying-native stage columns so the whiteboard has a single always-current replacement.

Acceptance criteria:

- Columns render in fixed order: `request, quoted, scheduled, fieldwork, drafting, review, delivered, invoiced`.
- Each card shows job number, client name, job type, county reference, due date.
- Cards display badges: **overdue** (due_date in the past, stage before `delivered`) and **prior-work-nearby** (≥1 other job in the same PLSS section or within 2 km).
- A card menu advances or regresses the job one stage; every transition writes a `job_events` row (actor, from_stage, to_stage, timestamp).
- Key transitions (at minimum: → `scheduled`, → `delivered`) also insert an `outbox` row addressed to the client contact.
- Board filters by job type and by crew; a list-view toggle shows the same data as a table.
- Board renders 25+ active jobs without perceptible lag (the "too slow" kill criterion).

US-2 — As Dana, I want stage regression to be possible (e.g., review sends a job back to drafting) so the board reflects reality, not an idealized flow.

- Regress is available from any stage except `request`; it writes a `job_events` row identical in structure to an advance.

4.2 Job detail (`/app/jobs/[id]`)

US-3 — As Marcus, I want one page with everything about a job so a status call is answerable in one glance (until the caller can be moved to the status link).

Acceptance criteria:

- Shows the full job record: number, client (linked), type, stage, quote amount, address, county/state, S-T-R + meridian when present, crew, due date, notes.
- Mini-map with the job's marker (Leaflet/OSM; if tiles fail offline, markers render on a plain coordinate-grid fallback — never a broken UI).
- Status timeline rendered from `job_events`, newest first.
- Stage controls identical in behavior to the board (events + outbox on key transitions).
- **Prior-work panel:** lists other jobs in the same section and jobs within 2 km, each with year, type, and original quote; same ranking logic as the Radar.
- **Copy client share link** button copies `/status/<share_token>` to the clipboard.
- Attachments list (metadata only in MVP) and editable notes.

4.3 Prior-Work Radar (`/app/radar`) — the wedge

US-4 — As Dana, I want to search by address or S-T-R when a request comes in, and instantly see every prior job on or near that ground, so I can quote in minutes with confidence.

Acceptance criteria:

- One search box accepts: a street address (matched against seeded geocodes in MVP), a `lat,lng` pair, or an S-T-R string in any of the supported formats (`T7N R69W Sec 14`, `T7N, R69W, S14`, `Township 7 North, Range 69 West, Section 14`, with optional aliquot quarters e.g. `NE1/4 SW1/4 S14 T7N R69W`).

- S-T-R input is parsed live by `lib/plss.ts` (SM-1); the parsed interpretation is echoed back to the user (township/range/section/meridian) so a misparse is visible before it misleads.
- If the parse is meridian-ambiguous (SM-2), the UI shows an explicit ambiguity warning and states which default was assumed.
- Results: a map with hit markers plus a ranked list — same-section matches first, then by distance — each row showing year, job type, deliverables, and original quote amount.
- Empty state explains the value proposition and shows a "try these" row of 3 example searches wired to seeded demo hooks (at least one returning ≥3 same-section hits).
- Address-first rule (SM-3): when a searched location has both geocode and S-T-R paths available, geocode-derived distance ranking is authoritative.

US-5 — As Marcus, I want the parser to fail loudly rather than guess, so a bad string never silently indexes the wrong ground.

- Unparseable input returns a clear "could not parse" message with format examples; it never returns a low-confidence match presented as fact.
- The parser has ≥12 vitest cases including failure cases (garbage input, missing section, out-of-range values like section 37).

4.4 Client status link (`/status/[token]`)

US-6 — As a title-company closer, I want a link I can refresh instead of calling, so I stop being the surveyor's interruption problem.

Acceptance criteria:

- Public page, **no authentication**, addressed by unguessable `share_token`.
- Read-only: firm-branded header, progress bar across the stage sequence, current stage, last-update timestamp, expected delivery date, office contact info.
- Exposes **no** pricing, no internal notes, no other jobs, no navigation into the app.
- An invalid token renders a friendly not-found page, not an error.
- (Production note: page footer reads "Tracked with Backsight" — the referral loop in the distribution plan.)

4.5 Dashboard (`/app`)

US-7 — As Dana, I want a morning view of what needs me, so nothing slips silently.

Acceptance criteria:

- KPI tiles: active jobs, overdue count, average days-in-stage, unbilled dollars (sum of quote amounts for `delivered`-but-not-`invoiced` jobs).
- Pipeline column counts mirroring the board.
- **Needs-attention list:** overdue jobs and jobs stuck in `review` longer than 10 days (the seeded bottleneck hook must appear here).
- Recent-activity feed from `job_events`.

4.6 Outbox (`/app/outbox`)

US-8 — As the evaluator of this demo, I want to see the notifications the system "sent," so the notification loop is proven without SMTP.

Acceptance criteria:

- Lists all `outbox` rows (timestamp, recipient, subject, body, linked job), newest first.
- New rows appear immediately after a key stage transition anywhere in the app.
- The page states plainly that this is a mocked email transport (see `MVP_SCOPE.md`).

4.7 Settings (`/app/settings`)

US-9 — As a demo user, I want an honest "what's real vs. mocked" panel, so the demo never oversells.

Acceptance criteria:

- Firm profile (name / logo text) editable.
- Demo-user switcher (Dana Whitfield, PLS / Marcus Lee) stored in a cookie; documented as mocked auth.
- A "What's real vs. mocked" panel matching the table in `MVP_SCOPE.md`: parser real; geocoding pre-seeded with `lib/geocode.ts` stub + documented TODO; email = outbox table; billing and auth mocked.

4.8 Cross-cutting MVP acceptance

- `npm install && npm run dev` on Node 22 with zero external credentials; `npm run build` clean; `npm test` green.
- `npm run seed` idempotently resets the demo database, including the four deliberate demo hooks (same-section radar hit, 2 overdue jobs, 1 job >10 days in review, several delivered-unbilled).
- No runtime network calls except OSM tiles, with a graceful offline fallback.

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5. Post-MVP roadmap

Ordered by dependency and sales impact. Timeline honesty per the implementability skeptic: a sellable production v1 is a **12–16 week** effort, not 6–8.

#	Feature	Description	Requirements & acceptance sketch
R-1	**CSV import wizard**	The onboarding wedge: upload the firm's historical job spreadsheet, map columns interactively, preview parsed/geocoded results on a map, confirm, import.	Column-mapping UI with saved mappings; per-row status (geocoded / S-T-R parsed / ambiguous / failed); **human map-confirmation step for every ambiguous or S-T-R-only row before it is indexed** (SM-2/SM-3); import must degrade gracefully to "address only" and to manual pin-drop; nothing enters the spatial index unconfirmed if flagged. Stress-test with 3–5 real firm spreadsheets before GA (skeptic directive).
R-2	**US Census Geocoder live integration**	Replace the seeded-geocode stub in `lib/geocode.ts` with the real Census Geocoder (free, no API key).	Batch endpoint for imports, single-address endpoint for new jobs; handles no-match and low-quality-match by flagging for pin-drop; results cached; retry/backoff; no hard runtime dependency (a geocoder outage degrades to manual placement, never blocks job creation).
R-3	**PLSS CadNSDI shapefile spatial join**	Import BLM's public-domain PLSS CadNSDI polygons (verified downloadable in bulk without credentials by adversarial review, https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons) so S-T-R strings resolve to real section polygons and point-in-section / adjacency queries are exact.	One-time ETL per state into PostGIS; parser output (SM-1/SM-2) joins to polygons within the resolved meridian only; adjacency = shared-boundary sections plus radius; per-state rollout starting with launch-ICP states. Buy-vs-build fallback exists (Township America sells a PLSS geocoding API covering all 37 meridians — https://townshipamerica.com/).
R-4	**Real email sending**	Outbox rows become actual emails (SES or equivalent) for client notifications and the weekly "stuck jobs" digest (delivered-but-uninvoiced first).	Outbox becomes a queue with send status/retries; per-client notification opt-out; digest scheduling; DKIM/SPF setup; email is B2B operational notification only — no SMS anywhere (concept constraint).
R-5	**Billing**	Stripe subscriptions for the three flat per-firm tiers ($79/$149/$249).	Trial (14 days) → paid conversion; seat counts enforced per tier; no per-user math anywhere customer-visible; dunning emails via R-4.
R-6	**Multi-firm auth**	Real authentication (Clerk or Auth.js) and hard multi-tenancy.	Every query firm-scoped; roles: owner, office, field; token status pages remain public but firm-scoped; session security review before first external customer. Includes the v1 **full data export** (CSV + attachments) skeptic directive.
R-7	**Mobile field view**	Responsive, thumb-first flow for crews: today's jobs, one-tap stage advance, note/photo attach from the truck.	Works on a phone browser over cell data (no native app in v1 — noting the competitive weakness the skeptic flagged: SurveyOps ships an iOS app); offline-tolerant form submission is a stretch goal.

Deliberately later / out of roadmap-v1: QuickBooks Online invoice push (public OAuth API verified feasible — https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ — but sequenced after R-1–R-6; the Intuit assessment form is submitted early regardless), Texas/GLO abstract data (phase-2 market per skeptic), GPS crew tracking, scheduling optimization, CAD/data-collector integration, payments, county-recorder enrichment.

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6. Non-functional requirements

- **Speed as a feature:** interactive pages respond <200 ms locally on seeded data; the board and radar must feel instant (the documented "too slow" churn cause in this vertical).
- **Trustworthiness of spatial claims:** no silent guesses anywhere — every ambiguity is surfaced (SM-2); every S-T-R interpretation is echoed back.
- **Offline-degradable demo:** the MVP runs with zero credentials and no network except optional map tiles.
- **Data stewardship (production):** as archive-of-record, Backsight carries contractual data-loss exposure; automated offsite backups plus a documented, rehearsed restore drill are required **before the first paying customer** (skeptic directive), plus customer-facing full export.
- **Privacy scope:** B2B business-contact data only; no consumer PII categories, no payments data in v1, no regulated deliverables (Backsight tracks the licensed-review stage; it never performs or certifies survey work).

7. Success metrics (production)

- Activation: firm imports ≥50 historical jobs and runs ≥1 radar search in week 1.
- Wedge proof: ≥1 radar hit used in a real quote within the 14-day trial (self-reported at trial-end check-in).
- Call deflection: ≥5 status-link views by external parties per firm per week by week 4.
- Revenue hygiene: delivered → invoiced median lag shrinking month-over-month per firm.
- Retention: logo churn <2%/mo after month 3; archive size (jobs indexed) growing monthly — the compounding moat metric.

Backsight MVP — Scope

Date: 2026-07-10 **Companion documents:** `PRD.md` (requirements), `MVP_BUILD_SPEC.md` (build-level spec, authoritative for implementation detail), `TECH_ARCHITECTURE.md`

The MVP is a **locally runnable demo application** (`product/app/`): `npm install && npm run dev` on Node 22, zero external credentials, everything demonstrable offline except optional map tiles. Its job is to prove the wedge and the retention loop end-to-end to a stranger in under ten minutes — not to be sellable software. The skeptic-verified production timeline (12–16 weeks to sellable v1) starts from this base.

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1. In / Out

Area	In MVP	Out of MVP (roadmap ref in `PRD.md` §5)
Pipeline	Kanban board with the 8 surveying stages, advance/regress, filters, list view, overdue + prior-work badges	Drag-and-drop reordering, custom stages, scheduling optimization
Job detail	Full record, mini-map, event timeline, stage controls, prior-work panel, share-link copy, notes, attachment metadata	Real file upload/storage (R-6/S3), CAD or data-collector integration
Prior-Work Radar	Search by address / lat,lng / S-T-R string; live license-clean parse with meridian defaults + ambiguity flag; ranked same-section-then-distance results on a map	Exact PLSS polygon spatial join (R-3), adjacency by shared section boundary, Texas/GLO abstract search
Client status link	Public tokenized read-only page: progress bar, current stage, last update, contact	Client-branded custom domains, client request-submission portal
Dashboard	KPI tiles, pipeline counts, needs-attention list (overdue, >10 days in review), activity feed	Configurable reports, revenue analytics, weekly digest email (R-4)
Notifications	Outbox table capturing every "sent" client notification on key stage transitions; `/app/outbox` viewer	Actual email delivery (R-4), SMS (never — concept constraint), digests
Data ingestion	Seeded database (~85 jobs, 12 clients, demo hooks) via `npm run seed`	CSV import wizard with human map-confirmation (R-1) — the production onboarding wedge
Geocoding	Pre-seeded lat/lng on all jobs; `lib/geocode.ts` stub with documented TODO	Live US Census Geocoder integration (R-2), pin-drop fallback UI
PLSS	From-scratch parser (`lib/plss.ts`) with state→meridian defaults and `ambiguous` flag; ≥12 vitest cases	CadNSDI shapefile ETL and PostGIS point-in-section queries (R-3)
Auth	Cookie-based demo-user switcher (Dana / Marcus), documented as mocked	Real auth + multi-firm tenancy (R-6)
Billing	Pricing page copy on the landing page only	Stripe subscriptions, trials, dunning (R-5)
Integrations	None	QuickBooks Online invoice push (post-roadmap-v1; Intuit assessment form submitted early)
Mobile	Responsive layout	Field-optimized mobile flow (R-7), native apps
Landing	Marketing page: hero, problem bullets, 3-tier flat pricing, FAQ incl. "what's mocked?"	SEO content, free S-T-R lookup lead magnet (dropped — slot occupied per adversarial review, https://www.earthpoint.us/Townships.aspx)

2. Real vs. mocked vs. pending

Component	Status	Detail
PLSS parser (`lib/plss.ts`)	**REAL**	Original, license-clean code (no pyTRS — its license prohibits commercial use, verified by adversarial review, https://github.com/JamesPImes/pyTRS). Parses the S-T-R grammar variants in `MVP_BUILD_SPEC.md` , resolves principal meridian from a state-default table (CO → 6th PM), sets `ambiguous: true` when the state is unknown. Fully unit-tested. This exact code ships to production.
Spatial ranking (same-section + distance)	**REAL** (method), simplified (data)	Section-string equality + haversine distance over stored coordinates. Real production logic upgrades to polygon containment/adjacency via CadNSDI (R-3); ranking semantics are unchanged.
Pipeline, events, dashboard, status page	**REAL**	Actual database reads/writes; the event log and outbox insertions are the production mechanism.
Geocoding	**MOCKED (pre-seeded) / stub PENDING**	All seed jobs carry realistic Fort Collins-area coordinates. `lib/geocode.ts` exposes the production interface with a documented TODO to wire the US Census Geocoder (free, no key). Address search in the Radar matches seeded addresses.
Email	**MOCKED**	Notifications insert rows into the `outbox` table; `/app/outbox` renders them. No SMTP anywhere. Production swaps the transport (SES) behind the same insertion point.

Component	Status	Detail
Auth	**MOCKED**	Cookie-based demo-user switcher; no passwords, no tenancy. Production: Clerk/Auth.js + firm scoping (R-6).
Billing	**MOCKED**	Pricing display only.
Attachments	**MOCKED**	Metadata rows only (filename, label); no file bytes stored. Production: S3 presigned uploads.
Map tiles	Real (OSM) with **offline fallback**	If tiles fail, markers render on a plain coordinate-grid background — never a broken UI.

The `/app/settings` page renders this same table in-product ("What's real vs. mocked" panel) so the demo can never oversell — a PRD acceptance criterion (US-9).

3. Demo narrative the MVP must support

Seven steps, one sitting, a stranger driving. Seed data guarantees every hook (`npm run seed` restores them idempotently).

1. **Landing page** (`/`) — the pitch in 30 seconds: "Your firm has surveyed this ground before. Backsight remembers." Flat pricing visible; FAQ answers "what's mocked in this demo?"
2. **Dashboard** (`/app`) — Whitfield Land Surveying's morning: active-job count, **2 overdue jobs**, **unbilled dollars tile** (several delivered-but-uninvoiced), needs-attention list including **1 job stuck in review >10 days**.
3. **Pipeline board** (`/app/jobs`) — the whiteboard replacement: 25 active jobs across 8 surveying-native stages; overdue and prior-work badges visible; advance a job to `delivered` from the card menu.
4. **Outbox** (`/app/outbox`) — the transition just fired a client notification: a new row addressed to the job's client sits at the top. Proof of the notification loop without SMTP.
5. **Prior-Work Radar** (`/app/radar`) — the wedge: click the first "try these" example (an incoming request-stage job). **≥3 historical jobs light up in the same section**, ranked above nearby others, each with year, type, and original quote. Type a raw S-T-R string (`Township 7 North, Range 69 West, Section 14`) and watch it parse live, meridian resolved to 6th PM with the interpretation echoed back; type a malformed string and see it fail loudly instead of guessing.
6. **Job detail** (`/app/jobs/[id]`) — one job's whole story: timeline of stage events, mini-map, prior-work panel, notes. Click **copy client share link**.
7. **Client status page** (`/status/[token]`) — paste the link in a new window: the read-only, no-login page a title company refreshes instead of calling. Progress bar, current stage, expected delivery, office contact — and nothing else exposed.

Close of demo: settings page (`/app/settings`) shows the real-vs-mocked panel — the honest accounting of what production still requires.

4. Explicit non-goals of the MVP

- Proving import quality on messy real-world spreadsheets (that is R-1, and the skeptic mandates stress-testing with 3–5 real firm spreadsheets before GA).
- Exact PLSS polygon geometry (section-string match + radius stands in for the CadNSDI join, R-3).
- Any claim of production security posture, tenancy, or backup discipline (R-6 and the pre-first-customer backup/restore drill).
- Sales collateral for non-PLSS states — the wedge demo is honest about covering PLSS ground (verified limitation, https://en.wikipedia.org/wiki/Texas_land_survey_system).

Backsight — Technical Architecture

Date: 2026-07-10 **Scope:** the local MVP (`product/app/`) and its evolution path to production. **Companion documents:** `MVP_BUILD_SPEC.md` (authoritative build spec), `PRD.md` , `MVP_SCOPE.md` .

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1. Stack and rationale

Layer	MVP choice	Why
Framework	**Next.js 14+ (App Router) + TypeScript**	One codebase for the marketing page, the authenticated app, and the public status page; server components + route handlers remove the need for a separate API service; TypeScript keeps the parser and data layer honest; it is the stack Claude Code iterates on fastest, which matters because the builder is a non-specialist (Decision D-011).
Styling	**Tailwind CSS**	Fast, consistent, no design-system bikeshedding; the "surveying = precision" visual language (dark slate + orange accent) is a token set, not a component library.
Database	**SQLite via `better-sqlite3`** (file `data/backsight.db`)	Zero-install, zero-credential, synchronous API that suits server components; the whole demo database is one file a stranger can delete and re-seed. Deliberately *not* an ORM: plain SQL keeps the schema visible and the Postgres migration mechanical.
Maps	**Leaflet via `react-leaflet`** , OpenStreetMap tiles	No API key (the no-credential constraint), permissive usage at demo scale, marker/popup primitives are all the MVP needs. **Offline fallback is mandatory:** if tiles fail, job markers render on a plain CSS coordinate-grid background — never a broken UI.
Testing	**vitest**	Fast TS-native unit tests; the parser test corpus is the highest-value test asset in the repo.
Auth	**None (mocked)**	Cookie-based demo-user switcher (Dana / Marcus). Documented as mocked in `/app/settings` .

Key stack constraint honored throughout: **no external network calls at runtime except OSM tiles; no API keys anywhere** (per `MVP_BUILD_SPEC.md`).

2. Data model

Five tables (SQLite; identical logical model in production Postgres):

```
clients ──< jobs ──< job_events
               |──< attachments
               |──< outbox (nullable job_id)
```

ER description:

- **`clients`** — one row per client organization/person. `id, name, kind (title_co|builder|homeowner|attorney|government), contact_email, phone` . A client has many jobs.
- **`jobs`** — the core entity; one row per survey job. `id, job_number ("2026-0142"), client_id → clients, type (boundary|alta|topo|construction_staking|subdivision_plat|elevation_cert), stage (request|quoted|scheduled|fieldwork|drafting|review|delivered|invoiced), quote_amount, address, county, state, lat, lng, plss_trs (nullable, "T7N R69W S14"), plss_meridian (nullable, "6th PM"), crew (nullable), due_date, created_at, delivered_at (nullable), notes, share_token (unique)` . The spatial columns implement the address-first rule: `lat/lng` is the primary index; `plss_trs` + `plss_meridian` are enrichment. `share_token` addresses the public status page.
- **`job_events`** — append-only status history. `id, job_id → jobs, at, actor, from_stage, to_stage, note` . Written on every stage transition; drives the job timeline and the dashboard activity feed. Never updated or deleted.
- **`attachments`** — deliverable/field-note metadata. `id, job_id → jobs, filename, label` . Metadata only in MVP (no file bytes).
- **`outbox`** — the mocked email transport. `id, at, to_email, subject, body, job_id → jobs` . Every "sent" notification is a row here; `/app/outbox` renders it.

Seed data (via `npm run seed` , idempotent): fictional firm **Whitfield Land Surveying** (Fort Collins, Larimer County, CO — 6th Principal Meridian); 12 clients; ~85 jobs 2019–2026 (~60 delivered/invoiced historical, ~25 active); coordinates scattered realistically around Fort Collins; ~70% of jobs carry area-consistent `plss_trs` values (T6N–9N, R68W–70W, sections 1–36) and the rest are address-only to exercise the fallback; plus the four demo hooks (same-section radar hit, 2 overdue, 1 stuck-in-review >10 days, several delivered-unbilled).

3. Core library: the PLSS parser (`lib/plss.ts`)

The wedge's hardest component, and the one under a hard legal constraint: **written from scratch, no pyTRS** — the only mature OSS parser bans commercial use under its Modified Academic Public License (verified by adversarial review, https://github.com/JamesPImes/pyTRS).

- Input grammar: `"T7N R69W Sec 14"` , `"T7N, R69W, S14"` , `"Township 7 North, Range 69 West, Section 14"` , optional aliquot prefixes (`"NE1/4 SW1/4 S14 T7N R69W"`).

- Output: `{township, townshipDir: 'N'|'S', range, rangeDir: 'E'|'W', section, quarters?: string[]} | null`.
- **Meridian disambiguation (skeptic-mandated):** an S-T-R tuple is not globally unique — it repeats across principal meridians. A state→default-meridian table (CO → 6th PM, etc.) resolves it; when the state is unknown, the result carries `ambiguous: true` and every consumer (Radar, import) must surface the flag rather than silently match. Silent wrong-section matches are the product's defined worst failure mode.
- Deterministic, dependency-free, ≥12 vitest cases including malformed input and out-of-range values. This exact module ships to production.

`lib/geocode.ts` is the geocoding seam: the production interface (address in, `{lat,lng,quality}` out) with a stub implementation and a documented TODO for the US Census Geocoder (free, no API key — chosen precisely because it keeps production credential-light).

4. Key flows

4.1 Stage transition → event + outbox

```
UI (board card menu / job detail controls)
  → server action: transition(jobId, toStage, actor)
      1. validate transition (any adjacent advance/regress; regress not from `request`)
      2. UPDATE jobs SET stage = ?, delivered_at = now() when entering `delivered`
      3. INSERT job_events (job_id, at, actor, from_stage, to_stage)
      4. if toStage ∈ {scheduled, delivered}:  -- key client-visible transitions
           INSERT outbox (to_email = client.contact_email, subject, body, job_id)
  → revalidate affected pages (board, job detail, dashboard, outbox)
```

One write path serves the board, the job page, the dashboard feed, the public status page (which reads `jobs.stage` + latest event), and the outbox — the single-source-of-truth property Marcus's persona demands. In production the outbox INSERT is unchanged; a queue worker drains it through SES (§6).

4.2 Radar search → parse/geocode → spatial rank

```
query string
  ├─ looks like S-T-R?  → lib/plss.parse()
  │     ├─ null → loud "could not parse" + format examples (never a guess)
  │     └─ parsed → resolve meridian via state default; set/display `ambiguous` flag
  │          → candidate jobs WHERE plss_trs matches (same meridian) → rank tier 1
  ├─ looks like "lat,lng"? → parse directly
  └─ otherwise address     → lib/geocode.lookup() (MVP: seeded-address match)
  → spatial ranking:
      tier 1: same PLSS section (string equality on normalized T/R/S + meridian)
      tier 2: haversine distance from the resolved point, ascending, cutoff 2 km
  → response: ranked list (year, type, deliverables, original quote) + map markers
```

Address-first rule (skeptic-mandated): when both a geocode and an S-T-R path are available, the geocoded point is authoritative for ranking. In production, tier 1 becomes point-in-polygon and polygon-adjacency against CadNSDI geometry (§6), with identical ranking semantics.

4.3 Public status page

`/status/[token]` → `SELECT ... FROM jobs WHERE share_token = ?` → render progress bar over the stage enum + latest `job_events.at` as "last updated." No session, no cookies, no data beyond that one job's client-safe fields. Invalid token → friendly not-found.

5. Security & privacy posture

MVP (demo): explicitly not hardened — mocked auth, local single-tenant SQLite, no PII beyond fictional seed data. The `/app/settings` real-vs-mocked panel discloses this in-product.

Production posture (requirements, not options):

- **Tenancy:** every query firm-scoped at the data layer; the tokenized status page stays public by design but exposes only client-safe fields of a single job (no pricing, notes, or navigation).
- **Tokens:** `share_token` generated with a CSPRNG, ≥128 bits, revocable/regenerable per job.
- **Data classification:** B2B business-contact data and job/site records; no consumer-PII categories, no payment data in v1, no regulated deliverables (Backsight tracks the licensed-review stage; the professional act stays with the customer — verified as carrying no software licensing regime by the adversarial review).
- **Archive-of-record liability (skeptic-flagged):** losing a firm's 500-job institutional memory is a business-ending grievance for the customer and a contractual (not regulatory) exposure for Backsight. Mitigations required before the first paying customer: automated offsite backups, a documented and rehearsed restore drill, ToS liability caps, and a customer-facing **full export** (CSV + attachments) — which also disarms the lock-in sales objection.
- Standard web hygiene: parameterized SQL throughout (already the pattern via `better-sqlite3`), no secrets in the repo, dependency audit in CI.

6. Production evolution path

The MVP is architected so each mock is a seam, not a rewrite:

Seam	MVP	Production step
Database	SQLite file	**Postgres (+ PostGIS)**. Schema ports table-for-table; SQL is already plain and parameterized. PostGIS is required by the CadNSDI join below.
Auth	Cookie demo-switcher	**Clerk or Auth.js**: real identities, firm-scoped tenancy, roles (owner / office / field). The demo-switcher's user object already shapes the session contract.
Geocoding	Seeded coords + `lib/geocode.ts` stub	**US Census Geocoder** live behind the same interface (free, no key): batch for imports, single for new jobs, cache + flag-for-pin-drop on no-match.
PLSS spatial join	Section-string equality + haversine	**CadNSDI import pipeline:** BLM publishes national PLSS CadNSDI polygons (townships + first-division/sections) in bulk, public-domain, no credentials — verified by adversarial review, https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons. ETL per launch state → PostGIS; parser output joins to real section polygons within the resolved meridian; adjacency = shared-boundary + radius queries. (~2.6M national section polygons fit comfortably in PostGIS per the same review. Buy-option fallback: Township America's commercial PLSS geocoding API, https://townshipamerica.com/.)
Email	`outbox` table	Outbox becomes a durable queue; a worker drains it via **SES** (DKIM/SPF, retries, per-client opt-out). The insertion point in the transition flow is unchanged — this is why the mock is a table, not a no-op.
Attachments	Metadata rows	**S3 presigned uploads**, virus scanning, size limits; attachment metadata schema is already in place.
Billing	Pricing copy	**Stripe** subscriptions for the flat per-firm tiers.
Hosting	`npm run dev`	Vercel/Fly/Render + managed Postgres; CI runs build + vitest + seed smoke.
QBO invoice push	absent	Public OAuth2 API; Intuit's production-app assessment is a ~30-minute form with near-immediate approval (verified by adversarial review, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ) — submit it in week 1 of production work; budget for the 100-day refresh-token lifecycle.

Sequencing follows `PRD.md` §5 (R-1 CSV import wizard → R-2 geocoder → R-3 CadNSDI → R-4 email → R-5 billing → R-6 auth/tenancy → R-7 mobile field view). Skeptic-verified schedule honesty: 12–16 weeks to a sellable v1, not the concept's original 6–8.

7. Known limitations

1. **PLSS coverage is not national.** Texas and the ~20 colonial metes-and-bounds states are outside PLSS; the S-T-R wedge is inert there and address/subdivision search must carry those markets (verified by adversarial review, https://en.wikipedia.org/wiki/Texas_land_survey_system). Launch GTM targets PLSS states; Texas is a phase-2 market via GLO abstract data.
2. **MVP spatial matching is approximate.** Section-string equality + haversine radius, not polygon geometry; edge cases (jobs near section corners, elongated parcels) mis-rank until R-3.
3. **Parser coverage is deliberately narrow.** The constrained grammar in §3 covers common firm-spreadsheet notation, not the full wildness of recorded legal descriptions; even mature parsers require human proofreading on real-world text (per the adversarial review of pyTRS's own documentation, https://github.com/JamesPImes/pyTRS) — hence the human map-confirmation step in the R-1 import wizard.
4. **Geocoding quality ceiling.** Rural addresses — common in surveying — geocode poorly; the pin-drop fallback (R-1) is a correctness feature, not a convenience.
5. **Demo data is synthetic.** Radar hit-rates on the seed corpus say nothing about hit-rates on a real firm's messy spreadsheet; the skeptic directive to stress-test with 3–5 real spreadsheets before GA stands.
6. **Single-writer SQLite.** Fine for a demo; concurrency, tenancy, and durability all arrive with Postgres (R-6 at the latest).
7. **No offline field capability.** Crews on cell-dead sites can't update stages until back in coverage; offline-tolerant submission is an R-7 stretch goal.
8. **Competitive non-uniqueness of individual features.** Adversarial review found vertical competitors already shipping map views, S-T-R search claims, and client portals (e.g., https://survey-ops.com/land-surveying-job-management-software); the architecture's defensible asset is the *imported-history* radar pipeline (parse → geocode → confirm → index) plus speed — which is why the parser, the import seams, and page performance get the engineering attention.

Backsight — Claude Code Implementation Plan

Date: 2026-07-10 **Audience:** a non-specialist (comfortable running terminal commands, not a professional developer) continuing this build with Claude Code. **Companion documents:** `MVP_BUILD_SPEC.md` (paste-ready build spec), `PRD.md`, `MVP_SCOPE.md`, `TECH_ARCHITECTURE.md`.

The working method throughout: give Claude Code the relevant spec file plus a milestone-sized instruction, let it build, then **verify with the commands listed before moving on**. Never accept a milestone whose verification step fails.

---

1. Environment setup

1. Install **Node 22** (https://nodejs.org or via `nvm install 22`). Verify: `node --version` → v22.x.
2. Install Claude Code (per its official docs) and open a session in `backsight-venture/product/`.
3. No accounts, API keys, or databases to install — SQLite is a file, map tiles need no key, and the MVP makes no other network calls. This is by design (`MVP_BUILD_SPEC.md` quality bar).

First prompt of the project — orientation, not code:

> Read product/MVP_BUILD_SPEC.md, product/PRD.md, product/MVP_SCOPE.md, and product/TECH_ARCHITECTURE.md. Summarize the build plan back to me and list any contradictions you see between them before writing any code.

2. Build sequence (ordered milestones)

Each milestone is independently verifiable. Expected wall-clock for the full MVP with Claude Code: a few working days for a non-specialist.

M0 — Scaffold

Prompt Claude Code to: create the Next.js 14+ App Router project in `product/app/` with TypeScript, Tailwind, `better-sqlite3`, `react-leaflet`, and vitest configured; add `npm run seed` and `npm test` scripts as stubs; commit nothing outside `product/app/`. **Verify:** `npm install && npm run dev` serves a placeholder page; `npm run build` passes.

M1 — Schema + seed

Prompt Claude Code to: implement the five tables (`clients`, `jobs`, `job_events`, `attachments`, `outbox`) exactly as specified in MVP_BUILD_SPEC.md, with auto-create on first run, and write `scripts/seed.ts` generating Whitfield Land Surveying's data: 12 clients, ~85 jobs (2019–2026, Fort Collins coordinates, ~70% with consistent PLSS values), and the four demo hooks — at least 3 historical jobs sharing a section with one request-stage job, 2 overdue jobs, 1 job in review >10 days, several delivered-but-uninvoiced. Seed must be idempotent. **Verify:** run `npm run seed` twice; row counts identical both times; spot-check the demo hooks with a SQL query Claude writes for you.

M2 — PLSS parser (the license-sensitive milestone)

Prompt Claude Code to: write `lib/plss.ts` from scratch per MVP_BUILD_SPEC.md — and state explicitly in the prompt: **"Do not use, port, or consult pyTRS or any code derived from it; its license prohibits commercial use. Write original code."** Require: the four input format families, the output shape, the state → default-principal-meridian table (CO → 6th PM), the `ambiguous: true` flag when state is unknown, and ≥12 vitest cases including failures (garbage input, missing section, section 37, township 0). **Verify:** `npm test` green; hand it three S-T-R strings from the seed data and one typo'd string and check outputs yourself. This module ships to production unchanged — spend review time here.

M3 — Pipeline board + stage-transition engine

Prompt Claude Code to: build `/app/jobs` (kanban per stage, cards with job #, client, type, county, due date, overdue and prior-work-nearby badges, type/crew filters, list-view toggle) and the single server-side transition function: validate → update stage → insert `job_events` → insert `outbox` row on transitions to `scheduled` and `delivered` (per TECH_ARCHITECTURE.md §4.1). **Verify:** advance a job on the board; confirm a new `job_events` row and (for key transitions) a new `outbox` row exist; regress it back.

M4 — Job detail + client status page

Prompt Claude Code to: build `/app/jobs/[id]` (full record, mini-map with offline grid fallback, event timeline, stage controls reusing the M3 engine, prior-work panel with same-section + within-2km lists, copy-share-link button, attachments list, notes) and the public `/status/[token]` page (read-only progress bar, current stage, last update, expected delivery, office contact; friendly not-found on a bad token; no pricing/notes/navigation exposed). **Verify:** copy a share link, open it in a private browser window (no cookies), confirm it renders and exposes nothing internal.

M5 — Prior-Work Radar

Prompt Claude Code to: build `/app/radar` per PRD US-4/US-5: one search box handling address (seeded match), `lat,lng`, and live S-T-R parsing via `lib/plss.ts`; echo the parsed interpretation back; show the ambiguity warning when flagged; rank results same-section first then haversine distance; map markers + ranked list (year, type, deliverables, original quote); empty state with value explanation; a "try these" row of 3 example searches wired to seed hooks; loud failure on unparseable input. **Verify:** the first "try these" example returns ≥3 same-section hits; a malformed string produces the error message, never a guess.

M6 — Dashboard + outbox + settings + landing

Prompt Claude Code to: build `/app` (KPI tiles: active, overdue, avg days-in-stage, unbilled $; pipeline counts; needs-attention list incl. stuck-in-review >10 days; activity feed), `/app/outbox` (all rows newest-first, mocked-transport disclosure), `/app/settings` (firm profile, Dana/Marcus cookie switcher, the "What's real vs. mocked" panel mirroring MVP_SCOPE.md §2), and the `/` landing page per MVP_BUILD_SPEC.md (hero, problem bullets, $79/$149/$249 flat pricing, FAQ incl. "what's mocked?", dark-slate + orange design, responsive). **Verify:** dashboard tiles match hand-computed values from the seed; the settings panel matches MVP_SCOPE.md.

M7 — Geocode stub, polish, README

Prompt Claude Code to: add `lib/geocode.ts` with the production interface and a documented US-Census-Geocoder TODO; sweep for the quality bar (clean `npm run build`, green `npm test`, no runtime network calls except OSM tiles, offline map fallback actually renders); write `product/app/README.md` with prerequisites, 3-command quickstart, the 7-step demo walkthrough from MVP_SCOPE.md §3, the mocked list, and a project-structure map. **Verify:** run the entire 7-step demo narrative yourself, offline if possible, from a fresh `npm run seed`.

3. Testing plan

Layer	Tool	What
Parser unit tests	vitest (`lib/plss.test.ts`)	≥12 cases: each grammar family, aliquot quarters, whitespace/comma variants, meridian default resolution, `ambiguous` flag on unknown state, and failures (garbage, missing section, out-of-range section/township). The highest-value test asset — it guards the wedge and the license-clean rewrite.
Smoke flow test	vitest against the transition function + queries	One scripted flow on a seeded test DB: create/read a job → advance `request→quoted→…→delivered` → assert a `job_events` row per hop and an `outbox` row on key transitions → radar query for the demo-hook section returns ≥3 hits ranked same-section-first → status-page query by token returns client-safe fields only.
Ranking unit tests	vitest	Same-section beats nearer different-section; 2 km cutoff honored; address-first rule when both geocode and S-T-R exist.
Build gate	`npm run build` + `npm test`	Both must pass before any milestone is "done"; make Claude Code run them itself at the end of every milestone.
Manual demo pass	you	The 7-step narrative in MVP_SCOPE.md §3 after M7 and before showing anyone.

Prompt pattern: "Write the tests first for , show me the failing run, then implement until green."

4. Environment variables

MVP: none. No keys, no secrets, no `.env` — this is a spec requirement, not an accident.

Future (production roadmap; create `.env.example` when each arrives):

Variable	Arrives with	Purpose
`DATABASE_URL`	R-6 / Postgres migration	Managed Postgres (+PostGIS) connection
`AUTH_SECRET` / Clerk keys	R-6	Real authentication
`AWS_ACCESS_KEY_ID` / `AWS_SECRET_ACCESS_KEY` / `S3_BUCKET`	attachments	Presigned uploads
`SES_REGION` / SES credentials	R-4	Real email transport
`STRIPE_SECRET_KEY` / `STRIPE_WEBHOOK_SECRET`	R-5	Billing
`QBO_CLIENT_ID` / `QBO_CLIENT_SECRET`	QBO integration	Intuit OAuth (submit the app-assessment form early — it's a ~30-minute questionnaire, https://help.developer.intuit.com/s/article/New-app-assessment-process-FAQ)

Note the US Census Geocoder (R-2) needs **no key** — that's why it was chosen.

5. Mocking strategy

Principle: **every mock is a seam with the production interface, never a dead end** (details in MVP_SCOPE.md §2 and TECH_ARCHITECTURE.md §6).

- **Email → `outbox` table.** Code "sends" by inserting a row; production adds a worker that drains the same table through SES. Never scatter send-calls — the insertion point is single.
- **Geocoding → `lib/geocode.ts` stub.** Callers use the real interface today; swapping in the Census Geocoder changes one file.
- **Auth → cookie switcher.** UI reads a `currentUser` object shaped like the future session; replacing it with Clerk/Auth.js doesn't touch page code.
- **Attachments → metadata rows.** Schema is production-shaped; S3 upload lands later without migration.
- **Spatial join → string-match + haversine.** Ranking semantics are the contract; CadNSDI polygons (R-3) upgrade precision behind the same result shape.
- **Honesty rule:** anything mocked is listed in `/app/settings` and the README. Instruct Claude Code: "never add a fake 'success' path that pretends an external service was called — insert a row or call the stub."

6. First 10 real-world hardening tasks (post-MVP, pre-first-customer)

Ordered; sourced from the PRD roadmap and the adversarial-review directives.

1. **Migrate SQLite → Postgres (+PostGIS)** and stand up managed hosting with CI (build + tests + seed smoke). Everything below assumes it.
2. **Real auth + firm tenancy (Clerk or Auth.js):** every query firm-scoped, roles (owner/office/field), CSPRNG share tokens with per-job revoke/regenerate.
3. **Automated offsite backups + a rehearsed restore drill,** documented — required before the first paying customer (skeptic directive; archive-of-record liability).
4. **Full data export (CSV + attachments) in the product UI** — mitigates data-loss exposure and kills the lock-in objection in sales.
5. **CSV import wizard (R-1)** with saved column mappings, per-row status, and **mandatory human map-confirmation for every ambiguous or S-T-R-only row**; then stress-test with 3–5 real firm spreadsheets and measure what fraction of rows parse/geocode (skeptic directive — this decides how the radar is pitched).
6. **Live US Census Geocoder integration (R-2):** batch + single, caching, no-match → pin-drop flag, and a manual pin-drop UI.
7. **CadNSDI PLSS pipeline (R-3):** ETL BLM's public-domain shapefiles for the launch states into PostGIS (https://gbp-blm-egis.hub.arcgis.com/datasets/BLM-EGIS::blm-national-plss-public-land-survey-system-polygons); point-in-section + adjacency queries within the resolved meridian; regression-test parser→polygon agreement on the seed corpus.
8. **Real email (R-4):** SES with DKIM/SPF, outbox-drain worker with retries, per-client opt-out, and the weekly stuck-jobs digest (delivered-but-uninvoiced first).
9. **Performance pass under load:** seed 5,000 jobs and keep board/radar interactions fast (indexes, pagination, query plans) — "too slow" is the documented churn cause in this vertical (KudurruStone abandonment, per adversarial review).
10. **Billing (R-5) + legal basics:** Stripe flat per-firm tiers with 14-day trial; lawyer-reviewed ToS with liability caps appropriate to an archive-of-record product; submit the Intuit app assessment now so QBO invoice push is unblocked when its turn comes.

7. Working rules with Claude Code (learned constraints)

- **The license rule is absolute:** any prompt touching PLSS parsing must restate "no pyTRS, no derived code." Review the parser diff yourself for tell-tale ported structure.
- **One milestone per session-thread;** start each by having Claude re-read the spec files. Contradictions between docs: `FINAL_CONCEPT.md` and `MVP_BUILD_SPEC.md` are authoritative.
- **Make Claude prove it:** end every milestone with "run `npm run build` and `npm test` and paste the output."
- **Never let it invent data or URLs** in user-facing copy; marketing claims come only from `business/FINAL_CONCEPT.md` and the citation-checked research files.

Backsight MVP — Build Specification

Target: a locally runnable demo app in `backsight-venture/product/app/` that a stranger can start with `npm install && npm run dev` (Node 22). No external credentials required; everything demonstrable offline except map tiles (graceful fallback required).

Stack (fixed)

- Next.js 14+ (App Router, TypeScript), Tailwind CSS
- SQLite via `better-sqlite3` (file `data/backsight.db`, auto-created + seeded on first run or via `npm run seed`)
- Leaflet via `react-leaflet` for maps, OpenStreetMap tiles; if tiles fail to load (offline), the map area must still render job markers on a plain coordinate grid background (CSS fallback), never a broken UI
- No auth provider: a demo-user switcher (owner "Dana Whitfield, PLS", office manager "Marcus Lee") stored in a cookie. Documented as mocked.

Data model (SQLite)

- `clients` (id, name, kind: title_co|builder|homeowner|attorney|government, contact_email, phone)
- `jobs` (id, job_number e.g. "2026-0142", client_id, type: boundary|alta|topo|construction_staking|subdivision_plat|elevation_cert, stage: request|quoted|scheduled|fieldwork|drafting|review|delivered|invoiced, quote_amount, address, county, state, lat, lng, plss_trs nullable e.g. "T7N R69W S14", plss_meridian nullable e.g. "6th PM", crew nullable, due_date, created_at, delivered_at nullable, notes, share_token unique)
- `job_events` (id, job_id, at, actor, from_stage, to_stage, note) — status history
- `attachments` (id, job_id, filename, label) — metadata only, no real files needed
- `outbox` (id, at, to_email, subject, body, job_id) — “sent” notifications land here (mocked email)

Seed data (`scripts/seed.ts`, run via `npm run seed`)

Fictional firm: **Whitfield Land Surveying, PLS** — Fort Collins, Larimer County, Colorado (PLSS state, 6th Principal Meridian).

- 12 clients (2 title companies, 3 builders, 1 municipality, 5 homeowners, 1 attorney)
- ~85 jobs spanning 2019–2026: ~60 delivered/invoiced historical, ~25 active spread across all stages. Coordinates: realistic lat/lng scatter around Fort Collins (40.35–40.75 N, -105.30 – -104.90 W). ~70% of jobs have plss_trs values consistent with that area (Townships 6N–9N, Ranges 68W–70W, sections 1–36); the rest address-only (tests the fallback).
- Deliberate demo hooks: (a) at least 3 historical jobs share the same section as one incoming `request`-stage job so Prior-Work Radar shows a hit; (b) 2 active jobs overdue (due_date past); (c) 1 job sitting in `review` >10 days (bottleneck callout); (d) unbilled: several `delivered` but not `invoiced`.

Core lib: PLSS parser (`lib/plss.ts`) — LICENSE-CLEAN, written from scratch

Parse strings like "T7N R69W Sec 14", "T7N, R69W, S14", "Township 7 North, Range 69 West, Section 14", "NE1/4 SW1/4 S14 T7N R69W" → `{township: 7, townshipDir: 'N', range: 69, rangeDir: 'W', section: 14, quarters?: string[]}` or null. Include a state→default principal meridian table (CO→6th PM etc.) and an `ambiguous: true` flag when state unknown. Unit tests with vitest (≥12 cases incl. failures). Do NOT use or port pyTRS (license-restricted); write original code.

Pages (App Router)

1. `/` — **Landing page** (marketing): hero ("Your firm has surveyed this ground before. Backsight remembers."), problem bullets, product screenshots placeholders (styled divs), 3-tier pricing ($79 / $149 / $249 flat per firm), FAQ (incl. "what's mocked in this demo?"), CTA → `/app`. Professional, restrained design (surveying = precision: dark slate + orange accent), responsive.
2. `/app` — **Dashboard**: KPI tiles (active jobs, overdue, avg days in stage, unbilled $), pipeline column counts, "needs attention" list (overdue, stuck-in-review), recent activity feed from job_events.
3. `/app/jobs` — **Pipeline board**: kanban columns per stage, cards (job #, client, type, county ref, due date, badges for overdue/prior-work-nearby), advance/regress via card menu (writes job_events, inserts outbox row notifying client on key transitions). Filter by type/crew. A list-view toggle.
4. `/app/jobs/[id]` — **Job detail**: full record, mini-map, status timeline, stage controls, prior-work panel (same-section + within-2km list), copy client-share-link button, attachments list, notes.
5. `/app/radar` — **Prior-Work Radar** (the wedge): search box accepting an address (matches seeded addresses / lat,lng) OR an S-T-R string (parsed live with lib/plss). Results: map with hit markers + ranked list (same section first, then distance), each showing year, type, deliverables, original quote. Empty-state explains the value. Include a "try these" row of 3 example searches wired to demo hooks.
6. `/status/[token]` — **Public client status page** (no auth): firm-branded read-only tracker — progress bar across stages, current stage, last update, expected delivery, office contact. This is the page a title company refreshes instead of calling.
7. `/app/settings` — firm profile (name/logo text), user switcher, and a **"What's real vs. mocked"** panel (parser: real; geocoding: pre-seeded, Census Geocoder integration stubbed at `lib/geocode.ts` with documented TODO; email: outbox table; billing/auth: mocked).
8. `/app/outbox` — list of "sent" notifications (proves the notification loop without SMTP).

Quality bar

- `npm run build` must pass clean; `npm test` (vitest) green.
- Seed idempotent (`npm run seed` resets).
- README.md in `product/app/` : prerequisites, 3-command quickstart, demo script walkthrough (7 steps hitting each page's hook), what's mocked, project structure map.
- No external network calls at runtime except OSM tiles (with fallback). No API keys anywhere.

Decision Log

Every executive decision made autonomously during this venture build, with reasoning. Newest entries at the bottom. All times UTC, 2026-07-10.

D-001 — Project location and structure

Decision: Build the venture in `painpoint-venture/` at the repo root (working name; will be renamed to the product name once selected), with subfolders `research/`, `business/`, `product/`, `decisions/`. **Reasoning:** The repo is a sandbox monorepo already containing multiple independent project folders (`bookshelf/`, `pinfactory/`, `storyforge/` …), so a self-contained top-level folder matches the established convention and keeps the Definition of Done ("a stranger can open the project folder") satisfiable with one directory.

D-002 — Research strategy: parallel hunting grounds before any ideation

Decision: Run 8 parallel research agents over distinct "hunting grounds" (SMB trades, niche professions, new 2024–26 regulations, AI-era ops pain, creators/solopreneurs, e-commerce sellers, devtools/B2B ops, hated incumbents) before generating a single product idea. **Reasoning:** Idea-first processes anchor on the first plausible concept. Evidence-first fan-out maximizes surface area and forces every later candidate to trace back to a cited pain. The 8 grounds were chosen to be mutually exclusive and to span the three classic sources of opportunity: underserved niches, newly created pain (regulation, AI), and incumbent failure.

D-003 — Hard buildability constraints enforced at research time, not later

Decision: Every scout carries the implementation constraint (no licensed professions, no gated APIs, no proprietary data, no hardware, no fragile scraping as core value) as a hard filter on what it may return. **Reasoning:** Filtering at the source is cheaper than discovering at the tournament stage that a winning idea is unbuildable by a non-specialist. This also matches the prompt's requirement that skeptics later re-verify implementability — two independent layers of defense.

D-004 — Citation discipline

Decision: Scouts must return structured evidence objects (`claim` + `source_url`) for every problem, with explicit instructions to never invent sources; claims that later fail spot-checks get flagged or dropped in the adversarial phase. **Reasoning:** The brief demands "no inventing — research and verify every fact." Structured citations make the adversarial verification phase mechanical instead of hopeful.

D-005 — Rate-limit recovery via workflow resume, not restart

Decision: When 6 of 8 scouts failed on a session rate limit (reset 04:30 UTC), the partial results (SMB-trades and niche-pros scouts, 8 cited problems) were saved to `research/raw-scan-partial-1.json` and the workflow was resumed after the reset with `resumeFromRunId`, replaying the 2 completed scouts from cache. **Reasoning:** Avoids re-spending ~175k tokens of completed research and keeps evidence identical across the retry — no silent divergence between what was saved and what feeds the tournament.

D-006 — Tournament shortlist: 12 of 34 problems

Decision: Shortlisted 12 candidates from the 34 scouted problems (`research/tournament-shortlist.json`), merging two independent-scout convergences into single candidates: COI tracking (found by both smb-trades and niche-pros scouts) and per-client AI cost attribution (found by both ai-era-ops and devtools-b2b scouts). **Selection criteria:** (a) pain denominated in hard dollars or legal risk, not convenience; (b) evidence quality — named prices, named incumbents, documented failures; (c) buildability without gated access; (d) diversity across the 8 hunting grounds so the tournament compares genuinely different theses. **Dropped notables and why:** interpreter-agency back-office (marketplace-ops heavy, 24/7 operational burden), HOA treasurer-in-a-box (money movement/fiduciary exposure needs ACH rails and trust the MVP can't demonstrate), pet-boarding software (crowded category with several well-funded challengers already circling the Gingr rollup), support-AI answer QA (core value requires access to customers' agent transcripts — an enterprise-ish integration the constraint forbids), youth-sports league management (seasonal sales cycle, volunteer buyers with near-zero budgets), surveyor job tracking (weaker evidence of software WTP vs hardware spend).

D-007 — Tournament design: 3 persona judges + weighted aggregate + dual skeptics on top 4

Decision: Judge panel = three personas with deliberately conflicting priors (seed VC, bootstrapped indie hacker, pragmatic CTO), each scoring all 12 concepts on 8 criteria (1–10). Aggregate = mean across judges of the weighted criterion sum, with weights emphasizing the brief's constraints: buildability ×1.5, claude_code_fit ×1.5, willingness_to_pay ×1.3, pain ×1.2, urgency/differentiation ×1.1, market/distribution ×1.0. Top 4 then face two independent skeptics each (implementability, market) doing FRESH web research including spot-checks of the original citations. **Reasoning:** Persona diversity catches different failure modes (VC catches vitamin products, indie catches unreachable niches/ops burden, CTO catches gated-API fantasies). Weights encode the prompt's hard constraint that Claude Code buildability dominates. Skeptics get web access precisely because the ideators did not — claims survive only if independently checkable.

D-008 — Reject the entire judge top 4; escalate ranks 5–8 to a second skeptic round

Decision: All four judge favorites (chargeback-evidence 72.2, coi-tracking 68.5, threepl-audit 68.0, commission-recon 68.0) were rejected after market skeptics found disqualifying facts the original scouts missed: (1) chargeback — Stripe Smart Disputes (Nov 2025, on by default) and Shopify Automated Dispute Response do the core loop natively for free, and ChargePay already sells flat-fee AI chargeback automation from $19.99/mo; (2) COI — TrustLayer Starter and bcs now give away free AI-parsing COI tracking tiers, destroying the "incumbents can't serve the low end" premise; (3) 3PL audit — the dim-weight recalculation wedge is technically unsound (billable weight comes from the 3PL's cubiscanned carton, not SKU dimensions) and ≥3 direct competitors exist including free-first-audit contingency players; (4) commission recon — zero-template AI parsing is commoditized (Fintary $199/mo, unLocked 332 carrier feeds) and every evidence source was vendor content-marketing. Ranks 5–8 (eaa-accessibility, hvac-agreements, membership-refuge, autorenewal-compliance) now face the identical dual-skeptic gauntlet before any winner is declared. **Reasoning:** A pattern emerged: judge scores rewarded "hard-dollar recovery" theses, but those same money-recovery categories attract fast-following competition and platform-native features — exactly what fresh adversarial research exposed. Selecting a "seriously_weakened" winner because it out-scored other weakened candidates would optimize the scoreboard, not the business. The tournament's integrity requires the next tier to face equal scrutiny (same prompts, same effort) rather than winning by default.

D-009 — Round 2 killed ranks 5–8 too; codify the kill taxonomy and run an engineered revival round

Decision: Round-2 skeptics seriously weakened all four of ranks 5–8: eaa-accessibility (category occupied at SMB price by Eye-Able/AudioEye/$49/mo and first EAA court rulings going to defendants), hvac-agreements (standalone architecture contradicts its own wedge; Jobber Core $39/mo already auto-schedules recurring visits; anchor evidence traced to a programmatic SEO farm), membership-refuge (MemberDay is a near-clone at $29/mo; Zeffy is free; SERPs saturated), autorenewal-compliance (the recurring deliverable is legal judgment — FTC v. DoNotPay exposure — and Recharge ships mandated reminders natively). Rather than promote ranks 9–12 (which scored lowest for good reasons), I codified the seven verified kill patterns from all 8 autopsies into a "kill taxonomy" and launched a final revival round: 4 candidates engineered against that taxonomy — three revived unskepticized problems from the original 34 (EU STR registration compliance for professional co-hosts, land-surveyor job tracking, interpreter-agency back-office) plus one skeptic-prescribed pivot (HVAC membership radar re-architected as integration-first/read-only, no SMS, no payments). Each gets a high-effort ideator plus the same dual skeptics. **Kill taxonomy (verbatim criteria given to ideators):** (1) no platform-native substitute risk; (2) no free-tier floor from funded players; (3) wedge technically sound and locally demonstrable; (4) evidence must trace to primary sources, never vendor content-marketing; (5) no regulated deliverables (legal verdicts, core SMS, payment migration); (6) no saturated panic-SEO as the distribution plan; (7) no structural churn (self-extinguishing value or fast buyer graduation). **Reasoning:** Two full adversarial rounds produced a validated failure taxonomy — that is itself a research asset. Designing candidates against known kill patterns and re-testing them adversarially is the tournament equivalent of a regression suite: the final winner must survive attacks that killed eight predecessors. The EU STR candidate is especially promising a priori: EU Regulation 2024/1028 became applicable 20 May 2026 (seven weeks ago), creating brand-new recurring obligations — demand created by law, not contested from competitors.

D-010 — Winner: Backsight (surveyor job tracking + prior-work intelligence)

Decision: Backsight is the selected venture. It was the only concept of 12 whose market skeptic returned zero fatal flaws; its implementability skeptic returned "survives_with_fixes" with a concrete, fully addressable fix list (license-clean parser instead of pyTRS, principal-meridian disambiguation, address-first ingestion, honest ~~7,000-firm TAM, repricing vs. KudurruStone). The a-priori favorite of round 3 — Registry Radar (EU STR compliance) — was killed by facts that would have destroyed the business post-launch: Conforme.info already ships the identical "regulatory system of record" positioning, Spain's Supreme Court annulled the national registry (judgment 620/2026, 21 May 2026), and the launch markets' registrations mostly have no renewal deadlines — the proposed retention engine tracked obligations that don't exist. **Reasoning:** Preferring the concept with surviving-but-modest economics (~~$590K ARR at 5% of ~7,000 firms, flat-priced) over concepts with bigger claimed TAMs and fatal flaws is exactly the trade the brief's "defensible with evidence, not safest" instruction demands: Backsight's risks are execution risks (competitor micro-vendors, TAM ceiling) rather than structural falsehoods (free-tier floors, platform-native substitutes, evidence built on SEO farms). It also best satisfies the buildability constraint: the wedge (parse → geocode → spatial join → radar UI) is deterministic, fully demonstrable offline with sample data, and involves zero gated access, zero licensing exposure, zero regulated deliverables.

D-011 — Build-phase architecture: parallel doc writers + one MVP builder, then completeness critic

Decision: Deliverables are produced by a final workflow: five parallel document-writer agents (research brief/persona/validation; competitor+gap analysis; tournament+adversarial review compilation; PRD+MVP scope+architecture+implementation plan; GTM bundle: landing copy, pricing, acquisition, sales/onboarding, metrics, risks), each reading the committed research JSONs and writing directly into `backsight-venture/`, plus one builder agent implementing `product/MVP_BUILD_SPEC.md`. The main session then runs and verifies the app, applies fixes, and a final completeness-critic agent audits everything against the Definition of Done. **Reasoning:** Doc writing is parallelizable with no shared files (distinct paths per agent); the MVP is a single coherent codebase, so one builder with a tight spec beats several colliding ones. Verification stays in the main session because it owns the toolchain and can iterate fastest on failures. Citation discipline is enforced by instructing writers to cite only URLs present in the research/skeptic JSONs (each of which was produced or spot-verified by web-researching agents).

D-012 — Completeness review passed; all defects fixed; venture package closed out

Decision: The independent completeness critic audited all 24 deliverables against the Definition of Done and returned PASS with 7 minor defects (1 medium: the review itself was not yet committed; 3 low: a wrong job count in the demo script, a wrong step pointer, a broken pricing table row; 1 info: missing "Tracked with Backsight" referral footer on the status page; 2 nits: missing engines field, unexpanded jargon). All seven were fixed the same day and verified with a clean production build; the review plus a fixes-applied addendum is committed at `decisions/FINAL_COMPLETENESS_REVIEW.md`. The venture package is complete. **Reasoning:** A completeness gate is only credible if its findings are (a) preserved verbatim, including the misses, and (b) demonstrably acted on. Fixing defect 5 in code (rather than leaving it a "production note") also makes the GTM referral loop demonstrable in the local demo, which strengthens deliverable coherence between the business plan and the product.

Final Completeness Review

Independent audit by a dedicated completeness-critic agent (2026-07-10), run after all deliverables and the MVP were finished. The critic read all ~2,800 lines of documentation, machine-checked every relative cross-reference, ran the test suite, booted the dev server and smoke-tested all 9 demo routes against a fresh seed, traced 16 cited URLs into the raw research JSONs, and swept every markdown file for dead evidence.

Overall verdict: PASS with minor defects — all of which were fixed immediately after this review (see addendum at the bottom).

Deliverable scorecard (24 items)

#	Deliverable	Verdict	Location
1	Executive summary	PASS	`README.md` §Executive summary
2	Market research brief w/ citations	PASS	`research/MARKET_RESEARCH_BRIEF.md` (all claims cited or labeled)
3	Customer persona	PASS	`business/CUSTOMER_PERSONA.md` (2 personas + evidence-traceability table)
4	Problem validation	PASS	`research/PROBLEM_VALIDATION.md` (flagged sources, interview gate, pre-registered pass/fail heuristics)
5	Competitor analysis	PASS	`research/COMPETITOR_ANALYSIS.md` (6 vertical vendors, verification caveats explicit)
6	Gap analysis	PASS	`research/GAP_ANALYSIS.md` (VERIFIED vs HYPOTHESIS labeling throughout)
7	Idea tournament with scores	PASS	`research/TOURNAMENT.md` (12 concepts, 3 full per-judge scorecards, weights reproduce rankings)
8	Final selected concept	PASS	`business/FINAL_CONCEPT.md`
9	PRD	PASS	`product/PRD.md` (user stories, acceptance criteria, skeptic-mandated requirements)
10	MVP scope	PASS	`product/MVP_SCOPE.md` (in/out table + real-vs-mocked table)
11	Technical architecture	PASS	`product/TECH_ARCHITECTURE.md` (data model, flows, production seams, 8 known limitations)
12	Claude Code implementation plan	PASS	`product/CLAUDE_CODE_IMPLEMENTATION_PLAN.md` (M0–M7 milestones, genuinely non-specialist-usable)
13	Local MVP	PASS	`product/app/` — 21/21 tests green; all 9 routes 200; radar returns the 3 same-section hits; garbage input fails loudly; bogus status token renders friendly not-found
14	Demo instructions	PASS*	`product/app/README.md` 7-step script — one numeric inaccuracy (defect 2, fixed)
15	Landing page copy	PASS	`business/LANDING_PAGE_COPY.md` (incl. do-not-fabricate social-proof guard)
16	Pricing strategy	PASS*	`business/PRICING_STRATEGY.md` — one broken table row (defect 4, fixed)
17	Go-to-market plan	PASS	`business/GO_TO_MARKET.md`
18	Customer acquisition strategy	PASS	`business/CUSTOMER_ACQUISITION.md` (funnel math labeled assumption-by-assumption; LTV/CAC arithmetic verified correct)
19	Sales & onboarding flow	PASS	`business/SALES_ONBOARDING_FLOW.md`
20	Metrics & success criteria	PASS	`business/METRICS_SUCCESS_CRITERIA.md` (north star, M3/M6/M12 thresholds, 7 numeric kill gates)
21	Risk register	PASS	`business/RISK_REGISTER.md` (16 risks, each with mitigation + early-warning signal)
22	Decision log	PASS	`decisions/DECISION_LOG.md`
23	Adversarial review	PASS	`research/ADVERSARIAL_REVIEW.md` (all 12 verdicts, per-claim evidence-check tables, kill taxonomy; discloses the one failed skeptic run)
24	Final completeness review	FAIL (absent) → **fixed**	This document

Defects found, ordered by severity

1. **[MEDIUM — missing deliverable]** No committed final completeness review; the README folder guide had no entry for it.
2. **[LOW — factual inaccuracy]** `product/app/README.md` demo step 2 claimed "~29 active jobs"; a fresh seed yields **25** by the dashboard's own definition (21 in-flight + 4 delivered; 85 total, 60 invoiced).
3. **[LOW — wrong step pointer]** Root `README.md` said the Radar aha moment is demo step 4; it is step 5.
4. **[LOW — broken Markdown table]** `business/PRICING_STRATEGY.md` had a stray hard line break inside the "Annual prepay" cell, splitting the discounting table.
5. **[INFO — disclosed demo/GTM gap]** The "Tracked with Backsight" status-page footer is load-bearing in the GTM referral loop but was absent from the MVP status page (PRD had labeled it a production note).
6. **[NIT]** `product/app/package.json` lacked an `engines` field despite the stated Node prerequisite.

7. **[NIT — jargon]** "PLSS" used in the root README before expansion; "ALTA" never expanded anywhere in the package.

Checks that came back clean (verified, not assumed)

- **Quickstart accuracy:** every command in both READMEs (`install` , `seed` , `dev` , `test` , `build`) exists in `package.json` ; seed is idempotent (85 jobs/12 clients on re-run); `data/` correctly gitignored.
- **Demo script vs code:** all named hooks exist and work — the request-stage job with the "Prior work nearby" badge, 3 same-section historical jobs in T7N R69W S14, the job stuck in review 15 days, 2 overdue jobs, $14,075 unbilled, the `demo-status` share token, all 3 radar "try these" chips, outbox inserts on stage transitions, the Dana/Marcus user switcher, the real-vs-mocked settings panel, and the offline coordinate-grid map fallback.
- **Cross-references:** every relative Markdown link and backticked file path across all 22 docs resolves (checked programmatically; zero broken).
- **Claims hygiene (16 URLs spot-checked):** every "verified by adversarial review" citation traces to an actual `research/*.json` file; unverifiable numbers are consistently labeled estimate/assumption; LTV ($4,760), CAC ($240–320), and penetration arithmetic all check out.
- **Honesty checks:** mocks (auth, email, geocoding, billing, attachments) disclosed in 4 places including in-product; "25,000" appears only as the corrected-from figure; the discredited $30k HVAC figure appears only inside the adversarial autopsy flagged as SEO-farm evidence; pyTRS appears only as prohibited/never-use.

Critic's overall verdict (verbatim)

> **PASS with minor defects.** The package meets the Definition of Done: a stranger can run the demo (verified end-to-end), evaluate the opportunity from unusually honest, citation-traced documents, and continue building via a concrete milestone plan with no credentials or specialist expertise. The one true completeness gap is self-referential — the final completeness review itself is not yet a file in the folder — plus three small accuracy/cosmetic fixes that are each one-line edits. Nothing found contradicts the package's claims; notably, the strictest checks (dead evidence, citation tracing, demo-hook numbers) all passed except the single "~29 vs 25" figure.

Addendum: post-review fixes applied (same day)

All seven defects were fixed immediately after this review and verified with a clean `npm run build` :

1. This document committed at `decisions/FINAL_COMPLETENESS_REVIEW.md` and added to the README folder guide.
2. Demo step 2 corrected to "~25 active jobs".
3. Root README pointer corrected to step 5.
4. Pricing table row repaired.
5. "Tracked with Backsight" footer added to the public status page (`app/status/[token]/page.tsx`), closing the referral-loop demo gap.
6. `"engines": {"node": ">=20"}` added to `package.json` .
7. PLSS expanded at first use in the root README; ALTA expanded at first use in `FINAL_CONCEPT.md` .